Masked, Not Missing: Mid-Depth Retrieval Collapse in T5-Family Encoders

Anonymous ACL submission

Abstract

Practitioners often read intermediate layers of off-the-shelf encoders, check them for anisotropy with mean pairwise cosine, and correct the embedding space post hoc. We test this practice on medieval Latin canon law, where a scholarly edition labels which manuscript copies (witnesses) share a source. In the raw T5 encoders LaTa, PhilTa and mT5-base, 26 of 36 layers rank same-source pairs near chance, although their first and last layers retrieve well. Mean pairwise cosine gives no reliable warning of this collapse, while the share of variance on the first principal component is high at every collapsed layer. The signal is masked, not missing: a projection fit on training embeddings alone brings all 100 layers of our six-encoder panel to AUROC 0.962–0.987. Of the six raw T5-family encoders we test, four collapse: these three and T5-v1.1-base. All four share the T5 v1.1 layout; encoder-only siblings pretrained on the same corpora and two checkpoints with the original T5 layout do not. Removing about three principal directions restores every collapsed layer, while zeroing the ten largest coordinates restores none. After the projection the six encoders route witnesses to their source or to "new" about equally well, but no frozen encoder at its train-selected layer beats a character n-gram reference.

1 Introduction

Off-the-shelf encoders are a common tool for finding reused text in digital-humanities collections (e.g., Manjavacas et al., 2019; Schelb et al., 2026). Practitioners often read an intermediate layer, because intermediate layers can outperform the final one (Skean et al., 2025). They check the space for anisotropy, usually with the mean cosine between embeddings (Ethayarajh, 2019). They then correct it, for example by removing the top principal components (Mu and Viswanath, 2018) or by whitening (Su et al., 2021). This practice assumes that the check warns when a layer retrieves poorly. Mean cosine is already known to be a poor isotropy measure (Rudman et al., 2022); we ask whether it, or any label-free statistic, warns where retrieval fails.

Medieval Latin suits this test. The Carolingian Canon Law project (CCL; Firey, 2009) keys each transcribed manuscript unit to the source it copies, with expert labels that no model shaped. Latin also offers T5 and encoder-only models from the same authors and pretraining corpora. Our testbed holds 1,705 *witnesses*, manuscript copies of 840 sources. We evaluate two tasks: *ranking* pairs of witnesses by cosine, scored by AUROC, and *routing* each witness to its source or to "new". Because witnesses copy their source's wording, a character n-gram reference ranks pairs at 0.987 AUROC, so a retrieval failure points to the representation.

Three raw T5 encoders, trained with no embedding objective, fail at mid-depth (Figure 1). In LaTa, PhilTa and mT5-base (Xue et al., 2021), 26 of 36 layers rank same-source pairs at AUROC 0.50 to 0.66, near chance. Their first and last layers still retrieve. Three embedding-trained encoders, which complete our six-model panel, never collapse. The usual check, mean pairwise cosine, gives no reliable warning in either direction. The share of variance on the first principal component (top-PC share) is high at every collapsed layer: necessary for collapse in our pretrained models, but not sufficient.

The signal is masked, not missing. All-but-the-Top (ABTT; Mu and Viswanath, 2018), a linear projection fit on training embeddings only, removes the mean and the top principal components from every embedding, with no parameter update. It brings all 100 panel layers to AUROC 0.962 to 0.987, the 26 collapsed layers included. Because the projection sees no test label, the ranking it recovers is already present in each collapsed layer.

Among the six raw T5-family encoders we test, the collapse tracks the T5 v1.1 layout: a gated-

GELU feed-forward block (Shazeer, 2020) and untied input and output embeddings (Google Research, 2020). LaBERTa and PhilBERTa, encoder-only siblings of LaTa and PhilTa, never fall below 0.826 and 0.883. T5-v1.1-base collapses on the same Latin text (lowest AUROC 0.489), while the original T5-base (Raffel et al., 2020) never falls below 0.816. T5-base also saw supervised tasks in pretraining, so we add T5-efficient-base (Tay et al., 2022), with T5-base's layout but pretrained on C4 alone, without dropout and on about 30 times fewer tokens; it does not collapse either. Of the six raw T5-family encoders we test, the four that collapse all have this layout. We do not isolate which change matters.

Inside the collapsed layers, about three directions mask the signal. Removing three principal components brings all 26 layers to AUROC 0.914 or above, and mT5-base keeps improving up to ten. Zeroing the ten largest coordinates restores none of them, although the first principal component (the dominant direction) sits mostly on such coordinates. In LaTa, pooling without the comma, the period and </s> restores every collapsed layer; in mT5-base, dropping up to its top 100 token types does not.

This paper makes four contributions. (1) A per-layer retrieval profile of T5 encoder states, the first we know of; it shows a mid-depth collapse that mean cosine misses, top-PC share flags and ABTT reverses (Section 4). (2) Controls showing that the collapse tracks the T5 v1.1 layout: T5-v1.1-base collapses where the original T5-base does not, encoder-only siblings on the same corpora do not, and the high-share middle layers recur on English text (Section 5). (3) A localization of the mask to about three directions, which zeroing the ten largest coordinates does not remove, with a token-level account in LaTa and none in mT5-base (Section 6). (4) A measure of what the projection gains for routing and where it stops, set against a character n-gram reference (Section 7). We will release the frozen testbed, split, evaluation code and character n-gram reference scores. We localize and repair the collapse but do not explain why it tracks the T5 v1.1 layout.

2 Related Work

Anisotropy statistics. Contextual embeddings occupy a narrow cone, usually checked by mean cosine over random pairs (Gao et al., 2019; Ethayarajh, 2019), which is not an appropriate measure of isotropy (Rudman et al., 2022). For pooled output embeddings, the variance share of the largest coordinate or top principal direction predicts how much rank-based similarity gains over cosine (Parupudi, 2026), and a topological signature led by effective rank moderately predicts retrieval (Rottach et al., 2025). Yet geometric statistics do not reliably predict downstream performance (Nastase and Merlo, 2025; Kulkarni et al., 2026). We relate mean cosine and top-PC share to retrieval at every layer (Section 4).

Outlier dimensions and their carrier tokens. One to three rogue dimensions dominate cosine similarity in pretrained transformers (Timkey and van Schijndel, 2021). In BERT they are tied to a few LayerNorm weights and to token frequency (Kovaleva et al., 2021; Puccetti et al., 2022). Clipping them improves mean-pooled sentence embeddings (Luo et al., 2021), though multilingual BERT is anisotropic without them (Rajaee and Pilehvar, 2022). Most BERT outliers sit on [SEP], the period and the comma (Bondarenko et al., 2023), and removing punctuation and special tokens before pooling improves sentence similarity (Fuster Baggetto and Fresno, 2022). Decoder language models carry massive activations on starting and delimiter tokens (Sun et al., 2024). The original T5's encoder carries outliers that grow with depth (Zhao et al., 2025), and outliers persist through fine-tuning (Rudman et al., 2023). On pooled T5 encoder states we compare coordinate zeroing, direction removal and token dropping (Section 6).

Layerwise analysis and post-hoc correction. Prior T5 embedding work reads the final layer (Ni et al., 2022) or the first and last layers averaged (Huang et al., 2021); Godey et al. (2024) trace anisotropy through T5-base's encoder and decoder layers but do not measure retrieval. Large decoder language models dip at mid-depth in retrieval and similarity and recover near the top, while two decoder embedding models stay roughly flat there before rising (Rizwan et al., 2026). Post-hoc corrections remove the mean and top principal components (Mu and Viswanath, 2018), whiten (Su et al., 2021), or down-weight frequent tokens (Arora et al., 2017). Timkey and van Schijndel (2021) apply them at every layer of four models and find that rogue dimensions obscure representational quality; our reading of the collapse as masked follows theirs, and we use ABTT as a probe. Whiten-

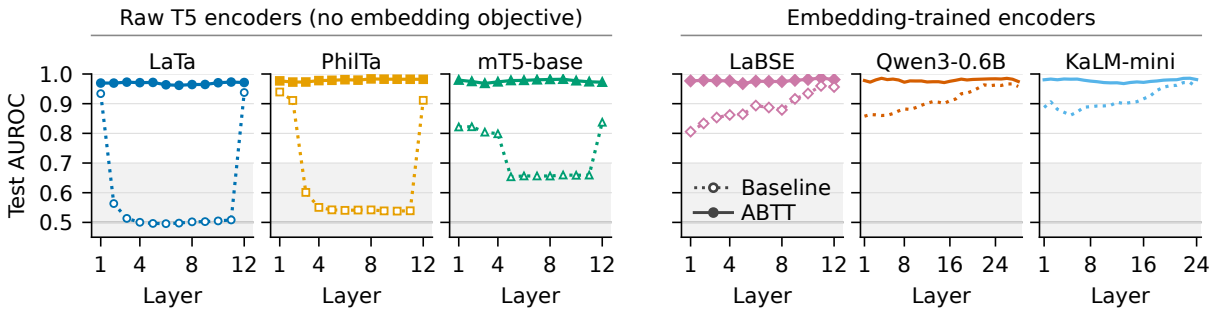


Figure 1: Test ranking AUROC at every layer of the six panel models. Dotted: baseline (mean-pooled hidden states). Solid: ABTT (remove the training mean and the top $D \leq 10$ principal components), fit on training embeddings only. Gray band: AUROC below 0.70, the collapsed range; gray line: chance. The three raw T5 encoders collapse at mid-depth; after ABTT every layer scores 0.962 to 0.987.

ing beats zeroing outlier dimensions in a multilingual encoder (Hämmerl et al., 2023). Centering alone gives embedding models gains under one point (Ren et al., 2026), and isotropy corrections stop helping after fine-tuning (Rajaee and Pilehvar, 2021). For routing, the nearest precedent is out-of-distribution detection in supervised classifiers, where low-variance components carry much of the signal (Kamoi and Kobayashi, 2020; Podolskiy et al., 2021), and nearest-neighbor scores depend on feature geometry (Sun et al., 2022). Lexical methods match or beat dense retrievers on Latin intertextuality (Schelb et al., 2026), while dense encoders reportedly help on paraphrastic reuse (de la Selle, 2026).

3 Experimental Setup

Corpus and labels. The testbed comes from CCL (Firey, 2009), an open-access edition of early medieval, mostly ninth-century, canon-law manuscripts. A *witness* is one manuscript's copy of a source text, kept as one file. Witnesses that copy the same source share a *directory*; the testbed holds 1,705 witnesses in 840 directories. The labels are expert-assigned by the project and checked by collation in its parallel display (Appendix A).

Split and train-only rule. We split the witnesses into 847 training and 858 test witnesses, with 565 and 596 same-source pairs; pairs form only within a split (Appendix B). The layer, routing threshold and ABTT are chosen or fit on the training split. The test split selects nothing.

Tasks and metrics. *Ranking* scores every pair of test witnesses by the cosine of their mean-pooled embeddings. We report AUROC with same-source pairs as positives. *Routing* is open-set: each test witness goes to an existing directory or to "new",

Model	Type	L	$\text{AUROC}_{\text{min}}$ (ℓ)	PC1_{max}	Coll.
Raw T5 encoders, T5 v1.1 layout					
LaTa	T5	12	0.496 (6)	0.952	10
PhilTa	T5	12	0.538 (10)	0.858	9
mT5-base	T5	12	0.654 (5)	1.000	7
T5-v1.1-base	T5	12	0.489 (2)	0.975	10
Raw T5 encoders, original layout					
T5-base	T5	12	0.816 (11)	0.545	0
T5-efficient-base	T5	12	0.736 (11)	0.627	0
Raw encoder-only siblings					
LaBERTa	Enc.	12	0.826 (1)	0.254	0
PhilBERTa	Enc.	12	0.883 (6)	0.162	0
Embedding-trained					
LaBSE	Enc.	12	0.806 (1)	0.524	0
Qwen3-0.6B	Dec.	28	0.858 (1)	0.183	0
KaLM-mini	Dec.	24	0.861 (5)	0.419	0

Table 1: The eleven main-text models, read as mean-pooled hidden states with no correction. Type: T5 encoder (T5), encoder-only (Enc.) or decoder (Dec.; KaLM-mini with bidirectional attention). L: blocks read. $\text{AUROC}_{\text{min}}$: lowest test ranking AUROC over layers, at layer ℓ. PC1_{max}: peak top-PC share of training embeddings. Coll.: collapsed layers (AUROC below 0.70). Only the four raw T5 encoders with the T5 v1.1 layout (gated-GELU feed-forward, untied embeddings) collapse.

and DirAcc@1 is the share of correct decisions (Section 7 gives the procedure).

A *collapsed layer* has *baseline* (uncorrected) test AUROC below 0.70. The cutoff only describes layers; it selects no reported number. Two label-free statistics, computed on the training embeddings, describe each layer. The *mean pairwise cosine* is the usual anisotropy check (Ethayarajh, 2019). The *top-PC share* is the share of variance on the first principal component of the centered embeddings.

Models. Of the eleven models in Table 1, six form the *panel*, analyzed in full over 100 layers. Three are raw T5 encoders with no embedding objective: the Latin LaTa (Riemenschnei-

der and Frank, 2023d), its Ancient Greek, Latin and English sibling PhilTa (Riemenschneider and Frank, 2023a), and mT5-base (Xue et al., 2021). Three are embedding-trained: LaBSE (Feng et al., 2022), Qwen3-Embedding-0.6B (Qwen3-0.6B; Zhang et al., 2025) and KaLM-mini (Hu et al., 2025). The other five are raw: T5-efficient-base and four controls, each matched to another model in Section 5: the encoder-only siblings LaBERTa (Riemenschneider and Frank, 2023c) and PhilBERTa (Riemenschneider and Frank, 2023a), and the English T5 encoders T5-v1.1-base (Google Research, 2020) and the original T5-base (Raffel et al., 2020). We mean-pool the hidden states of every transformer block, dropping only padding and tokens that decode to no text, such as SentencePiece's bare word-boundary piece; special tokens such as </s> and punctuation are kept. Appendix J compares last-token pooling. Table 6 lists all thirteen models, adding Sentence-T5 (Ni et al., 2022) and SPhilBERTa (Riemenschneider and Frank, 2023b).

Post-processing. ABTT (Mu and Viswanath, 2018) subtracts the training mean and removes the top D principal components of the centered training embeddings from every vector. We choose $D \in \{1, 2, 3, 5, 7, 10\}$ per layer by training DirAcc@1. ABTT is fit on training embeddings only and updates no parameter; we also call it *the projection*. We compare SIF frequency weighting (Arora et al., 2017) and reduced-rank whitening (Su et al., 2021) in Appendix K.

4 Collapse by Depth

In LaTa, PhilTa and mT5-base most middle layers rank near chance. Mean cosine gives no reliable warning, but top-PC share is high at every collapsed layer. ABTT restores ranking at every panel layer.

LaTa, PhilTa and mT5-base fall to near chance at mid-depth. Figure 1 gives baseline ranking AUROC at every layer of the six panel models. In LaTa, PhilTa and mT5-base, 26 of the 36 layers collapse, ranking at AUROC 0.50 to 0.66. They are layers 2–11 of LaTa, 3–11 of PhilTa and 5–11 of mT5-base. The lowest remaining panel layer ranks at 0.799 (mT5-base, layer 4), so any cutoff from 0.66 to 0.79 gives the same 26 layers. Training AUROC below 0.70 picks the same 26 layers. The first and last layers still retrieve; LaTa ranks at 0.934 at layer 1 and 0.938 at layer 12. The embedding-trained panel models never rank below 0.806.

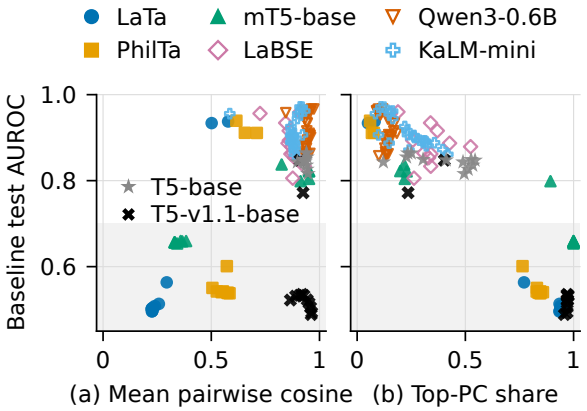


Figure 2: Baseline test AUROC against two label-free statistics of the training embeddings: (a) mean pairwise cosine and (b) top-PC share, the share of variance on the first principal component. One point per layer of the six panel models, T5-base and T5-v1.1-base (filled: raw T5-family; open: embedding-trained). Gray band: AUROC below 0.70, the collapsed range. Collapsed layers occur at both low and high mean cosine. Every collapsed layer has a top-PC share of at least 0.76, but a high share does not imply collapse (mT5-base layer 4: 0.893, AUROC 0.799).

A train-only projection restores ranking at every layer of the six panel models. ABTT (Section 3) lifts all 100 layers of the panel, the 26 collapsed layers included, to AUROC 0.962 to 0.987 (Figure 1, solid lines). Reduced-rank whitening (128 or 256 components), fit on the same training embeddings, ranks within 0.021 of ABTT at every panel layer (Appendix K). The narrow band holds for ranking only. Routing after ABTT still varies by layer, from 75.6 to 90.2 DirAcc@1 (Appendix I).

Mean pairwise cosine gives no reliable warning. Figure 2a plots training-set mean pairwise cosine against baseline test AUROC. Collapsed panel layers have a lower mean cosine (0.22 to 0.58) than any embedding-trained panel layer (0.59 to 0.97). But LaTa's healthy layers 1 and 12 sit in the collapsed range (0.50 and 0.58). T5-v1.1-base, which also collapses (Table 1; Section 5), sits at 0.87 to 0.96 at every layer, collapsed or not. Neither a high nor a low mean cosine marks a collapsed layer.

Top-PC share is high at every collapsed layer of the pretrained models. Top-PC share needs no labels either (Figure 2b). It is at least 0.76 at every collapsed layer of the panel and of T5-v1.1-base. In the three embedding-trained panel models it peaks at 0.524 (LaBSE). A high share is necessary for collapse in our pretrained models but

not sufficient: of the 88 healthy layers in Figure 2, only mT5-base layer 4 reaches 0.76 (0.893, AUROC 0.799). Sentence-T5 (Ni et al., 2022), an embedding-trained T5 encoder, has a share of 0.76 or more at layers 2–9 (peak 0.94) yet ranks at 0.800 or above at every layer (Table 6). Among collapsed layers a higher share does not predict a lower AUROC (Appendix C).

5 Collapse Across Models

Of the eleven models in Table 1, four collapse: LaTa, PhilTa, mT5-base and T5-v1.1-base. All four have the *T5 v1.1 layout*: a gated-GELU feed-forward (Shazeer, 2020) and untied input and output embeddings (Google Research, 2020) (Table 6). None of the other seven collapses at any layer. T5-base and T5-efficient-base keep a ReLU feed-forward and tied embeddings.

Encoder-only siblings pretrained on the same corpora do not collapse. LaBERTa and PhilBERTa share authors and pretraining corpora with LaTa and PhilTa (Riemenschneider and Frank, 2023d,c,a) and, like them, have no embedding objective. LaTa and PhilTa fall to 0.496 and 0.538 AUROC; the siblings never fall below 0.826 (LaBERTa) and 0.883 (PhilBERTa). Their top-PC share peaks at 0.254 and 0.162 (vs. 0.952 and 0.858 in LaTa and PhilTa). Shared pretraining text and a missing embedding objective do not produce the collapse on their own. Architecture, layout included, pretraining objective and tokenizer vary together within each pair, so the pairs cannot say which matters (Appendix D).

T5-v1.1-base collapses; T5-base and T5-efficient-base, with the original layout, do not. T5-v1.1-base and T5-base share their tokenizer, the span-corruption objective and C4 pretraining text (Raffel et al., 2020; Google Research, 2020). On Latin text, T5-v1.1-base falls to 0.489 at layer 2, and its layers 2 to 11 collapse. T5-base stays at 0.816 or above at every layer, with top-PC share peaking at 0.545. The two differ in layout and in pretraining mix: T5-base also saw supervised tasks (Raffel et al., 2020). To separate layout from that mix we add T5-efficient-base (Tay et al., 2022): T5-base's layout, but pretrained like T5-v1.1-base on C4 alone and without dropout, and on about 30 times fewer tokens than either. At that budget it does not collapse: its lowest AUROC (0.736) and its peak top-PC share (0.627) both fall at layer 11 (Table 8). On the four controls in Table 1, ABTT lifts every layer to AUROC 0.927 or above, the ten collapsed layers of T5-v1.1-base included, but eight of the 48 (seven in T5-v1.1-base) stay below the panel's band (0.962–0.987; Table 9). Among the six raw T5-family encoders we test, the collapse tracks the T5 v1.1 layout; we do not isolate which change is responsible.

In three models with the T5 v1.1 layout, high top-PC share in the middle layers appears on English text too. T5-v1.1-base was pretrained on English alone, so its high-share layers on Latin could reflect a language mismatch. We read mT5-base, PhilTa and T5-v1.1-base on 847 English passages of US court opinions (Harvard Law School Library Innovation Lab, 2024; Kandpal et al., 2025), paired with the Latin training witnesses and matched in mT5 token length. On English, in all three models, layers 4 to 11 have a top-PC share of at least 0.82 and the first and last layers stay at or below 0.34 (Table 10, Figure 7). At layers 2 to 11 of T5-v1.1-base, top-PC share is 0.931 to 0.950 on English and 0.958 to 0.975 on Latin. The check covers geometry only; without labels, it says nothing about English retrieval.

In LaTa, contrastive fine-tuning leaves the middle layers collapsed. In a small test, we fine-tune LaTa on 499 training pairs, adding the contrastive embedding objective it lacks (Appendix H). Fine-tuning lifts the last layer from 0.938 to 0.984 AUROC but leaves layers 2 to 11 at 0.50 to 0.57. Peak top-PC share barely moves (0.952 before, 0.945 after; Table 11). Hämmerl et al. (2023) report the same split: their sentence-tuned multilingual encoder keeps a large outlier dimension at a middle layer but none at its output layer.

6 Localizing the Collapse

In the collapsed layers about three directions mask the retrieval signal. Zeroing the ten largest coordinates does not remove this mask, and a few token types feed it in LaTa but not in mT5-base.

We run each test at the 26 collapsed layers, fit every intervention on training embeddings, and read test ranking AUROC. A layer counts as restored at AUROC 0.90 or above. Appendix E gives further tables and controls.

Zeroing the ten largest coordinates restores no collapsed layer. In pretrained transformers a few outlier coordinates dominate cosine similarity

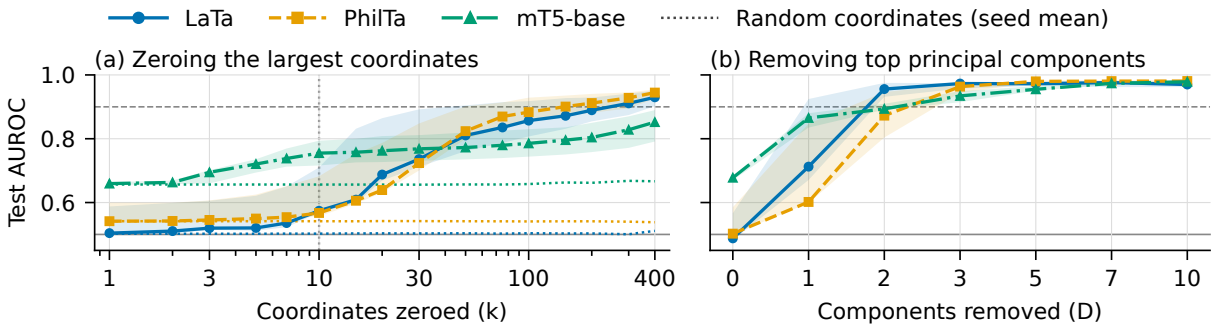


Figure 3: Test ranking AUROC at the 26 collapsed layers of LaTa (10 layers), PhilTa (9) and mT5-base (7); lines: median over a model's collapsed layers; bands: their range; dashed line: AUROC 0.90. (a) Zeroing the k coordinates of largest mean absolute value; dotted: k random coordinates (mean of five draws). Up to $k = 10$ (vertical line) the best layer reaches 0.794. (b) Removing the top D principal components ($D = 0$: centering only; ticks follow the selection grid, not evenly spaced in D). From $D = 3$ every layer passes 0.90. All fits use training embeddings.

(Timkey and van Schijndel, 2021). Zeroing them improves pooled similarity in pretrained encoders (Luo et al., 2021), and whitening does better still (Hämmerl et al., 2023). In our encoders, zeroing up to ten coordinates, ranked on training embeddings by variance or by mean absolute value, restores none of the 26, and the best layer reaches 0.794 (Figure 3a). The dominant direction loads mostly on these coordinates: the ten of highest variance carry a median 92% of its squared loading. Yet after zeroing them, its remainder still holds a median 72% of the remaining variance. Zeroing far more coordinates does restore LaTa and PhilTa. Ranked by variance, recovery starts at 50 coordinates, and their 19 collapsed layers all recover by 300 (by 400 under the mean-absolute-value ranking of Figure 3a). No mT5-base layer recovers even at 400. The dominant direction loads mostly on a few coordinates but acts beyond them, which extends the outlier-dimension account.

Removing three directions restores every collapsed layer. ABTT removes whole directions instead of coordinates, so we vary the number D of principal components it removes (Figure 3b). Centering alone ($D = 0$) does not help: the median AUROC over the collapsed layers falls from 0.541 to 0.503. At these layers the first component holds a median 94% of the centered training variance. It is not mainly a passage-length direction: $|\rho| \leq 0.44$ with log token count at all 26 layers. Removing it alone recovers a median 45% of the AUROC gain from removing ten components, and 80% or more at only one of the 26 layers. Removing three components brings all 26 layers to 0.914 or above. mT5-base keeps improving up to ten components (median 0.934 at three, 0.978 at ten). The ten removed components rank at 0.489, the next ten at 0.919, and the retained subspace at 0.977 (Table 14). At these layers the signal lies outside the removed directions.

In LaTa, three token types feed the removed directions. A pooled vector is the mean of its token vectors, so a few token types with large values along the removed directions could dominate a witness's scores on those directions. We rank token types by their share of the variance of training-witness scores on the top three components, then pool without the top m types, for m up to 100. We ask whether some m restores the collapsed layers. In LaTa, $m = 3$ does. Pooling without the comma, the period and </s> drops 11% of the test tokens and restores all 10 collapsed layers (median 0.919). Random token types of the same training share leave the median at 0.500. The tokens are carriers, not a cause, since LaBERTa also pools commas, periods and </s> and does not collapse (Section 5). PhilTa needs more types: dropping its top 30, about a third of its tokens, restores all 9 of its collapsed layers on test passages (settings chosen per layer on training AUROC restore every collapsed layer of both models, medians 0.926 and 0.928).

In mT5-base, dropping up to its top 100 token types restores no collapsed layer. Dropping up to 100 token types, ranked the same way, restores none of its 7 collapsed layers, the best reaching 0.748. mT5-base also resists coordinate zeroing, yet removing three to ten directions repairs it. Its mask sits neither in a few coordinates nor in a few token types, and we have no token-level account.

7 Routing and Limits

At their train-selected layers after ABTT, the six frozen encoders route about equally well, but none beats a character n-gram reference.

Of the 858 test witnesses, 535 have a same-source partner in test (*existing*) and 323 do not (*new*). Each test directory scores a witness by the highest cosine between the witness and any other member. A pseudo-directory `NEW` scores a fixed threshold τ, chosen to maximize training F1 at separating same-source from different-source pairs. The witness goes to the highest-scoring directory. DirAcc@1 counts a decision as correct if it names the labeled directory, or `NEW` for a new witness. Assignment accuracy scores only the existing-versus-new decision; always answering existing scores 62.4%. Each headline number reads a model at its *train-selected* layer: the one of highest training AUROC for ranking, or of highest training DirAcc@1 for routing (Table 21). The task is lexically solvable by design, since the witnesses of one source copy its wording. Our *character n-gram reference*, TF-IDF over character 3–5-grams fit on training witnesses, reaches 0.987 AUROC and 89.9 DirAcc@1 (Table 20).

At train-selected layers, routing after the projection converges across the six encoders. At the train-selected layers of Table 2, the DirAcc@1 spread across the six encoders falls from 39.3 points without correction (95% CI 34.6 to 43.7) to 3.3 after ABTT (1.8 to 6.0). Uncorrected, they run from 46.6 (mT5-base) to 85.9 (KaLM-mini); after ABTT, from 86.1 to 89.4. Every encoder's routing gain has a bootstrap interval excluding zero, from 3.5 points for KaLM-mini (1.0 to 6.0) to 41.8 for mT5-base (Table 17).

For the embedding-trained encoders, the routing gain does not depend on the threshold. LaBSE, Qwen3-0.6B and KaLM-mini already rank well without correction (AUROC 0.956 to 0.972), so their routing gain cannot come from a ranking repair. ABTT still raises their DirAcc@1 by 7.0, 9.1 and 3.5 points. Because the gain could come from a better-placed threshold τ, we also score the existing-versus-new decision without one, as the AUROC of each test witness's best-match cosine, existing against new. ABTT raises it by 0.037 for LaBSE, 0.055 for Qwen3-0.6B and 0.027 for KaLM-mini, and every paired interval excludes zero (Appendix G; per-cell values in Table 18).

	Ranking AUROC		Routing DirAcc@1	
Model	Base	ABTT	Base	ABTT
LaTa	0.938 [.910,.963]	0.971$^{\downarrow}$ [.948,.991]	72.1 [68.4,75.8]	86.1$^{\downarrow}$ [83.2,88.9]
PhilTa	0.939 [.918,.958]	0.983 [.969,.993]	69.3 [65.2,73.5]	88.3 [85.6,90.9]
mT5-base	0.838 [.810,.866]	0.975 [.954,.992]	46.6 [42.5,51.0]	88.5 [85.7,91.2]
LaBSE	0.956 [.932,.977]	0.987 [.977,.995]	81.5 [78.1,84.7]	88.5 [85.8,90.9]
Qwen3-0.6B	0.966 [.950,.981]	0.973$^{\downarrow}$ [.948,.990]	80.3 [76.9,83.6]	89.4 [86.7,92.0]
KaLM-mini	0.972 [.955,.986]	0.981 [.963,.993]	85.9 [82.9,88.7]	89.4 [86.7,91.9]
Spread	0.134	0.016	39.3	3.3
Fine-tuned; Base = fine-tuned, ABTT = fine-tuned + ABTT				
LaTa	0.984	0.970	81.6	85.2
Qwen3-0.6B	0.996	0.994	90.8	91.4
KaLM-mini	0.997	0.994	91.7	92.5
Char. n-grams	0.987 [.976,.996]		89.9 [87.3,92.3]	

Table 2: Ranking (test AUROC) and routing (test DirAcc@1, %) without correction (Base) and after ABTT, each at its train-selected layer (Table 21); ABTT is fit on training embeddings only. Fine-tuned: contrastive fine-tuning on 499 training pairs (intervals in Table 16). Char. n-grams: TF-IDF over character 3–5-grams. Brackets: 95% bootstrap intervals over the 514 test directories. Spread: largest minus smallest printed value across the six encoders. $\downarrow$: below the n-gram reference, paired interval excluding zero (Table 17); overlapping brackets do not imply a tie.

ABTT's gain in assignment accuracy also holds at each setting's best test threshold and under a finer threshold grid (Appendix F). For these three encoders, centering alone ($D = 0$) scores 12 to 26 points lower in assignment accuracy than ABTT at the same layer (Table 17). We do not test what in the removed components produces this gain.

No frozen encoder at its train-selected layer beats a character n-gram reference. The best ABTT results in Table 2 tie the reference: LaBSE ranks at 0.987, and Qwen3-0.6B and KaLM-mini route at 89.4, with every difference interval including zero. Two ranking cells fall below it with intervals excluding zero, LaTa and Qwen3-0.6B (by 0.016 and 0.014 AUROC); mT5-base falls 0.012 below, its interval touching zero (0.000 to 0.027). LaTa's routing cell falls 3.7 points below it (Table 17). Our claims thus concern where the signal sits in the encoders and what masks it.

Fine-tuning Qwen3-0.6B and KaLM-mini adds about two routing points over the projection. As a supervised reference, we fine-tune LaTa,

Qwen3-0.6B and KaLM-mini contrastively on 499 training pairs (Appendix H). For routing, fine-tuning plus ABTT beats zero-shot ABTT by 1.8 points for Qwen3-0.6B (92.3 against 90.5) and by 2.1 for KaLM-mini (93.0 against 90.9); on LaTa the two tie at 87.7. These are means over five re-seedings, each a new random draw of which test witnesses are queries and which are the references they are routed against (Table 19). In the single-split cells of Table 2 the gains are 2.0 and 3.1 points (only KaLM-mini's interval excludes zero), and LaTa falls 0.9. These references are flattered: 206 of the 535 existing test witnesses (38.5%) sit in a directory that supplied a training pair. Fine-tuned Qwen3-0.6B and KaLM-mini rank above the n-gram reference in point estimate (0.996 and 0.997 AUROC). After fine-tuning, ABTT no longer helps ranking: AUROC falls slightly in point estimate in all three models (by 0.002 to 0.014; every interval includes zero), in line with Rajaee and Pilehvar (2021).

8 Discussion

Before reading an intermediate layer of a frozen encoder, fit a projection on training embeddings. In the six panel models it scored above the baseline in point estimate at all 100 layers in ranking and routing (Appendix I), which makes it cheap insurance. Routing after it still varies by layer, so the layer still matters. To screen uncorrected states, check top-PC share, not mean pairwise cosine. In our models mean cosine gave no reliable warning in either direction, while every collapsed layer of the pretrained models had a top-PC share of at least 0.76 (Section 4). The 0.76 is the lowest share we observed at a collapsed layer of a pretrained checkpoint, not a cutoff validated on held-out models. A high share is necessary for collapse here but not sufficient, and does not order layers inside the collapsed regime. Choosing among layers still needs a train-only task signal, such as training AUROC.

Why models with the T5 v1.1 layout collapse remains open. Gated feed-forward blocks are reported to amplify outlier activations in decoder language models (Fishman et al., 2025; Yang et al., 2025), but the original T5 also carries large outliers (Zhao et al., 2025). Our localization names carriers in LaTa but no cause. Concurrently, Mikkelsen (2026) reports a mid-depth retrieval trough on clinical text; retrieval-trained encoders recover in the last layers, and encoders without a retrieval objective recover only partly. In our models, by contrast, neither the embedding-trained encoders nor the encoder-only siblings collapse, and it tests no T5 model.

For the editor of a text collection, ABTT makes the choice among these six encoders matter little (Section 7; maps in Appendix L). The existing-versus-new decision is still wrong for about one test witness in ten (assignment accuracy 88.5 to 91.7; Table 16). The final call stays with the scholar. Because the task is lexically solvable, a practical tool can start from character n-grams; dense encoders are reported to help where reuse is paraphrastic (de la Selle, 2026), a case we do not test.

Limitations

Four factors are uncontrolled. (i) We use one labeled corpus in one language and genre, early medieval Latin canon law; its 1,705 witnesses are many for this field but few by current NLP standards. (ii) Within each sibling pair, architecture, pretraining objective and tokenizer vary together, so the pairs do not say which of them matters. That LaTa and LaBERTa share authors and pretraining corpora rests on their model cards (Riemenschneider and Frank, 2023d,c). (iii) The T5 v1.1 changes (gated-GELU feed-forward with a narrower inner width, untied embeddings) vary together, so we do not say which of them matters. T5-efficient-base has the original layout without T5-base's supervised pretraining, but it is a single checkpoint pretrained on about 30 times fewer tokens; it separates layout from pretraining mix only at that budget. (iv) The token ablation accounts for the masking directions in LaTa with three token types and in PhilTa only with 30, about a third of its tokens; in mT5-base dropping up to its top 100 token types restores no collapsed layer, so its mask has no token-level account.

The scope of three results is narrow. We claim the narrow band for the six-model panel only: on the four controls and T5-efficient-base ABTT lifts every layer to AUROC 0.927 or above, but nine of their 60 layers, seven of them in T5-v1.1-base, stay below the panel's band (0.962 to 0.987). The English check measures geometry only; we have no labels for English retrieval. The task is lexically solvable by design, and a character n-gram reference ties the best encoders after ABTT, so we do not claim that encoders beat surface overlap on this corpus.

The fine-tuning test is small (499 pairs, at most eight epochs), and its references are flattered by directory overlap with the training pairs (Section 7). Our intervals resample test directories with the layer, threshold and ABTT fit held fixed; they cover test sampling and not the choice of split. ABTT needs enough text to estimate principal components, so smaller archives need separate study.

The labels are expert-assigned, but by one project. Its director or supervised graduate students assigned the keys, and verification is by collation in the project's parallel display, not by independent double annotation. No inter-annotator agreement figure therefore exists for this testbed. Collation needs at least two witnesses of a key, and 545 of the 840 directories hold only one in this testbed.

Ethical Considerations

The testbed holds early medieval legal texts, transcribed from manuscripts and early printings, and contains no personal data. The Carolingian Canon Law project publishes its material under a Creative Commons Attribution-NonCommercial-ShareAlike 4.0 license (CCL, 2026a). The project has confirmed in writing that the license covers the transcriptions, annotations and markup in the exports we use and that, in the view of the project's Advisory Board, attribution to the project suffices; the project's interface credits the transcriber and proofreader of each unit. The project's corpus keeps growing and being corrected, so the testbed is a dated snapshot (version 1), fixed by a sha256 manifest of its 1,705 labeled files, and every number in this paper comes from it. We will release it under the same license, in coordination with the project's periodic versioned exports, and later versions will follow those exports. The intended use is research on retrieval and on assisted source identification: the system ranks candidates for an expert to judge and does not identify sources on its own. The English control sample consists of public-domain US court opinions from the Caselaw Access Project (Harvard Law School Library Innovation Lab, 2024), which name litigants; we use it only for label-free geometry statistics and do not release it. We used AI assistants for writing and checking code; the authors verified all code and numbers.

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A Corpus Provenance and Derivation

This appendix records where the testbed text comes from and how our files were derived from it, in the

detail that Section 3 leaves out.

The project. CCL (Firey, 2009) is an open-access digital edition founded in 2009; Eichbauer (2014) reviews it.[1] Its texts are early medieval canon-law manuscripts, most of them ninth-century and never edited to modern standards. Each manuscript is identified by its siglum (Place-Library-Shelfmark), and a witness is one manuscript's copy of a source text, transcribed as one unit (one canon or excerpt).

Transcription. Transcription is manual, from manuscript images or, for some texts, eighteenth-century printings, with one exception: HTR has been attempted on one manuscript (Paris, BnF, lat. 1454; 142 of the 1,705 witnesses), and its transcription is still in proofreading. Before 2019 transcribers worked from purchased microfilm into word-processed files, which CCL contributors converted to TEI-P5 (the Text Encoding Initiative's XML standard for scholarly editions). Since 2019 they transcribe in the CCL Transcription Desk beside a IIIF image of the folio (a zoomable web-served scan of the manuscript leaf); an approved proofreader on the Proofreading Desk then checks the text against the image to the letter and adds TEI markup (CCL, 2026b). That proofreading stage belongs to the post-2019 workflow. The interface credits the transcriber and proofreader of each unit (CCL, 2026a).

Identifiers. The project has two identifier systems, both designed at its launch in 2009 by its director in consultation with a mathematician, a software developer and a second scholar of early medieval canon law. The first is a location identifier. Every manuscript is a unique object, so every unit has exactly one place in one manuscript, and the identifier records it as the manuscript siglum, the folium, and the unit's position on that folium. Our file names are these identifiers. The second is the source key. Many units in different manuscripts copy the same prescription issued by the same authority, and the key names that common source. It encodes the type of source (a church council, a papal decretal, a royal capitulary, a theologian, a penitential), the point of origin (a particular council, pope or author), the date of origin, and the item's reference in modern scholarship, such as a canon, letter or register number. The key is designed so that a researcher can find the modern authoritative edition of the source where one exists.

Labels. CCL keys a unit to its original source, such as a fourth-century council or a papal decretal, once that source is identified, and our directories reuse that key as supplied. In a label, a four-letter code carries the type of source and its point of origin, followed by the year and the canon or chapter number, as in `CTOU.567.16`; some labels carry a trailing annotation, as in `CTOU.567.16 (15)` or `CARL.501?.18`. A label counts as key-named when it begins with a four-letter code, a period, a year and a period, whatever follows; by this rule 689 of the 840 directory labels are source keys. The other 151 are modern edition citations, the Apostolic Canons (`Can.apost.`N), references to other collections such as the Capitula Martini, biblical references, and a handful of unidentified sources; each unidentified source is a singleton directory and contributes no positive pairs.

Assignment and verification. Keys were assigned by the project's director, a specialist in early medieval canon law, or by graduate students working under the director's supervision, who had enough Latin to check a manuscript version against the modern authoritative version of the source. Verification, and occasional correction, runs through the project's parallel display, which sets side by side all units that share a key and shows how far their wording agrees. A misidentified unit stands out there as a text that cannot be a version of the others. This is verification by collation (see Limitations). The project supplied two label corrections during this study, and both are logged with the frozen testbed. We assigned no label ourselves and changed only those two. What remains unidentified is large: across the several hundred manuscripts that contain the project's texts, thousands of units still carry no key, and they are the queries that routing is meant to serve.

Derivation. We hold the witnesses as plain text derived from CCL TEI-P5 exports. The rule is one file per unit: the directory is the source key from the unit's `scholSource` division, which is stripped from the text so that no file carries its own label; the text is the manuscript rubric followed by the canon text, with inline markup flattened

[1] Founding year, pre-2019 workflow, proofreading standard, the one manuscript transcribed by handwritten text recognition (HTR), the design of the identifiers, the assignment and verification of keys, and the extent of unidentified material: unpublished documentation and correspondence supplied by the project, 2026.

(editorial notes kept, deletions dropped) and layout whitespace collapsed; a unit without a source key receives no directory. The rule is scripted. On a current export of one manuscript (Paris, BnF, lat. 2123), it reproduces all 279 units present in both the export and the corpus character for character after whitespace normalization, and places every one in the same directory.

B Experimental Setup Details

This appendix gives the parts of the setup that Sections 3 and 7 leave out: how the split treats directories of each size, the further layer statistics, the SIF weighting, and how the lexical reference fares by degree of overlap.

Split. Singleton directories (545 of 840) are divided file by file between the two splits. Doubletons go whole to one side. Larger directories are divided internally, so that every size class yields training positives. Pairs form only within a split.

Further layer statistics. Let $p_i = \lambda_i / \sum_j \lambda_j$ for the covariance eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots$ of the centered training embeddings; the top-PC share is p_1. The *effective rank* is $\exp(-\sum_i p_i \log p_i)$, the entropy effective rank of Roy and Vetterli (2007) applied to the covariance eigenvalues. It is 1 when one direction holds all the variance. The *cosine gap* g is the mean same-directory test cosine minus the mean different-directory test cosine. The *standardized gap* g/s divides it by the standard deviation s of all pairwise test cosines, which puts models with different cosine spreads on one scale. The two gaps need labels; the effective rank, like the top-PC share and the mean pairwise cosine, does not. Our mean pairwise cosine averages over all pairs of pooled passages, an adaptation of the word-level statistic of Ethayarajh (2019), which uses random pairs.

SIF. SIF (Arora et al., 2017) reweights each token by $a/(a + p(w))$ before pooling, with $a = 10^{-3}$ and $p(w)$ estimated on training texts. It gives special tokens, such as the end-of-sequence token `</s>`, zero weight. We use only this weighting step of SIF, not its common-component removal. SIF+ABTT applies both; ABTT-only marks ABTT alone where we contrast the two.

Software. We extract hidden states with PyTorch (Paszke et al., 2019) and Hugging Face Transformers (Wolf et al., 2020), fit whitening and the character TF-IDF reference with scikit-learn (Pedregosa et al., 2011), and compute principal components and the remaining statistics with NumPy (Harris et al., 2020). BM25 is our own implementation.

Lexical reference by overlap. We cut the positive test pairs into terciles of n-gram overlap, using the TF-IDF score itself. In every tercile, the character n-gram reference ranks a witness's partner first at least as often as every baseline and ABTT encoder configuration at its train-selected layer. Table 20 gives all three lexical scorers.

C Collapse by Depth: Further Diagnostics

This appendix adds detail to the four findings of Section 4. It gives the geometry of each panel model over its layers, the correlations behind the statement that top-PC share marks the collapsed regime without ordering layers inside it, two label-based gaps, the spread of cosines at the collapsed layers, the effect of SIF there, and the score distributions before and after ABTT. Every statistic is computed on baseline mean-pooled hidden states unless a paragraph says otherwise.

Geometry by model. Table 3 summarizes each panel model over its layers. At the 26 collapsed layers, effective rank lies between 1.0 and 4.6; in the embedding-trained models it ranges from 16 to 150. At mT5-base layers 5–11 one direction holds all the variance to three decimals. The table reads mean cosine on test passages, while Figure 2 reads it on training passages. On test passages the collapsed layers span 0.23 to 0.62, so the PhilTa layers overlap the bottom of the embedding-trained range. ABTT with ten components, fit on training vectors, removes the concentration on held-out passages too: at the collapsed layers, test top-PC share falls to 0.03–0.05 and effective rank rises to 148–173. Table 5 shows this at the most anisotropic layer of each T5 encoder.

Correlations with ranking. Table 4 correlates each statistic with baseline AUROC, one observation per model-layer. Across all 100 panel layers, top-PC share and effective rank reach Spearman -0.77 and $+0.77$. Both values are inflated by the two clusters of collapsed and healthy layers, and layers within a model are not independent. Within the 26 collapsed layers, top-PC share carries no ordering (Spearman $+0.24$, $p = 0.23$). Within the 74 other layers it orders them only moderately

			Geometry (baseline)				Baseline test AUROC		
Model	**Type**	L	Mean cos.	Cos. SD	Max PC1	Min rank	ℓ^*	at ℓ^*	Min (layer)
LaTa	T5 raw	12	0.23–0.58	0.10–0.63	0.952	1.36	4	0.500	0.496 (6)
PhilTa	T5 raw	12	0.53–0.71	0.09–0.46	0.858	1.81	6	0.541	0.538 (10)
mT5-base	T5 raw	12	0.30–0.95	0.05–0.76	1.000	1.00	5	0.654	0.654 (5)
LaBSE	BERT emb.	12	0.73–0.93	0.04–0.10	0.524	16.19	8	0.879	0.806 (1)
Qwen3-0.6B	Decoder emb.	28	0.85–0.97	0.01–0.04	0.183	71.60	16	0.910	0.858 (1)
KaLM-mini	Decoder emb.	24	0.58–0.93	0.02–0.07	0.419	23.83	5	0.861	0.861 (5)

Table 3: Geometry and retrieval regimes of the six encoders under baseline mean pooling (no post-hoc correction). Type: T5 raw is a T5 encoder pretrained without a sentence-embedding objective; emb. marks a BERT encoder or a decoder trained with one. L: number of transformer layers. Mean cos. and Cos. SD: range over layers of the mean and the standard deviation of the cosine over all pairs of test passages. Max PC1: peak top-PC share over layers, the largest share of variance on the first principal component of the centered training embeddings. Min rank: smallest entropy effective rank of the same centered training embeddings over layers. All four geometry columns are label-free. ℓ^*: layer where the PC1 share peaks. The last two columns give the baseline test pairwise AUROC of cosine at ℓ^* and its minimum over layers, with the layer in parentheses. Layer 1 is the first transformer block.

	All (100)		**T5** (36)		**Non-T5** (64)		**LOMO** ρ	
Statistic	r	ρ	r	ρ	r	ρ	min	max
Target: baseline AUROC								
Mean pairwise cosine	0.83	0.55	0.61	0.67	−0.06	0.10	0.42	0.67
Top-PC share	−0.92	−0.77	−0.86	−0.50	−0.47	−0.43	−0.83	−0.70
Effective rank	0.80	0.77	0.87	0.53	0.48	0.40	0.70	0.85
Raw cosine gap	0.24	0.22	0.63	0.72	0.15	−0.03	−0.10	0.48
Standardized gap	0.86	0.97	0.87	0.96	0.89	0.93	0.96	0.98
Target: ABTT AUROC gain								
Mean pairwise cosine	−0.83	−0.54	−0.60	−0.66	0.07	−0.07	−0.66	−0.41
Top-PC share	0.92	0.76	0.86	0.50	0.45	0.41	0.69	0.83
Effective rank	−0.80	−0.76	−0.88	−0.52	−0.44	−0.37	−0.84	−0.68
Raw cosine gap	−0.23	−0.23	−0.61	−0.71	−0.16	0.00	−0.48	0.09
Standardized gap	−0.85	−0.97	−0.87	−0.95	−0.88	−0.92	−0.98	−0.96

Table 4: Correlation between per-layer geometry and retrieval, one observation per model-layer, baseline mean pooling. r: Pearson; ρ: Spearman. Columns pool all 100 model-layers, the 36 layers of the three T5 encoders, or the 64 layers of the three embedding-trained models; LOMO gives the minimum and maximum Spearman over the six leave-one-model-out subsets. Mean pairwise cosine is on test passages; top-PC share and effective rank are on centered training embeddings. These three statistics are label-free. The two gaps are label-based separation indices. Raw cosine gap is the mean test cosine of same-directory pairs minus that of different-directory pairs; standardized gap divides it by the standard deviation of all test pairwise cosines at that layer. Targets are the baseline test AUROC and the ABTT gain (AUROC with ABTT at its train-selected D minus baseline AUROC). The standardized gap is a signal-to-noise ratio of the kind AUROC summarizes, so it tracks AUROC by construction. ABTT AUROC varies little (0.962–0.987), so the gain correlations mirror the baseline ones. The pooled Pearson and Spearman values are both inflated by the two clusters of collapsed and healthy layers, and layers within a model are not independent.

(-0.49). Mean pairwise cosine correlates positively with AUROC within the three panel T5 encoders ($+0.67$ on test passages, $+0.72$ on training passages). This pattern does not carry over to T5-v1.1-base, whose collapsed layers sit at mean cosines of 0.87 to 0.96 on training passages, around the 0.91 and 0.92 of its two healthy layers; within that model the correlation is -0.64. The lower block of the table adds no separate evidence. ABTT lifts every panel layer into a narrow band (0.962 to 0.987), so its gain is nearly a constant minus the baseline, and the gain correlations mirror the baseline ones. The absence of a reliable ordering agrees with reports that geometric statistics do not reliably predict downstream performance (Kulkarni et al., 2026; Nastase and Merlo, 2025).

Raw and standardized gap. The raw cosine gap, the margin a routing threshold works with, tracks ranking poorly: across the panel it correlates with AUROC at Spearman $+0.22$ (Figure 4). The standardized gap reaches $+0.97$, but it is a label-based signal-to-noise ratio close to the d' from which a binormal model computes AUROC. It tracks AUROC by construction, so we do not count it as a predictor. It explains one pattern in mT5-base (Figure 5). There the raw gap rises from about 0.03 to

Model	Most anisotropic layer	PC1 before	PC1 after	Eff. rank before→after
LaTa	8	0.956	0.036	1.34→168.40
PhilTa	6	0.862	0.039	1.83→155.02
mT5-base	5	1.000	0.044	1.00→151.74

Table 5: Most anisotropic layers, by top-PC share on the test split, for the three T5 encoders. PC1: top-PC share. PC1 and effective-rank values compare baseline geometry with ABTT-D10 at the most anisotropic layer, on the test split. Table 3 and the main text read top-PC share on the training split, where LaTa peaks at layer 4 instead of 8; LaTa's training curve is flat, with a top-PC share of 0.93 to 0.95 over layers 3–11.

0.21–0.25 exactly where ranking fails. Over the same layers the standard deviation of all pairwise cosines grows from 0.05–0.07 to about 0.75. Divided by that spread, the gap falls to 0.28–0.33, below the 0.41–0.49 of layers 1–3 and the 0.78 of layer 12. The collapsed layers thus spread all cosines widely, and same-source pairs gain little relative to that spread. mT5-base layer 4 is an exception: its standardized gap is the lowest of the model (0.19), yet it ranks at 0.799.

Cosine spread at the collapsed layers. At the collapsed layers, test cosines spread across both signs. At mT5-base's collapsed layers the 5th percentile lies between -0.83 and -0.77 and the median between 0.84 and 0.90, with a standard deviation of about 0.75. At LaTa's collapsed layers the 5th percentile lies between -0.50 and -0.26, and the standard deviation reaches 0.63. In the embedding-trained models the standard deviation stays between 0.007 and 0.10 at every layer. Pooled passages at a collapsed layer therefore lie on both sides of one dominant direction, which Section 6 examines.

SIF at the collapsed layers. SIF pooling (Arora et al., 2017) down-weights frequent tokens and drops special tokens (Appendix B). At the collapsed layers it recovers about 80 percent of the ABTT gain in LaTa and PhilTa, and under 10 percent in mT5-base. This split agrees with the token ablation of Section 6: in LaTa, mean pooling without the comma, the period and </s> restores every collapsed layer, while in mT5-base dropping up to its top 100 token types, ranked on training passages, does not. SIF stays below ABTT at every collapsed layer, and its full sweeps are in Appendix K.

Score distributions. Figure 6 restates the collapse and the repair as score distributions rather than as summary metrics. At PhilTa's collapsed layer 8 the cosines of same-source and different-source pairs overlap almost completely, and ABTT separates them.

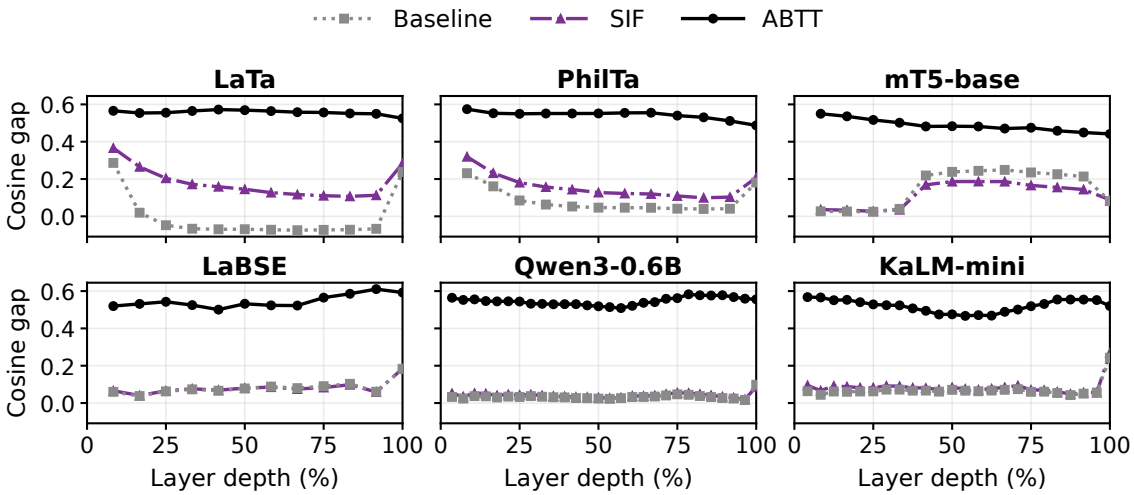


Figure 4: Raw cosine gap against normalized layer depth for the six panel models, test split. The gap is the mean cosine of same-directory pairs minus that of different-directory pairs, the margin a threshold works with. Gray dotted: baseline mean pooling. Purple dash-dot: SIF-weighted pooling. Black solid: ABTT, with D chosen per layer on training data. Baseline gaps stay below 0.29 at every layer of every model. In the middle layers the LaTa and PhilTa gaps stay within 0.09 of zero, while mT5-base's gap rises as its ranking fails (compare Figure 1). Figure 5 resolves the mT5-base case by dividing the gap by the spread of all cosines. ABTT raises every model's gap to between 0.44 and 0.61 at every depth. SIF raises the LaTa and PhilTa gaps and leaves the other four near or below baseline.

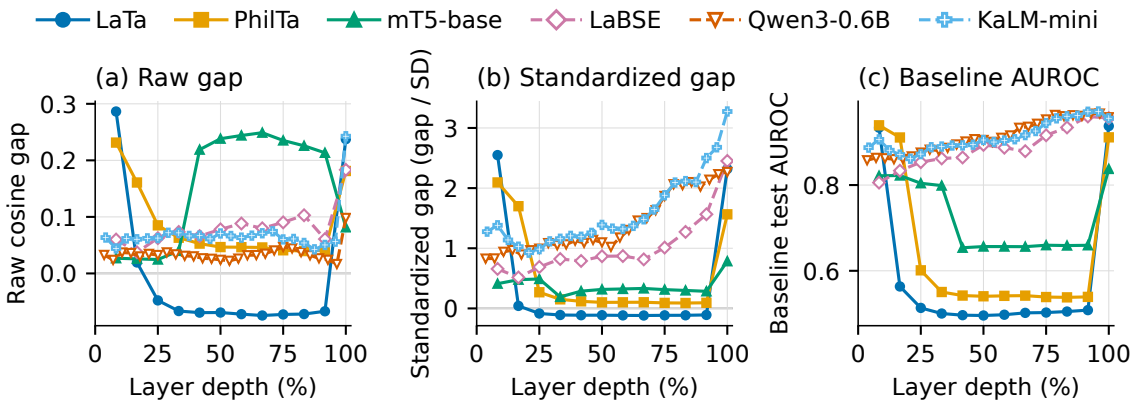


Figure 5: Cosine gap and retrieval against normalized layer depth for the six panel models, baseline mean pooling, test split. Solid lines with filled markers: T5 encoders. Dashed lines with hollow markers: embedding-trained models. (a) Raw gap: mean cosine of same-directory pairs minus that of different-directory pairs. (b) Standardized gap: the raw gap divided by the standard deviation of all pairwise test cosines at that layer. (c) Baseline test ranking AUROC. In the middle layers of mT5-base the raw gap rises to 0.21–0.25 while AUROC falls to 0.654, and the standardized gap falls with AUROC. Across all 100 model-layers the standardized gap tracks AUROC at Spearman 0.97, and the raw gap at 0.22.

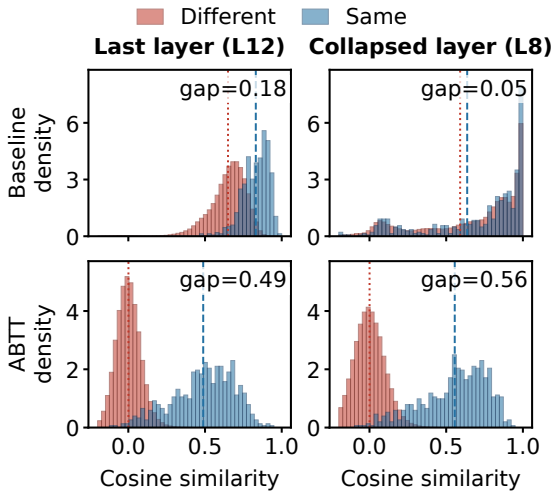


Figure 6: Cosine similarity distributions for PhilTa at its last layer and at the collapsed layer 8 (test AUROC 0.542, close to chance), before and after ABTT. Same: pairs that share a directory; Different: pairs that do not; dashed and dotted vertical lines: the Same and Different means; gap: their difference. Layer 8 has the highest training AUROC of PhilTa's layers 4 to 8 (30 to 70% of depth), all of them collapsed, so it is not PhilTa's worst layer: that is layer 10, at baseline test AUROC 0.538. Before ABTT the two distributions sit almost on top of each other, and ABTT separates them.

D All Thirteen Models

This appendix supports Section 5. Table 6 extends Table 1 in two ways. It adds two embedding-trained models that Table 1 leaves out, Sentence-T5 (Ni et al., 2022) and SPhilBERTa (Riemenschneider and Frank, 2023b). It also gives each checkpoint's feed-forward type and width, whether its embeddings are tied, its lowest effective rank and its range of mean pairwise cosine. None of the added models collapses. Every checkpoint has at most 0.6B parameters, and each is released under the Apache 2.0 license, as its Hugging Face model card states. The rest of the appendix gives the per-layer readouts behind the sibling pairs, T5-efficient-base, the projection on the four controls, the English-text check and the fine-tuned LaTa model.

The sibling pairs. Both encoder-only siblings stay above 0.80 AUROC at every layer (Table 7). LaBERTa clears that floor by 0.026, at layer 1, and rises to 0.910 at layer 12; PhilBERTa stays between 0.883 and 0.923. Effective rank falls to 1.36 in LaTa and 1.81 in PhilTa, and never below 57.87 in LaBERTa or 82.12 in PhilBERTa. Within each pair, several factors differ together. The T5 member is the encoder of an encoder-decoder, pretrained with span corruption; the sibling is an encoder-only RoBERTa-style model, pretrained with masked language modeling (Riemenschneider and Frank, 2023a). The T5 members use a SentencePiece tokenizer and the siblings a byte-level BPE tokenizer, so the tokens that enter the mean also differ. By their checkpoint configurations, the T5 members also use relative position biases, a gated-GELU feed-forward block and untied embeddings, and the siblings learned absolute positions, a GELU feed-forward block and tied embeddings. The pairs therefore rule out the shared pretraining text and the missing embedding objective as sufficient causes, and they cannot say which within-pair difference matters.

T5-base and T5-v1.1-base. The two checkpoints share their tokenizer, which splits the Latin corpus identically, the span-corruption objective, relative position biases and pre-layer RMS normalization, at the same depth and width (Raffel et al., 2020; Google Research, 2020). They differ in feed-forward block (ReLU with inner width 3072, against gated GELU with 2048), in embedding tying, and in pretraining. T5-base was pretrained on C4 mixed with supervised tasks, with dropout (Raffel et al., 2020); T5-v1.1-base on C4 alone, without dropout by its release notes (Google Research, 2020), although its released pretraining configuration sets a dropout rate of 0.1. Dropout does not separate the collapsing models from the others, because PhilTa was also pretrained with dropout (Riemenschneider and Frank, 2023a). T5-base shows a weak form of the same depth profile: its top-PC share rises to 0.545 and its effective rank falls to 14.85 at layer 8, while its AUROC stays between 0.816 and 0.864. T5-v1.1-base recovers only partly at its last layer, to 0.772. T5-efficient-base (Tay et al., 2022) has T5-base's ReLU feed-forward with inner width 3072, tied embeddings, vocabulary, depth and width, and T5-v1.1-base's C4-only pretraining data; it was pretrained without dropout and on a much shorter run: 524,288 steps of 65,536 tokens, about 1/30 of the tokens of either, by the released training configurations. On the Latin corpus it never falls below 0.736 (layer 11), and its top-PC share stays at or below 0.627 (Table 8). It ranks below T5-base at every layer, by 0.016 to 0.080, and its layer 11 has effective rank 8.1, a mild form of the low-rank profile. Its tokenizer adds one whitespace token to 471 passages, which the pooling filter drops; reading it with T5-base's tokenizer changes no AUROC by more than 0.002 and no top-PC share by more than 0.07 (peak 0.615). Its mean pairwise cosine is 0.97 to 0.995 at layers 1 to 11, so a high mean cosine also flags a model that does not collapse.

The projection on the controls. Table 9 applies ABTT to the four controls with the panel's settings. LaBERTa (0.963 to 0.977) and PhilBERTa (0.974 to 0.980) stay inside the panel's band at every layer, T5-base misses it at layer 11 (0.959), and T5-v1.1-base reaches 0.927 to 0.970, below the band at layers 6 to 12. T5-efficient-base reaches 0.959 to 0.975 (Table 8), below the band at layer 11 only. $D = 10$, the top of the selection grid, is selected at 46 of the 48 control layers, so a larger D might narrow the gap; we did not test this.

Sentence-T5. Sentence-T5 has the original T5 layout and a contrastive embedding objective. At its layers 2 to 11, top-PC share is 0.659 to 0.943 while AUROC stays between 0.800 and 0.891. Sentence-T5 thus shows that a high top-PC share is not sufficient for collapse. It is necessary in our data: every one of the 36 collapsed layers of the pretrained checkpoints in Table 6 (26 in the panel, 10 in T5-v1.1-base) has a share of at least 0.76.

			Checkpoint			Over layers				Layers	
Model	Type	Emb. obj.	Feed-forward	d_{ff}	Tied	$\text{AUROC}_{\text{min}}$	PC1_{max}	Rank_{min}	Mean cos.	Coll.	PC1$\geq$0.76
Main text											
LaTa	T5	none	gated GELU	2048	no	0.496 (6)	0.952 (4)	1.36 (8)	0.22–0.58	2–11	2–11
PhilTa	T5	none	gated GELU	2048	no	0.538 (10)	0.858 (6)	1.81 (6)	0.51–0.71	3–11	3–11
mT5-base	T5	none	gated GELU	2048	no	0.654 (5)	1.000 (5)	1.00 (5)	0.33–0.95	5–11	4–11
T5-v1.1-base	T5	none	gated GELU	2048	no	0.489 (2)	0.975 (9)	1.14 (9)	0.87–0.96	2–11	2–11
T5-base	T5	none	ReLU	3072	yes	0.816 (11)	0.545 (8)	14.85 (8)	0.90–0.95	none	none
T5-efficient-base	T5	none	ReLU	3072	yes	0.736 (11)	0.627 (11)	8.10 (11)	0.89–0.99	none	none
LaBERTa	Enc.	none	GELU	3072	yes	0.826 (1)	0.254 (9)	57.87 (9)	0.70–0.91	none	none
PhilBERTa	Enc.	none	GELU	3072	yes	0.883 (6)	0.162 (6)	82.12 (11)	0.84–0.96	none	none
LaBSE	Enc.	contrastive	GELU	3072	n/a	0.806 (1)	0.524 (8)	16.19 (8)	0.73–0.93	none	none
Qwen3-0.6B	Dec.	contrastive	gated SiLU	3072	yes	0.858 (1)	0.183 (16)	71.60 (16)	0.84–0.97	none	none
KaLM-mini	Dec.	contrastive	gated SiLU	4864	yes	0.861 (5)	0.419 (5)	23.83 (5)	0.59–0.93	none	none
Appendix only											
Sentence-T5	T5	contrastive	ReLU	3072	n/a	0.800 (11)	0.943 (6)	1.56 (6)	0.72–0.98	none	2–9
SPhilBERTa	Enc.	distillation	GELU	3072	n/a	0.892 (1)	0.127 (11)	64.67 (11)	0.61–0.95	none	none

Table 6: All thirteen models, read on the Latin corpus as mean-pooled hidden states with no post-hoc correction. Type: the encoder of a T5 encoder-decoder (T5), an encoder-only model (Enc.), or a decoder (Dec.). Emb. obj.: the embedding objective trained after pretraining, if any. Checkpoint: the feed-forward activation and inner width d_{ff}, and whether input and output embeddings are tied, as the checkpoint's configuration file states them (T5-base, T5-efficient-base, LaBERTa and PhilBERTa leave tying at the library default, tied; n/a: the checkpoint has no output head). Over layers: the lowest test ranking AUROC, the peak top-PC share and the lowest entropy effective rank, each with its layer, and the range of mean pairwise cosine; top-PC share, effective rank and mean cosine are computed on the training embeddings. Layers: the collapsed layers (Coll.), with AUROC below 0.70, and those with top-PC share of at least 0.76. Every one of the 36 collapsed layers (26 in the panel, 10 in T5-v1.1-base) has a top-PC share of at least 0.76, but a high share also occurs without collapse (mT5-base layer 4, Sentence-T5). Checkpoints: LaTa, `bowphs/LaTa`; PhilTa, `bowphs/PhilTa`; mT5-base, `google/mt5-base`; T5-v1.1-base, `google/t5-v1_1-base`; T5-base, `google-t5/t5-base`; T5-efficient-base, `google/t5-efficient-base`; LaBERTa, `bowphs/LaBerta`; PhilBERTa, `bowphs/PhilBerta`; LaBSE, `sentence-transformers/LaBSE`; Qwen3-0.6B, `Qwen/Qwen3-Embedding-0.6B`; KaLM-mini, `KaLM-Embedding/KaLM-embedding-multilingual-mini-instruct-v2.5`; Sentence-T5, `sentence-transformers/sentence-t5-base`; SPhilBERTa, `bowphs/SPhilBerta`.

English text. Table 10 and Figure 7 give the label-free check of Section 5. The English sample comes from the public-domain US court opinions of the Caselaw Access Project (Harvard Law School Library Innovation Lab, 2024), in the redistribution of the Common Pile (Kandpal et al., 2025); every sampled document is marked public domain. Legal prose is the nearest openly licensed English register to canon law. Each non-empty Latin passage receives one English passage from a distinct opinion, and the two empty Latin files receive an empty one. The English passage is a run of whole words from a random sentence start in the body text, with an mT5 token count within $\max(2, 3\%)$ tokens of the Latin passage's, and it inherits its partner's split. The two texts then have the same mT5 length distribution (median 98 tokens). The match holds only under the mT5 tokenizer: under PhilTa's tokenizer the English passages are about 40 percent longer, and under T5-v1.1's English-only tokenizer the Latin passages are about 55 percent longer. Re-pairing the training passages on each model's own token lengths (653 to 847 pairs) moves top-PC share by at most 0.010 at the layers that Table 10 lists. At mid-depth the top-PC share is above 0.8 in every cell. The effective rank is near 1 for mT5-base and T5-v1.1-base but not for PhilTa, whose effective rank is 1.82 to 2.30 at English layers 4 to 11, and 1.81 to 4.34 at Latin layers 3 to 11. The two texts differ in language, era and genre together, and the English sample covers one register.

Fine-tuned LaTa. Table 11 gives fine-tuned LaTa at every layer, with no post-hoc correction, beside the pretrained model. Fine-tuning lifts the last layer from 0.938 to 0.984 AUROC and leaves layers 2 to 11 at 0.499 to 0.569. Their top-PC share stays at 0.759 to 0.945, against 0.771 to 0.952 before fine-tuning, so the collapsed geometry stays in place (fine-tuned layer 2, at 0.759, sits just below the 0.76 screen). The fine-tuning runs at most eight epochs on 499 pairs (Appendix H). It is therefore a weak test of whether an embedding objective repairs the middle layers: small-scale supervision

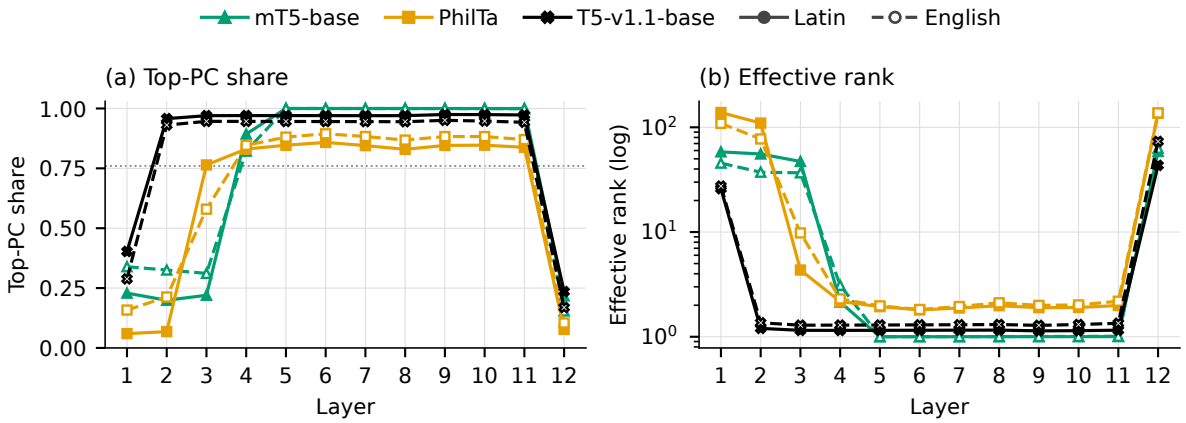


Figure 7: Label-free layer geometry of three T5 encoders on the Latin corpus (solid, filled markers) and on English US court opinions matched to the Latin passages in mT5 token length (dashed, hollow markers), read on 847 passages per cell. (a) Top-PC share, the share of centered variance on the first principal component (dotted line at 0.76, drawn as a visual guide); every collapsed Latin layer of the pretrained models has a share of at least 0.76. (b) Entropy effective rank, log scale. Every model is low-rank from layer 2 to 4 through layer 11 on both texts and recovers at layer 12.

	LaBERTa			PhilBERTa		
Layer	AUROC	PC1	Rank	AUROC	PC1	Rank
1	0.826	0.170	87.15	0.893	0.092	134.77
2	0.832	0.193	75.32	0.905	0.096	109.71
3	0.839	0.184	78.12	0.913	0.112	97.87
4	0.830	0.191	74.89	0.903	0.124	98.85
5	0.828	0.191	72.73	0.885	0.151	95.27
6	0.834	0.198	68.46	0.883	0.162	97.28
7	0.836	0.228	61.98	0.896	0.149	97.34
8	0.844	0.248	58.96	0.905	0.157	94.21
9	0.857	0.254	57.87	0.911	0.140	92.12
10	0.873	0.245	61.55	0.919	0.120	89.92
11	0.892	0.233	65.75	0.913	0.143	82.12
12	0.910	0.154	85.69	0.923	0.083	105.56

Table 7: Per-layer readouts of LaBERTa and PhilBERTa, the encoder-only siblings of LaTa and PhilTa in the matched pairs of Table 1, on the Latin corpus with no post-hoc correction. AUROC: test ranking AUROC of cosine on mean-pooled vectors. PC1: top-PC share, the share of centered variance on the first principal component of the 847 training passages. Rank: entropy effective rank of the same passages. Two facts about the inputs: LaBERTa's tokenizer encodes every space as its own token, so the indentation of the corpus files becomes tokens, which the pooling filter drops from the mean; PhilBERTa's published tokenizer adds no sequence-boundary tokens, so the two whitespace-only files of the corpus pool to zero vectors, and scoring without those two files changes no AUROC by more than 0.0003.

	T5-efficient-base					T5-base
Layer	AUROC	PC1	Rank	Cos	ABTT	AUROC
1	0.820	0.164	49.85	0.994	0.975	0.855
2	0.822	0.137	54.66	0.995	0.968	0.864
3	0.812	0.190	48.51	0.994	0.973	0.864
4	0.798	0.193	47.78	0.994	0.974	0.849
5	0.794	0.231	45.57	0.994	0.971	0.848
6	0.803	0.304	35.04	0.993	0.967	0.843
7	0.809	0.263	37.85	0.992	0.969	0.849
8	0.812	0.327	32.52	0.991	0.970	0.847
9	0.794	0.260	36.81	0.990	0.964	0.836
10	0.782	0.276	34.43	0.987	0.962	0.826
11	0.736	0.627	8.10	0.973	0.959	0.816
12	0.827	0.213	66.67	0.891	0.964	0.843

Table 8: T5-efficient-base, the original T5 layout (ReLU feed-forward, tied embeddings) pretrained on C4 alone without dropout for 524,288 steps of 65,536 tokens, on the Latin corpus. AUROC: test ranking AUROC of cosine on unmodified mean-pooled vectors. PC1: top-PC share, the share of centered variance on the first principal component of the 847 training passages. Rank: their entropy effective rank. Cos: their mean pairwise cosine. ABTT: test AUROC after ABTT fit on the training passages, with D chosen on training DirAcc@1. Last column: T5-base, which has the same layout and pools the same tokens of every passage and was pretrained on C4 mixed with supervised tasks, with dropout, for about 30 times as many tokens; its vectors come from the same extraction run.

at the output may leave them untouched whatever produces the collapse.

	LaBERTa		PhilBERTa		T5-base		T5-v1.1-base	
Layer	Base	ABTT	Base	ABTT	Base	ABTT	Base	ABTT
1	0.826	0.974	0.893	0.976	0.855	0.981	0.848	0.970
2	0.832	0.977	0.905	0.977	0.864	0.981	0.489	0.969
3	0.839	0.972	0.913	0.978	0.864	0.974	0.493	0.965
4	0.830	0.969	0.903	0.974	0.849	0.971	0.505	0.963
5	0.828	0.967	0.885	0.976	0.848	0.969	0.515	0.963
6	0.834	0.963	0.883	0.974	0.843	0.967	0.523	0.959
7	0.836	0.964	0.896	0.974	0.849	0.962	0.533	0.952
8	0.844	0.963	0.905	0.975	0.847	0.966	0.536	0.945
9	0.857	0.965	0.911	0.979	0.836	0.969	0.533	0.934
10	0.873	0.968	0.919	0.978	0.826	0.962	0.531	0.930
11	0.892	0.966	0.913	0.980	0.816	0.959	0.522	0.927
12	0.910	0.967	0.923	0.976	0.843	0.973	0.772	0.959

Table 9: Test ranking AUROC of the four control models on the Latin corpus, before (Base: unmodified mean-pooled vectors) and after ABTT fit on the training passages, with D chosen per layer on training DirAcc@1 from 1, 2, 3, 5, 7 and 10, as for the panel.

			Layers with PC1 ≥ 0.76		
Model	Text	$\mathrm{PC1}_{\max}$ (ℓ)	Layers	PC1	Eff. rank
mT5-base	Latin	1.000 (5)	4–11	0.893–1.000	1.00–2.15
	English	1.000 (5)	4–11	0.822–1.000	1.00–3.10
PhilTa	Latin	0.858 (6)	3–11	0.764–0.858	1.81–4.34
	English	0.894 (6)	4–11	0.846–0.894	1.82–2.30
T5-v1.1-base	Latin	0.975 (9)	2–11	0.958–0.975	1.14–1.20
	English	0.950 (9)	2–11	0.931–0.950	1.28–1.36

Table 10: Label-free layer geometry crossed by model and text. Each cell reads the mean-pooled hidden states of 847 passages: the Latin training passages, or the English passages paired with them one to one and matched in mT5 token length (US court opinions, Caselaw Access Project). $\mathrm{PC1}_{\max}$: peak top-PC share over the 12 layers, the share of centered variance on the first principal component, with its layer ℓ. The last three columns cover the layers whose top-PC share is at least 0.76, a share that every collapsed Latin layer of the pretrained models reaches, and give the range of top-PC share and of entropy effective rank over them. No labels are used, so the English rows say nothing about retrieval.

	AUROC		Top-PC share		Eff. rank	
Layer	Before	After	Before	After	Before	After
1	0.934	0.936	0.050	0.050	134.33	134.44
2	0.563	0.569	0.771	0.759	4.61	4.92
3	0.513	0.512	0.935	0.929	1.59	1.64
4	0.500	0.499	0.952	0.945	1.38	1.43
5	0.496	0.500	0.943	0.931	1.41	1.48
6	0.496	0.500	0.934	0.921	1.44	1.51
7	0.498	0.501	0.937	0.923	1.43	1.50
8	0.502	0.505	0.950	0.940	1.36	1.42
9	0.503	0.507	0.943	0.931	1.40	1.48
10	0.505	0.510	0.942	0.930	1.43	1.50
11	0.508	0.512	0.939	0.927	1.47	1.55
12	0.938	0.984	0.081	0.069	132.22	137.68

Table 11: LaTa before and after contrastive fine-tuning on training pairs, per layer, with no post-hoc correction. AUROC: test ranking AUROC of cosine on mean-pooled vectors. Top-PC share and effective rank: label-free geometry of the centered training vectors, which include the passages of the fine-tuning pairs. Fine-tuning lifts the last layer and leaves the middle layers collapsed.

E Localizing the Collapse: Details

This appendix gives the per-layer tables and the further controls behind Section 6. Collapsed layers are the 26 layers of LaTa, PhilTa and mT5-base with baseline AUROC below 0.70. Every coordinate ranking, component, token ranking and threshold is fit on training passages only, and every AUROC is test ranking AUROC. A layer counts as restored when it reaches 0.90.

Coordinate ablation. Table 12 reads each model at its lowest-AUROC baseline layer. The two coordinate rankings pick nearly the same coordinates at the collapsed layers: on average 9.1 of the top ten are shared. Massive coordinates alone do not mark a collapse. At LaTa layer 6, the lowest-AUROC layer of the panel, the three pooled coordinates of largest mean magnitude reach 720, 684 and 676, against a median of 8.5. At mT5-base layer 1, a healthy layer (AUROC 0.822), they reach 135, 131 and 63, against a median of 1.9. Per-coordinate

standardization with training statistics (Timkey and van Schijndel, 2021) is a partial repair. It recovers a median 79% of the gain from removing ten components, but it restores only 9 of the 26 layers (1 of 10 in LaTa, 5 of 9 in PhilTa, 3 of 7 in mT5-base). The embedding-trained models barely react to zeroing: at the train-selected layers of LaBSE, Qwen3-0.6B and KaLM-mini, zeroing up to ten coordinates changes AUROC by at most 0.007.

Zeroing more coordinates. Figure 8 extends the zeroing to 400 coordinates. Under the variance ranking, zeroing restores none of the 26 layers up to 30 coordinates, 7 at 100, and 19 at 300 and 400. The 19 are every collapsed layer of LaTa and PhilTa; no mT5-base layer is restored. Under the mean-absolute-value ranking of Figure 3a, the first layers are restored at 50 coordinates and all 19 at 400. Zeroing 400 random coordinates removes a median 55% of the training variance, yet the median AUROC stays at 0.538, against 0.541 uncorrected. Top-PC share falls before AUROC recovers: at 100 zeroed coordinates it is below 0.2 at 19 layers, while only 7 are restored.

What survives zeroing. A follow-up zeroes the ten highest-variance coordinates and asks which direction then dominates. These ten coordinates carry a median 92% of the first component's squared loading. The new dominant direction is the remainder of the first component on the other coordinates (median cosine 0.999 with that remainder). It holds a median 72% of the remaining variance. Removing the original second and third components while keeping the first restores none of the 26 layers. Zeroing followed by ABTT with one, two or three components restores 9, 21 and 26 layers, against 1, 13 and 26 without zeroing.

Number of directions. Figure 9 plots the sweep over D at each model's lowest-AUROC layer, and Table 13 gives medians over groups of layers. LaTa reaches the full gain of removing ten components with two (median 98%), and PhilTa with three (95%). mT5-base reaches a median 85% with three and keeps rising up to ten. The failure of a single component is not an estimation artifact. At every collapsed layer the first component fit on training passages and the one fit on test passages agree to $|\cos| \geq 0.994$. The selection grid stops at $D = 10$; extending it to $D = 50$ changes test AUROC by at most 0.023.

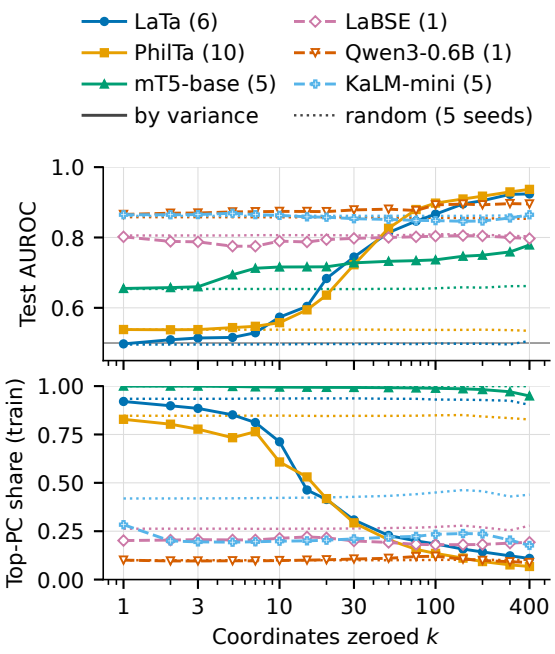


Figure 8: Test ranking AUROC (top) and top-PC share of the training embeddings (bottom) against the number k of zeroed residual coordinates on a logarithmic axis, one color per model at its lowest-AUROC baseline layer (in parentheses in the legend). Coordinates are ranked by variance across training passages; the dotted line in the same color zeroes k random coordinates instead (mean over five seeds). Rankings and top-PC share are fit on training embeddings only; the gray horizontal line marks chance (AUROC 0.5). Filled markers and solid lines: T5 encoders; hollow markers and dashed lines: embedding-trained models. LaTa and PhilTa pass AUROC 0.90 only after 200 and 150 coordinates are zeroed (0.924 and 0.937 at $k = 400$). mT5-base reaches only 0.779 at $k = 400$, where its top-PC share is still 0.949. At the plotted T5 layers random zeroing moves AUROC by at most 0.010, and zeroing by variance moves the embedding-trained models by at most 0.038.

Removed and retained subspaces. Table 14 ranks passages within the removed and within the retained subspace. At the collapsed layers the removed subspace holds a median 99% of the centered variance of vectors that already rank at chance, so its chance-level ranking there is close to a tautology. The next ten components alone rank at a median 0.919 at those layers, a dimension-matched control (Table 14). The healthy layers are more informative. At the 74 healthy panel layers the removed subspace still ranks at 0.736 to 0.937, so ABTT discards some signal. Yet at all 36 layers of the three pretrained T5 encoders the removed directions rank below the next D components. At LaTa's last layer, contrastive fine-tuning moves the

			Zeroed, ranked by mean $\lvert x\rvert$				Zeroed, ranked by variance							
Model	L	Base	1	3	5	10	1	3	5	10	Std.	D=0	ABTT$_{D=1}$	ABTT$_{D=3}$
Test ranking AUROC														
LaTa	6	0.496	0.498	0.514	0.512	0.574	0.498	0.514	0.516	0.574	0.822	0.482	0.671	0.973
PhilTa	10	0.538	0.537	0.540	0.540	0.557	0.538	0.538	0.544	0.558	0.907	0.501	0.602	0.964
mT5-base	5	0.654	0.655	0.660	0.694	0.716	0.655	0.660	0.694	0.716	0.806	0.670	0.835	0.914
LaBSE	1	0.806	0.804	0.793	0.794	0.788	0.802	0.788	0.775	0.790	0.907	0.867	0.886	0.929
Qwen3-0.6B	1	0.858	0.866	0.874	0.877	0.880	0.866	0.870	0.873	0.874	0.963	0.954	0.959	0.970
KaLM-mini	5	0.861	0.862	0.857	0.857	0.862	0.865	0.865	0.869	0.863	0.933	0.917	0.926	0.969
Top-PC share of the training vectors														
LaTa	6	0.934	0.921	0.885	0.859	0.712	0.921	0.885	0.851	0.712	0.211	0.934	0.712	0.071
PhilTa	10	0.847	0.827	0.796	0.760	0.580	0.828	0.777	0.733	0.608	0.143	0.847	0.763	0.088
mT5-base	5	1.000	1.000	1.000	0.998	0.995	1.000	1.000	0.998	0.995	0.910	1.000	0.458	0.138
LaBSE	1	0.263	0.263	0.267	0.258	0.199	0.202	0.206	0.205	0.213	0.189	0.263	0.138	0.077
Qwen3-0.6B	1	0.100	0.101	0.096	0.095	0.095	0.101	0.096	0.097	0.098	0.102	0.100	0.077	0.049
KaLM-mini	5	0.419	0.298	0.301	0.306	0.192	0.284	0.195	0.193	0.198	0.196	0.419	0.210	0.067

Table 12: Coordinate ablation at each model's lowest-AUROC baseline layer (L). Top block: test ranking AUROC for the unmodified mean-pooled vectors (Base); after zeroing the k residual coordinates with the largest mean absolute training value, or with the largest variance across training passages; after per-coordinate standardization with training statistics (Std.); after centering alone (D=0); and after ABTT with one or three components (ABTT$_{D=1}$, ABTT$_{D=3}$), the two reference repairs. Bottom block: top-PC share of the training vectors after the same interventions, the share of their centered variance on the first principal component; centering leaves it unchanged by definition. In the three T5 rows, AUROC stays at or below 0.716 after zeroing ten coordinates under either ranking, while standardization reaches 0.806 to 0.907 and ABTT with three components 0.914 to 0.973. At the collapsed T5 layers (baseline AUROC below 0.70, 26 layers), one component recovers a median 45 percent of the AUROC gain of D=10, and three recover at least 80 percent at every layer (Table 13). All statistics are fit on training embeddings only. Figure 8 extends the zeroing to $k = 400$.

top ten directions from below the next ten (0.868 against 0.907) to above them (0.953 against 0.900), consistent with Rajaee and Pilehvar (2021).

Token audit. Table 15 gives the token audit at each model's lowest-AUROC layer. In LaTa, a token mix with one mean contribution per token type, fit on training tokens, explains a median 0.985 of the variance of the first component's score on test passages. For random directions the same statistic is 0.18. At layer 6 the comma alone holds 0.962 of the first component's score variance over training passages. Ranking token types by the first component alone restores at most 2 of LaTa's 10 collapsed layers, so the second and third components matter too. Two pooling variants separate the two changes that SIF makes. In LaTa, dropping the 100 most frequent training tokens recovers at least 80% of SIF's gain at all 10 layers (median AUROC 0.901). Dropping special tokens alone recovers it at none (median 0.498). In PhilTa the token mix explains only a median 0.42 of the first component's score. Its ablation needs more types: on test passages, dropping the top 3 restores 1 of 9 layers, the top 10 restore 8, and the top 30 restore all 9 (median 0.925); on training passages the counts are 1, 6 and 7. With the ranking and m chosen per layer on training AUROC, every collapsed layer of both models is restored: test medians 0.926 (LaTa) and 0.928 (PhilTa), minima 0.921 and 0.908, against mass-matched random controls of 0.491 and 0.542. At $m = 30$, dropping random types of the same training mass stays at a median of 0.543. In mT5-base the token mix explains a median 0.066 of the first component's score, against a median 0.160 for random directions. Dropping its top 100 types lifts AUROC to 0.724 to 0.748, short of 0.90. In none of the three models does the first token carry the dominant direction (share at most 0.005). The same three delimiter roles carry outlier activations in BERT (Bondarenko et al., 2023), and delimiter tokens carry massive activations in large language models (Sun et al., 2024). Removing punctuation and special tokens before pooling also helps BERT (Fuster Baggetto and Fresno, 2022); here it separates chance from recovery.

			Removed components D										
Layers	n	raw	0	1	2	3	5	7	10	15	20	30	50
Lowest-AUROC baseline layer of each model													
LaTa (6)	1	0.496	0.482	0.671	0.940	0.973	0.973	0.977	0.964	0.973	0.966	0.981	0.986
PhilTa (10)	1	0.538	0.501	0.602	0.888	0.964	0.977	0.982	0.982	0.975	0.977	0.973	0.969
mT5-base (5)	1	0.654	0.670	0.835	0.892	0.914	0.951	0.971	0.978	0.972	0.965	0.965	0.967
LaBSE (1)	1	0.806	0.867	0.886	0.915	0.929	0.962	0.967	0.977	0.977	0.977	0.977	0.979
Qwen3-0.6B (1)	1	0.858	0.954	0.959	0.962	0.970	0.972	0.979	0.978	0.980	0.981	0.976	0.984
KaLM-mini (5)	1	0.861	0.917	0.926	0.969	0.969	0.983	0.982	0.983	0.978	0.974	0.974	0.976
Median over model-layers													
Collapsed T5	26	0.541	0.503	0.713	0.902	0.964	0.973	0.977	0.977	0.974	0.975	0.974	0.971
LaTa	10	0.502	0.488	0.713	0.956	0.973	0.972	0.976	0.970	0.971	0.969	0.980	0.983
PhilTa	9	0.542	0.502	0.602	0.872	0.963	0.980	0.981	0.981	0.975	0.978	0.973	0.970
mT5-base	7	0.657	0.678	0.865	0.894	0.934	0.955	0.974	0.978	0.975	0.974	0.965	0.962
Other T5	10	0.874	0.936	0.952	0.967	0.973	0.979	0.976	0.973	0.972	0.972	0.973	0.971
Embedding-trained	64	0.903	0.944	0.966	0.971	0.976	0.979	0.980	0.978	0.977	0.975	0.974	0.973
All	100	0.885	0.936	0.958	0.968	0.973	0.978	0.979	0.977	0.975	0.974	0.974	0.972

Table 13: Test ranking AUROC against the number D of top principal components that ABTT removes, fit on training embeddings only. Raw: mean-pooled vectors with no correction; $D = 0$: centering on the training mean alone. The top block reads each model at its lowest-AUROC baseline layer (layer in parentheses); the bottom block gives medians over the 26 collapsed T5 layers (baseline AUROC below 0.70), also split by model, the other 10 T5 layers, the 64 layers of the embedding-trained models, and all 100 model-layers. The paper's selection grid for D stops at 10.

				Full vector		**Removed subspace**			**Retained subspace**		
Model	ℓ	D	Var.	Raw	Cent.	PC1	PCs 2–D	PCs 1–D	Next D	Rand. D	All
Pretrained, train-selected layer and D											
LaTa	12	10	0.35	0.938	0.972	0.670	0.825	0.868	0.907	0.861	0.971
PhilTa	9	10	0.99	0.539	0.501	0.462	0.545	0.487	0.922	0.873	0.983
mT5-base	2	10	0.57	0.822	0.910	0.610	0.735	0.751	0.896	0.866	0.975
LaBSE	11	10	0.48	0.960	0.958	0.647	0.932	0.909	0.958	0.902	0.987
Qwen3-0.6B	2	10	0.44	0.863	0.940	0.607	0.792	0.798	0.906	0.882	0.973
KaLM-mini	1	10	0.41	0.887	0.958	0.518	0.846	0.819	0.910	0.880	0.981
Collapsed T5, median (26)	–	–	0.99	0.541	0.503	0.463	0.601	0.489	0.919	0.874	0.977
Fine-tuning contrast, same layer, $D = 10$											
LaTa, pretrained	12	10	0.35	0.938	0.972	0.670	0.825	0.868	0.907	0.861	0.971
LaTa, fine-tuned	12	10	0.33	0.984	0.985	0.697	0.935	0.953	0.900	0.874	0.970
Qwen3-0.6B, pretrained	27	10	0.41	0.967	0.969	0.643	0.913	0.917	0.909	0.884	0.982
Qwen3-0.6B, fine-tuned	27	10	0.31	0.993	0.996	0.684	0.980	0.983	0.969	0.927	0.992
KaLM-mini, pretrained	24	10	0.32	0.956	0.979	0.663	0.909	0.924	0.909	0.855	0.981
KaLM-mini, fine-tuned	24	10	0.23	0.997	0.996	0.699	0.977	0.978	0.967	0.918	0.994

Table 14: Test ranking AUROC of cosine within parts of the pooled vector. ABTT removes the top D principal components of the centered training embeddings; Var. is their share of the centered training variance. Raw: uncorrected vector; Cent.: centered on the training mean; PC1: projection onto the first component alone, where cosine reduces to a sign match; PCs 2–D and PCs 1–D: projections onto the rest of the removed subspace and onto all of it. Retained subspace, a dimension-matched control: Next D, PCs $D{+}1$ to $2D$; Rand. D, random D-dimensional projections of the retained vectors (mean over five seeds); All, the ABTT output. Top block: each pretrained model at its train-selected ABTT layer ℓ with D chosen on training DirAcc@1, and the median over the 26 collapsed T5 layers (baseline AUROC below 0.70), each at its own layer and D. Bottom block: each contrastively fine-tuned encoder at its train-selected ABTT layer, next to its pretrained model at the same layer, both with $D = 10$. All components are fit on training embeddings only.

		AUROC by pooling					Token ablation		Token-mix EV		Share by token group			
Model	L	Mean	No spec.	SIF+spec.	SIF	No freq.	Drop	Rand.	PC1	Rand.	Spec.	Freq.	Other	$\|\rho\|$
LaTa	6	0.496	0.493	0.860	0.878	0.904	0.922	0.483	0.98	0.19	0.01	0.97	0.02	0.38
PhilTa	10	0.538	0.496	0.677	0.879	0.692	0.917	0.540	0.44	0.14	0.23	0.75	0.02	0.11
mT5-base	5	0.654	0.653	0.675	0.673	0.726	0.724	0.656	0.07	0.24	0.00	0.53	0.47	0.20
LaBSE	1	0.806	0.809	0.812	0.818	0.816	0.827	0.800	0.36	0.73	0.09	0.37	0.53	0.95
Qwen3-0.6B	1	0.858	0.855	0.856	0.852	0.869	0.885	0.836	0.99	0.71	0.74	0.08	0.18	0.58
KaLM-mini	5	0.861	0.859	0.867	0.863	0.872	0.879	0.854	0.98	0.47	0.01	0.02	0.97	0.51

Table 15: Token audit at each model's lowest-AUROC baseline layer (L). Pooling: test ranking AUROC of mean pooling (Mean); mean pooling without special tokens (No spec.); SIF frequency weights with special tokens kept (SIF+spec.); SIF pooling with training-only token frequencies, which also drops special tokens (SIF); and mean pooling without the 100 most frequent training tokens (No freq.). Token ablation: test AUROC of mean pooling after dropping the m token types that contribute most to the top principal components (Drop), with m (1 to 100) and the ranking (one of two) chosen by training AUROC, and the mean test AUROC of five random sets of other token types, sampled in proportion to training count, that hold at least as many training tokens as the dropped types (Rand.). Token-mix EV: the share of the variance of the first principal component's score across test passages that is explained by which token types a passage contains, with one mean per token type fit on training tokens (PC1), next to the mean of the same quantity over 20 random directions orthogonal to the top ten principal components (Rand.). Share by token group: the covariance share of that variance contributed by special tokens, by the 100 most frequent tokens and by all other tokens; the three sum to one, but a single share can fall below zero or exceed one. $|\rho|$: absolute Spearman correlation between the PC1 score and the log token count of test passages (the sign of a principal component is a convention). Components, token frequencies and token rankings are fit on training passages only.

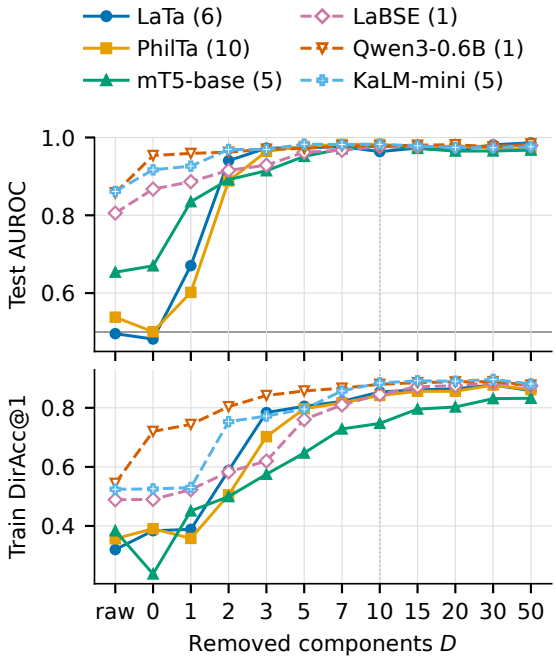


Figure 9: Test ranking AUROC (top) and training DirAcc@1 as a fraction (bottom) against the number D of removed principal components, one line per model at its lowest-AUROC baseline layer (in parentheses in the legend). The horizontal axis is categorical: raw is the uncorrected vector, $D = 0$ is centering on the training mean alone, and the ticks are not evenly spaced in D. ABTT is fit on training embeddings only; the dotted vertical line marks $D = 10$, the top of the selection grid, and the gray horizontal line chance (AUROC 0.5). Filled markers and solid lines: T5 encoders; hollow markers and dashed lines: embedding-trained models. At the plotted T5 layers one component restores part of the ranking. LaTa and PhilTa recover nearly all of it with two or three components; mT5-base keeps improving up to ten.

F Bootstrap Intervals and Routing Checks

This appendix gives the intervals behind Table 2 and the routing checks that Section 7 quotes.

Intervals. A bootstrap replicate draws the 514 test directories with replacement ($B = 10{,}000$, seed 233). A drawn directory keeps every file and pair: a directory drawn c times weights its files and within-directory pairs by c and its pairs with a directory drawn c' times by cc', and no pair is formed between two copies of one directory. The layer, threshold, ABTT components, D, and SIF weights stay fit on the training split, and every score is computed once on the full test set, so a replicate only reweights test pairs and files. Intervals are 95% percentiles; a paired difference recomputes both cells on each replicate. Before resampling, every cell was recomputed with the paper's evaluation code and matched its printed value. Table 16 gives every cell and Table 17 the contrasts quoted in Section 7.

Routing checks. Table 18 reports, at the headline routing cells, the AUROC of each test witness's maximum cosine for existing against new witnesses (no threshold), the gap between the best test threshold and the training threshold, the skewness of the 10-occurrence distribution (Radovanović et al., 2010), centering alone ($D = 0$), and assignment accuracy with τ refit as the exact best-F1 cut over all training pair scores.

Threshold grid. We keep the grid of 200 evenly spaced thresholds in $[0, 1]$ used for every printed number. Every finer grid optimizes the same training objective, the best F1 over same- against different-directory training pairs, and differs only in resolution. With the layer and D fixed, a grid of 10,001 values moves 45 of the 66 printed routing numbers, by a median of 3 files (0.35 points); ABTT, SIF+ABTT, and fine-tuned cells move by at most 0.8 points, and Base and SIF cells by up to 2.6 (mT5-base's baseline). The exact best-F1 cut moves 48 of them. Refitting would trade one set of differences of one to a few files for another; it changes no conclusion of Section 7.

		Ranking		Routing	
Model	**Setting**	AUROC	Cosine gap	Assignment acc.	DirAcc@1
LaTa	Base	0.938 [0.910, 0.963]	0.237 [0.222, 0.253]	73.8 [70.2, 77.3]	72.1 [68.4, 75.8]
	SIF	0.948 [0.915, 0.974]	0.366 [0.342, 0.390]	83.2 [80.1, 86.3]	81.8 [78.5, 85.1]
	ABTT	0.971 [0.948, 0.991]	0.525 [0.491, 0.559]	88.5 [85.8, 90.9]	86.1 [83.2, 88.9]
	SIF+ABTT	0.966 [0.938, 0.989]	0.580 [0.541, 0.617]	90.2 [87.6, 92.6]	88.2 [85.4, 90.9]
PhilTa	Base	0.939 [0.918, 0.958]	0.231 [0.217, 0.247]	70.9 [66.9, 74.9]	69.3 [65.2, 73.5]
	SIF	0.955 [0.935, 0.972]	0.319 [0.300, 0.339]	79.0 [75.6, 82.4]	77.7 [74.2, 81.3]
	ABTT	0.983 [0.969, 0.993]	0.540 [0.508, 0.571]	91.1 [88.9, 93.2]	88.3 [85.6, 90.9]
	SIF+ABTT	0.982 [0.969, 0.993]	0.573 [0.538, 0.608]	91.4 [89.1, 93.6]	88.9 [86.2, 91.5]
mT5-base	Base	0.838 [0.810, 0.866]	0.081 [0.072, 0.092]	47.2 [43.1, 51.6]	46.6 [42.5, 51.0]
	SIF	0.845 [0.816, 0.876]	0.036 [0.031, 0.042]	54.5 [50.3, 58.8]	54.1 [49.8, 58.4]
	ABTT	0.975 [0.954, 0.992]	0.536 [0.498, 0.574]	90.3 [87.9, 92.7]	88.5 [85.7, 91.2]
	SIF+ABTT	0.974 [0.950, 0.993]	0.519 [0.480, 0.559]	90.3 [88.0, 92.6]	87.5 [84.8, 90.2]
LaBSE	Base	0.956 [0.932, 0.977]	0.183 [0.171, 0.195]	83.3 [80.2, 86.4]	81.5 [78.1, 84.7]
	SIF	0.956 [0.931, 0.977]	0.184 [0.172, 0.197]	83.4 [80.2, 86.5]	81.9 [78.6, 85.2]
	ABTT	0.987 [0.977, 0.995]	0.611 [0.575, 0.646]	90.8 [88.5, 92.9]	88.5 [85.8, 90.9]
	SIF+ABTT	0.976 [0.960, 0.989]	0.580 [0.543, 0.617]	89.6 [87.2, 91.9]	86.9 [84.2, 89.5]
Qwen3-0.6B	Base	0.966 [0.950, 0.981]	0.024 [0.023, 0.026]	82.3 [79.1, 85.4]	80.3 [76.9, 83.6]
	SIF	0.942 [0.919, 0.963]	0.053 [0.048, 0.060]	76.0 [72.5, 79.4]	74.2 [70.6, 77.8]
	ABTT	0.973 [0.948, 0.990]	0.553 [0.513, 0.590]	91.5 [89.1, 93.7]	89.4 [86.7, 92.0]
	SIF+ABTT	0.972 [0.947, 0.991]	0.540 [0.499, 0.580]	90.8 [88.3, 93.1]	88.6 [85.8, 91.2]
KaLM-mini	Base	0.972 [0.955, 0.986]	0.056 [0.053, 0.060]	87.5 [84.8, 90.1]	85.9 [82.9, 88.7]
	SIF	0.964 [0.944, 0.982]	0.056 [0.051, 0.062]	85.7 [82.8, 88.5]	84.0 [80.9, 87.1]
	ABTT	0.981 [0.963, 0.993]	0.568 [0.529, 0.605]	91.7 [89.5, 93.8]	89.4 [86.7, 91.9]
	SIF+ABTT	0.981 [0.962, 0.995]	0.535 [0.496, 0.573]	90.8 [88.4, 93.0]	88.0 [85.2, 90.7]
LaTa (fine-tuned)	Base	0.984 [0.973, 0.992]	0.387 [0.368, 0.406]	83.4 [80.2, 86.5]	81.6 [78.1, 84.9]
	ABTT	0.970 [0.947, 0.990]	0.548 [0.511, 0.583]	87.8 [85.2, 90.2]	85.2 [82.3, 87.9]
Qwen3-0.6B (fine-tuned)	Base	0.996 [0.993, 0.999]	0.684 [0.653, 0.714]	91.4 [88.7, 93.8]	90.8 [88.1, 93.3]
	ABTT	0.994 [0.989, 0.998]	0.716 [0.680, 0.751]	92.1 [89.6, 94.3]	91.4 [88.8, 93.7]
KaLM-mini (fine-tuned)	Base	0.997 [0.993, 0.999]	0.636 [0.605, 0.666]	92.4 [90.1, 94.6]	91.7 [89.3, 94.0]
	ABTT	0.994 [0.988, 0.998]	0.651 [0.612, 0.687]	93.4 [91.2, 95.4]	92.5 [90.3, 94.7]
BM25 (word)	–	0.974 [0.956, 0.987]	0.243 [0.222, 0.266]	62.5 [58.2, 66.6]	56.3 [52.0, 60.5]
Char. n-grams	–	0.987 [0.976, 0.996]	0.555 [0.519, 0.591]	91.8 [89.7, 93.8]	89.9 [87.3, 92.3]
Levenshtein	–	0.952 [0.927, 0.975]	0.362 [0.332, 0.393]	87.8 [85.0, 90.4]	86.2 [83.2, 89.2]

Table 16: Every headline cell, those of Table 2 and the SIF and SIF+ABTT cells, with its 95% percentile intervals from a directory-level bootstrap ($B = 10{,}000$ resamples of the 514 test directories with replacement, seed 233). A replicate keeps every file and pair of a drawn directory; a directory drawn twice counts twice, and pairs between its two copies are not formed. The threshold τ, the ABTT components and D, the SIF weights and the layer stay fit on the fixed training split; only the test evaluation is resampled, so the intervals cover test-set sampling and not the choice of split. Each cell is read at its train-selected layer (Table 21); routing in percent. The lexical rows are the lexical reference systems of Table 20; Char. n-grams is TF-IDF over character 3–5-grams.

Contrast	**Model**	Δ AUROC	Δ Assignment acc.	Δ DirAcc@1
ABTT − Base	LaTa	+0.034 [0.017, 0.054]	+14.7 [11.1, 18.3]	+14.0 [10.4, 17.6]
	PhilTa	+0.043 [0.029, 0.060]	+20.3 [16.2, 24.5]	+19.0 [15.0, 23.1]
	mT5-base	+0.137 [0.109, 0.166]	+43.1 [38.6, 47.6]	+41.8 [37.2, 46.4]
	LaBSE	+0.030 [0.014, 0.049]	+7.5 [4.6, 10.4]	+7.0 [4.1, 10.0]
	Qwen3-0.6B	+0.006 [−0.010, 0.019]	+9.2 [6.3, 12.2]	+9.1 [6.1, 12.2]
	KaLM-mini	+0.009 [−0.002, 0.021]	+4.2 [1.7, 6.7]	+3.5 [1.0, 6.0]
Fine-tuned: ABTT − Base	LaTa	−0.014 [−0.032, 0.001]	+4.3 [1.7, 7.1]	+3.6 [1.0, 6.3]
	Qwen3-0.6B	−0.002 [−0.005, 0.001]	+0.7 [−0.7, 2.1]	+0.6 [−0.8, 2.0]
	KaLM-mini	−0.003 [−0.007, 0.001]	+0.9 [−0.3, 2.2]	+0.8 [−0.5, 2.1]
Fine-tuned ABTT − zero-shot ABTT	LaTa	−0.002 [−0.010, 0.005]	−0.7 [−2.6, 1.2]	−0.9 [−2.8, 1.0]
	Qwen3-0.6B	+0.022 [0.006, 0.044]	+0.6 [−1.6, 2.9]	+2.0 [0.0, 4.0]
	KaLM-mini	+0.013 [0.002, 0.029]	+1.6 [−0.3, 3.6]	+3.1 [1.2, 5.2]
Char. n-grams − ABTT	LaTa	+0.016 [0.003, 0.030]	+3.4 [1.1, 5.7]	+3.7 [1.4, 6.2]
	PhilTa	+0.004 [−0.002, 0.011]	+0.7 [−1.0, 2.4]	+1.5 [−0.2, 3.3]
	mT5-base	+0.012 [0.000, 0.027]	+1.5 [−0.6, 3.7]	+1.4 [−0.7, 3.6]
	LaBSE	0.000 [−0.004, 0.005]	+1.0 [−1.0, 3.1]	+1.4 [−0.8, 3.6]
	Qwen3-0.6B	+0.014 [0.002, 0.031]	+0.3 [−1.6, 2.3]	+0.5 [−1.4, 2.5]
	KaLM-mini	+0.006 [−0.002, 0.016]	+0.1 [−1.6, 1.9]	+0.5 [−1.3, 2.3]
Centering − Base (ABTT layer)	LaTa	+0.034 [0.021, 0.049]	−2.2 [−5.2, 0.9]	+6.8 [4.6, 9.1]
	PhilTa	−0.038 [−0.060, −0.016]	+12.9 [9.0, 16.9]	+12.4 [8.5, 16.3]
	mT5-base	+0.088 [0.066, 0.109]	+20.5 [16.0, 24.8]	+15.6 [11.3, 19.9]
	LaBSE	−0.002 [−0.009, 0.004]	−4.7 [−7.4, −2.0]	−4.7 [−7.3, −2.0]
	Qwen3-0.6B	+0.076 [0.061, 0.092]	+19.2 [15.4, 23.2]	+16.7 [12.7, 20.7]
	KaLM-mini	+0.070 [0.052, 0.089]	+19.3 [15.5, 23.2]	+16.9 [13.1, 20.7]
ABTT − centering (ABTT layer)	LaTa	−0.001 [−0.016, 0.012]	+40.4 [35.8, 44.8]	+49.2 [44.1, 53.9]
	PhilTa	+0.482 [0.460, 0.504]	+7.3 [4.9, 10.0]	+6.6 [4.1, 9.3]
	mT5-base	+0.065 [0.050, 0.081]	+24.5 [20.7, 28.3]	+27.7 [23.7, 31.7]
	LaBSE	+0.029 [0.013, 0.048]	+12.1 [8.8, 15.5]	+11.7 [8.4, 15.0]
	Qwen3-0.6B	+0.033 [0.018, 0.050]	+25.9 [22.0, 29.9]	+27.7 [23.5, 32.1]
	KaLM-mini	+0.023 [0.008, 0.040]	+12.0 [9.1, 15.1]	+13.8 [10.7, 17.0]
Spread over six models	Base	0.134 [0.110, 0.158]	40.3 [35.7, 44.7]	39.3 [34.6, 43.7]
	ABTT	0.015 [0.006, 0.034]	3.3 [1.8, 5.8]	3.3 [1.8, 6.0]

Table 17: Paired differences between headline cells, with 95% intervals from the same directory-level bootstrap replicates as Table 16 ($B = 10{,}000$, 514 test directories, seed 233). Both cells of a contrast are recomputed on each replicate, so the interval reflects their correlation. Each cell sits at its own train-selected layer. The two centering contrasts compare baseline, centering alone ($D = 0$) and ABTT at the layer of the ABTT cell, the ranking ABTT layer for Δ AUROC and the routing ABTT layer for the two routing columns. Spread: the largest minus the smallest of the six zero-shot models. Routing in points. Spreads and differences are computed before rounding, so they can differ by one unit in the last digit from the difference of the rounded cells. The threshold τ, the ABTT components and D, the SIF weights and the layer stay fit on the fixed training split; only the test evaluation is resampled, so the intervals cover test-set sampling and not the choice of split.

Model	**Setting**	**L**	Cos. SD	Exist./new AUROC	Assign.	Oracle gap	τ	Exact	Skew
LaTa	Base	1	0.112	0.869 [0.840, 0.896]	73.8	+5.6 [2.8, 9.4]	0.844	74.6	2.81
	Base @ ABTT L	8	0.632	0.621 [0.583, 0.658]	50.2	+16.4 [11.9, 21.9]	0.995	44.1	1.59
	Centering @ ABTT L	8	0.901	0.604 [0.564, 0.644]	48.0	+14.3 [9.4, 21.8]	0.995	42.8	1.97
	ABTT	8	0.100	0.943 [0.925, 0.960]	88.5	+0.6 [0.2, 1.7]	0.472	88.5	1.03
PhilTa	Base	1	0.111	0.843 [0.813, 0.873]	70.9	+7.0 [3.7, 11.5]	0.894	71.0	3.15
	Centering @ ABTT L	1	0.126	0.906 [0.882, 0.928]	83.8	+1.2 [0.6, 2.5]	0.548	84.4	1.78
	ABTT	1	0.086	0.960 [0.944, 0.974]	91.1	+0.7 [0.2, 1.8]	0.417	91.1	0.72
mT5-base	Base	12	0.104	0.685 [0.642, 0.726]	47.2	+19.6 [14.3, 26.8]	0.970	49.5	2.59
	Base @ ABTT L	1	0.065	0.696 [0.654, 0.737]	45.3	+23.1 [16.5, 30.2]	0.995	52.2	3.83
	Centering @ ABTT L	1	0.242	0.765 [0.729, 0.801]	65.9	+5.6 [2.0, 10.5]	0.688	66.0	3.29
	ABTT	1	0.081	0.952 [0.935, 0.967]	90.3	+0.9 [0.2, 2.4]	0.437	91.1	1.13
LaBSE	Base	11	0.039	0.922 [0.901, 0.942]	83.3	+1.0 [0.1, 3.5]	0.975	82.5	2.00
	Centering @ ABTT L	11	0.209	0.891 [0.866, 0.914]	78.7	+2.8 [0.7, 6.3]	0.704	79.0	1.84
	ABTT	11	0.102	0.960 [0.944, 0.973]	90.8	+0.6 [0.1, 1.6]	0.452	90.8	0.58
Qwen3-0.6B	Base	28	0.043	0.900 [0.874, 0.923]	82.3	+1.0 [0.2, 2.6]	0.945	82.6	2.51
	Base @ ABTT L	5	0.034	0.708 [0.666, 0.748]	46.4	+23.4 [17.7, 30.4]	0.990	53.0	2.49
	Centering @ ABTT L	5	0.171	0.802 [0.767, 0.835]	65.6	+9.6 [5.9, 14.5]	0.653	66.6	1.77
	ABTT	5	0.086	0.954 [0.935, 0.971]	91.5	+0.7 [0.2, 1.7]	0.417	91.5	1.07
KaLM-mini	Base	23	0.021	0.931 [0.910, 0.950]	87.5	0.0 [0.0, 1.1]	0.965	86.2	1.97
	Base @ ABTT L	3	0.054	0.756 [0.719, 0.792]	60.4	+11.3 [6.7, 17.0]	0.975	58.3	3.58
	Centering @ ABTT L	3	0.135	0.881 [0.853, 0.906]	79.7	+2.7 [1.2, 4.8]	0.558	79.5	1.75
	ABTT	3	0.085	0.958 [0.941, 0.973]	91.7	+0.2 [0.0, 1.6]	0.397	91.7	0.87

Table 18: Routing checks at the headline routing cells (Table 2) of the six zero-shot models: the Base and ABTT cells at their train-selected layers (L), and baseline and centering alone ($D = 0$, subtract the training mean) at the ABTT cell's layer (a second baseline row only where that layer differs). Cos. SD: standard deviation of all test pairwise cosines. Exist./new AUROC: threshold-free AUROC of each test file's maximum cosine for existing against new files. Assign.: assignment accuracy at the train-fit τ (the printed cell). Oracle gap: best assignment accuracy over every test threshold minus Assign. Exact: assignment accuracy with τ re-fit as the exact best-F1 cut over all training pair scores instead of the 200-point grid. Skew: skewness of the k-occurrence distribution N_{10} over test files (Radovanović et al., 2010). Brackets: 95% directory-bootstrap intervals ($B = 10{,}000$). Holding layer and D fixed, the fine grid (step 10^{-4}) moves 45 and the exact cut 48 of the 66 printed routing numbers; re-selecting layer and D under the new grid as well moves 42 (fine) and 44 (exact) of 60.

G Routing Gain Without a Threshold

This appendix gives the paired contrasts behind the second finding of Section 7, for the three embedding-trained encoders. Each contrast compares the ABTT cell with the baseline cell, each at its own train-selected routing layer, and its 95% interval comes from the directory-level bootstrap of Appendix F. Per-cell values are in Table 18.

No threshold. The existing-versus-new AUROC of each test witness's best-match cosine rises by 0.037 (0.021 to 0.055) for LaBSE, 0.055 (0.034 to 0.077) for Qwen3-0.6B, and 0.027 (0.012 to 0.043) for KaLM-mini.

Best test threshold. At the baseline cells the train-fit threshold comes within 1.0 point of the best test threshold. When both cells use their best test threshold, ABTT still raises assignment accuracy by 7.0 points (4.2 to 9.3) for LaBSE, 8.9 (6.1 to 11.7) for Qwen3-0.6B, and 4.4 (2.2 to 6.8) for KaLM-mini.

The training mean. Centering alone ($D = 0$) at the baseline cell's layer changes assignment accuracy by -4.7 points (-7.4 to -2.0) for LaBSE, $+2.6$ (-0.4 to 5.5) for Qwen3-0.6B, and -1.9 (-3.7 to -0.1) for KaLM-mini. At the ABTT cell's layer, the removed components add 12.1, 25.9 and 12.0 points on top of centering (Table 17).

Hubness. From the baseline cell to the ABTT cell, the skewness of the 10-occurrence distribution (Radovanović et al., 2010) falls from 2.0–2.5 to 0.6–1.1, while centering alone at the ABTT layer leaves it near 1.8. Across representations, low hubness does not track good routing. The character n-gram reference has a skewness of 10.6 and routes as well as the best projected encoder, and SIF raises skewness in LaTa and PhilTa while it raises their routing. We therefore report the drop in hubness as an observation and do not test it as a cause.

H Reference Systems

Table 2 carries four reference rows below the six encoders: one supervised fine-tuning ceiling for each of the three fine-tuned models, and the character 3–5-gram TF-IDF reference. This appendix describes them. Table 19 gives the three fine-tuning ceilings with the zero-shot rows they are compared against. Some routing numbers are means over *five reseedings*: each reseeding keeps the training/test split and redraws, within each directory's test witnesses, which serve as queries and which as references; queries are routed against the reference directories. Every model we fine-tuned is reported, on whichever side of its own zero-shot row it lands.

Three encoders are fine-tuned under one recipe, LaTa, Qwen3-0.6B and KaLM-mini. Fine-tuning only the Latin-pretrained model would leave the ceiling confounded with Latin pretraining, so the second model is a multilingual decoder-only encoder with no Latin-specific pretraining, Qwen3-0.6B. The third, KaLM-mini, has the best ABTT cell in the paper (91.7 assignment accuracy; 89.4 DirAcc@1, tied with Qwen3-0.6B), so it tests whether supervision can clear that cell.

All three are trained with symmetric InfoNCE over positive pairs and in-batch negatives (no two pairs from one directory in a batch), temperature 0.05, AdamW at learning rate 2×10^{-5} with weight decay 0.01, linear warmup over the first 10 percent

of steps then linear decay, gradient clipping at 1.0, 16 pairs per batch, up to 8 epochs with early stopping after 3 epochs without dev improvement, and seed 42. All three are mean-pooled with the same token filter as their zero-shot rows, so the fine-tuned vectors live in the space those rows are measured in; Qwen3-0.6B and KaLM-mini add gradient checkpointing, which recomputes activations rather than storing them and leaves the gradients unchanged.

The dev slice holds out 28 of the 190 train directories with at least two files (71 files), the same directories for every model, so each trains on 499 of the 565 positive train pairs, and the checkpoint is chosen by dev DirAcc@1 with dev AUROC as the tie-break. Accuracy differences at or below one dev file are treated as ties, because one file out of 71 is the pool's whole resolution; without that, a model whose zero-shot encoder already routes all 71 dev files can never select a trained checkpoint. A 71-file pool is the weak link in this protocol: one borderline file flipped between two otherwise identical Qwen3-0.6B runs at a training loss of 6×10^{-4}, which is GPU non-determinism rather than a difference in the model, and it is worth 1.4 points of the selection criterion. LaTa selects epoch 7, the last epoch run before early stopping fired, and its ABTT sweep selects $D = 10$, the top of the grid, at every layer, so the tuned and fixed ABTT variants coincide there. Qwen3-0.6B selects epoch 3 of 6 run, and its sweep is a genuine one, selecting the top of the grid at 12 of 28 layers and $D = 2$ at the layer its routing row reports. KaLM-mini selects epoch 1 of 4 run, the only model of the three whose zero-shot encoder leaves the dev pool room to improve, and its sweep selects the top of the grid at 19 of 24 layers including the layer its routing row reports. All three rows are ceilings at this training budget rather than asymptotes: training loss reaches 4×10^{-5} on Qwen3-0.6B by epoch 4 and 3×10^{-5} on KaLM-mini, so the 499 pairs are memorized and more epochs buy nothing.

Compute. Most hidden-state extraction and all fine-tuning ran on a single NVIDIA A100 GPU; the control-model extraction and the token audit ran on a single NVIDIA A40, and T5-efficient-base, with a re-extraction of the four controls for Table 9, on CPU. The whole project, exploratory runs included, used about 160 GPU-hours and under 200 CPU core-hours. Every post-hoc analysis (ABTT, whitening, zeroing, bootstrap intervals) ran on CPU from cached vectors.

Fine-tuning ties the projection on LaTa routing and adds about two points on Qwen3-0.6B and KaLM-mini. Fine-tuning LaTa on the positive training pairs gives an uncorrected ranking AUROC of 0.984, above every zero-shot baseline cell (0.838 to 0.972), and 83.4 assignment accuracy. With ABTT on top it reaches 87.8 assignment accuracy and 85.2 DirAcc@1. In point estimate that is below every zero-shot ABTT cell (88.5 to 91.7 and 86.1 to 89.4). Against LaTa's own zero-shot ABTT cell, however, the DirAcc@1 difference is -0.9 points (-2.8 to 1.0; Table 17), and over five reseedings both reach 87.7. ABTT lowers the fine-tuned row's AUROC from 0.984 to 0.970 in point estimate (-0.014, interval -0.032 to 0.001) while it widens the cosine gap from 0.387 to 0.548. On the other two encoders the same recipe goes further. With ABTT, fine-tuned Qwen3-0.6B reaches 92.1 assignment accuracy and 91.4 DirAcc@1, and fine-tuned KaLM-mini 93.4 and 92.5, both above every zero-shot ABTT cell in point estimate. On the single split only KaLM-mini's DirAcc@1 lead over its zero-shot ABTT cell excludes zero ($+3.1$, 1.2 to 5.2). Over five reseedings the fine-tuned encoder with ABTT beats its zero-shot ABTT row by 1.8 points on Qwen3-0.6B (92.3 against 90.5) and by 2.1 points on KaLM-mini (93.0 against 90.9). The tie with the projection is therefore a statement about LaTa, not about post-processing in general. All three ceilings are also flattered: 206 of the 535 test witnesses whose directory already exists (38.5 percent) sit in a directory that supplied a training pair, and witnesses inside a directory are near-duplicates, so each margin is an upper bound.

Lexical reference systems. Table 20 gives the three lexical reference systems (BM25, character 3–5-gram TF-IDF, and Levenshtein similarity) on both tasks; the character n-gram row is the reference quoted in Section 7.

System	Ranking		Routing		
	AUROC	Cosine gap	Assign. acc.	DirAcc@1	DirAcc@1 5 reseedings
LaTa (zero-shot)	0.938	0.237	73.8	72.1	73.1 ± 1.0
LaTa (zero-shot) + ABTT	0.971	0.525	88.5	86.1	87.7 ± 0.4
LaTa (fine-tuned)	0.984	0.387	83.4	81.6	83.4 ± 0.9
LaTa (fine-tuned) + ABTT	0.970	0.548	87.8	85.2	87.7 ± 0.8
Qwen3-0.6B (zero-shot)	0.966	0.024	82.3	80.3	82.4 ± 1.0
Qwen3-0.6B (zero-shot) + ABTT	0.973	0.553	91.5	89.4	90.5 ± 0.4
Qwen3-0.6B (fine-tuned)	0.996	0.684	91.4	90.8	91.7 ± 0.6
Qwen3-0.6B (fine-tuned) + ABTT	0.994	0.716	92.1	91.4	92.3 ± 0.2
KaLM-mini (zero-shot)	0.972	0.056	87.5	85.9	87.3 ± 0.7
KaLM-mini (zero-shot) + ABTT	0.981	0.568	91.7	89.4	90.9 ± 0.6
KaLM-mini (fine-tuned)	0.997	0.636	92.4	91.7	92.6 ± 0.8
KaLM-mini (fine-tuned) + ABTT	0.994	0.651	93.4	92.5	93.0 ± 0.5

Table 19: Supervised fine-tuning reference ceiling on LaTa, Qwen3-0.6B and KaLM-mini. Fine-tuning raises ranking AUROC for every model, and ABTT on top of it lowers AUROC again in point estimate; on routing over the five reseedings, the fine-tuned encoder with ABTT is level with its zero-shot ABTT row for LaTa and above it for Qwen3-0.6B and KaLM-mini. The fine-tuned encoders are trained contrastively on 499 of the 565 positive training pairs (in-batch negatives, symmetric InfoNCE), with a dev slice carved out of the training split by directory for model selection; the test split is untouched until this evaluation. Ranking columns are test AUROC and cosine gap at the layer chosen by training AUROC; routing columns are test assignment accuracy and DirAcc@1, in percent, at the layer chosen by training DirAcc@1, with τ learned on the training split. The selected layers are listed in Table 21. The last column repeats DirAcc@1 at the same layer as a mean $\pm$ standard deviation over five reseedings of the query and reference files, with the training/test split fixed. ABTT rows for LaTa sweep D per layer on the training split and select $D = 10$ everywhere, which is the top of the grid $\{1, 2, 3, 5, 7, 10\}$, so the tuned and fixed variants coincide. The selected checkpoint for LaTa is epoch 7, the terminal epoch of the sweep under the 8-epoch budget, so this is a ceiling at this training budget rather than an asymptote. ABTT rows for Qwen3-0.6B sweep D per layer on the training split over the grid $\{1, 2, 3, 5, 7, 10\}$; 12 of 28 layer rows select the top of the grid. The selected checkpoint for Qwen3-0.6B is epoch 3 of 6 run under the 8-epoch budget, so this is a ceiling at this training budget rather than an asymptote. ABTT rows for KaLM-mini sweep D per layer on the training split over the grid $\{1, 2, 3, 5, 7, 10\}$; 19 of 24 layer rows select the top of the grid. The selected checkpoint for KaLM-mini is epoch 1 of 4 run under the 8-epoch budget, so this is a ceiling at this training budget rather than an asymptote. This is a reference ceiling, not a proposed method: it consumes supervision the zero-shot pipeline never sees.

	Ranking		Routing	
System	AUROC	Gap	Assign.	DirAcc@1
BM25 (word)	0.974	0.243	62.5	56.3
TF-IDF char 3–5	0.987	0.555	91.8	89.9
Levenshtein	0.952	0.362	87.8	86.2

Table 20: Lexical reference systems on the same 50/50 split, the same held-out test set and the same evaluation code as the embedding models, so the numbers are directly comparable to the headline ranking and routing tables. Ranking is measured over the test $n \times n$ score matrix: AUROC treats same-directory test pairs as positives, and Gap is the mean same-directory minus the mean different-directory score. Routing is reported in percent: Assign. is the existing-versus-new decision against a threshold τ fitted on the training split, and DirAcc@1 additionally requires the correct directory. TF-IDF char 3–5 is the strongest of the three at 89.9 percent DirAcc@1; on this corpus of hand-copied witnesses the best projected encoder configurations at best tie it. BM25 separates pairs well but routes poorly: a raw BM25 score grows with query length, so each file's maximum score tracks document length as much as relatedness, and the τ fitted on the training split transfers badly. BM25 uses $k_1 = 1.5$, $b = 0.75$ with document frequencies, vocabulary and average document length taken from training files only; the character TF-IDF vectorizer is likewise fitted on training files only, and normalized Levenshtein has nothing to fit. Every score matrix is min-max rescaled using the training block; the map is monotone, so AUROC is unchanged, while the threshold τ is fitted on a fixed grid afterwards, which matters for BM25 (unrescaled, τ pins at the grid ceiling) and is the identity for char TF-IDF, whose training scores already span [0, 1]. Under five reseedings of the query and reference files, in which only the query/reference partition of the test half is redrawn and the training/test split stays fixed, DirAcc@1 is BM25 (word) 68.3 $\pm$ 7.1; TF-IDF char 3–5 91.1 $\pm$ 0.5; Levenshtein 87.2 $\pm$ 0.3.

I Per-Layer Results

This appendix gives the per-layer numbers behind Figure 1 and the routing results at every layer. Table 21 lists the train-selected layer behind every headline cell. Tables 22, 23 and 24 give ranking AUROC, routing assignment accuracy and routing DirAcc@1 at every layer of LaTa, PhilTa and mT5-base. Tables 25, 26 and 27 give the same for LaBSE, Qwen3-0.6B and KaLM-mini. Each compares baseline mean pooling with ABTT, its D tuned per layer on the training split. Fixing the number of removed components at ten gives the same range (AUROC 0.962 to 0.987 at all 100 panel layers), so the result does not rest on how that number is chosen.

	Ranking layer				**Routing layer**			
Model	Base	SIF	ABTT	SIF+ABTT	Base	SIF	ABTT	SIF+ABTT
LaTa	12	1	12	7	1	1	8	12
PhilTa	1	1	9	8	1	1	1	1
mT5-base	12	1	2	1	12	1	1	2
LaBSE	12	12	11	10	11	12	11	11
Qwen3-0.6B	26	22	2	2	28	27	5	5
KaLM-mini	23	23	1	2	23	22	3	1
LaTa (fine-tuned)	12	–	12	–	12	–	12	–
Qwen3-0.6B (fine-tuned)	28	–	27	–	28	–	27	–
KaLM-mini (fine-tuned)	24	–	24	–	24	–	24	–

Table 21: Train-selected layer behind every headline cell (Tables 2 and 16). For each model and post-processing setting, the ranking layer has the highest training-set AUROC and the routing layer the highest training-set DirAcc@1; the test scores in those tables are read at these layers, so no layer is chosen on test. Layer 1 is the first transformer block. The fine-tuned rows below the rule have no SIF setting. Table 19 reads its zero-shot rows at the Base and ABTT layers above the rule and its fine-tuned rows at the layers below it.

Model	Layer	Ranking (T5 encoders)			
		AUROC (Base)	AUROC (ABTT)	Cosine gap (Base)	Cosine gap (ABTT)
LaTa	1	0.934	0.970	0.286	0.566
	2	0.563	0.969	0.019	0.554
	3	0.513	0.973	-0.048	0.555
	4	0.500	0.971	-0.067	0.565
	5	0.496	0.972	-0.069	0.572
	6	0.496	0.964	-0.069	0.569
	7	0.498	0.962	-0.072	0.564
	8	0.502	0.965	-0.074	0.558
	9	0.503	0.965	-0.073	0.557
	10	0.505	0.971	-0.072	0.551
	11	0.508	0.973	-0.067	0.550
	12	0.938	**0.971**	0.237	**0.525**
PhilTa	1	0.939	0.977	0.231	0.574
	2	0.911	0.973	0.161	0.553
	3	0.601	0.973	0.085	0.549
	4	0.551	0.978	0.063	0.551
	5	0.542	0.979	0.053	0.551
	6	0.541	0.981	0.047	0.551
	7	0.542	0.979	0.046	0.554
	8	0.542	0.984	0.046	0.555
	9	0.539	**0.983**	0.041	**0.540**
	10	0.538	0.982	0.040	0.531
	11	0.539	0.982	0.041	0.511
	12	0.911	0.982	0.181	0.487
mT5-base	1	0.822	0.979	0.027	0.550
	2	0.822	**0.975**	0.026	**0.536**
	3	0.804	0.968	0.025	0.517
	4	0.799	0.974	0.040	0.502
	5	0.654	0.978	0.219	0.482
	6	0.657	0.978	0.238	0.483
	7	0.656	0.980	0.244	0.482
	8	0.656	0.982	0.249	0.471
	9	0.660	0.982	0.235	0.475
	10	0.659	0.977	0.226	0.459
	11	0.659	0.974	0.213	0.449
	12	0.838	0.972	0.081	0.442

Table 22: Per-layer ranking metrics for the three T5 encoders (LaTa, PhilTa, mT5-base). Base is mean pooling without correction; ABTT is applied without SIF weighting, with D tuned per layer on the training split. Both AUROC and cosine gap are computed over the test n×n cosine matrix with no threshold and no directory routing: AUROC treats same-directory test pairs as positives, and cosine gap is the difference between mean same-directory and mean different-directory cosine. Bold marks the layer chosen on the training split, the layer with the highest training-set AUROC under ABTT, and its ABTT values; the other columns of that row stay plain, because the layer is selected for ABTT only. It is the layer behind the ABTT cell of Table 16 (listed in Table 21), and not always the layer with the highest test score in this table.

Model	Layer	Routing (T5 encoders, file-level, τ-thresholded)					
		Existing (Base)	Existing (ABTT)	New (Base)	New (ABTT)	Overall (Base)	Overall (ABTT)
LaTa	1	0.630	0.910	0.916	0.907	0.738	0.909
	2	0.153	0.905	0.913	0.892	0.439	0.900
	3	0.256	0.882	0.842	0.950	0.477	0.908
	4	0.389	0.893	0.718	0.920	0.513	0.903
	5	0.374	0.892	0.715	0.913	0.502	0.900
	6	0.378	0.884	0.712	0.913	0.503	0.895
	7	0.372	0.875	0.715	0.916	0.501	0.890
	8	0.379	**0.884**	0.706	**0.885**	0.502	**0.885**
	9	0.366	0.886	0.724	0.873	0.501	0.881
	10	0.344	0.873	0.752	0.882	0.498	0.876
	11	0.310	0.867	0.786	0.901	0.490	0.880
	12	0.576	0.890	0.932	0.833	0.710	0.868
PhilTa	**1**	0.591	**0.918**	0.904	**0.901**	0.709	**0.911**
	2	0.479	0.895	0.867	0.885	0.625	0.892
	3	0.193	0.897	0.889	0.879	0.455	0.890
	4	0.148	0.886	0.923	0.873	0.439	0.881
	5	0.252	0.879	0.796	0.873	0.457	0.876
	6	0.338	0.905	0.700	0.817	0.474	0.872
	7	0.301	0.901	0.746	0.830	0.469	0.874
	8	0.262	0.875	0.808	0.885	0.467	0.879
	9	0.303	0.890	0.762	0.870	0.476	0.882
	10	0.299	0.908	0.780	0.793	0.480	0.865
	11	0.256	0.869	0.824	0.836	0.470	0.857
	12	0.338	0.880	0.898	0.876	0.549	0.879
mT5-base	**1**	0.135	**0.877**	0.981	**0.947**	0.453	**0.903**
	2	0.110	0.890	0.975	0.910	0.436	0.897
	3	0.161	0.794	0.926	0.941	0.449	0.850
	4	0.269	0.787	0.858	0.882	0.491	0.823
	5	0.422	0.746	0.659	0.864	0.512	0.790
	6	0.636	0.766	0.489	0.876	0.580	0.808
	7	0.594	0.793	0.517	0.851	0.565	0.815
	8	0.557	0.809	0.560	0.842	0.558	0.822
	9	0.364	0.841	0.746	0.814	0.508	0.831
	10	0.314	0.794	0.783	0.876	0.491	0.825
	11	0.297	0.763	0.780	0.854	0.479	0.797
	12	0.198	0.834	0.926	0.799	0.472	0.821

Table 23: Per-layer routing accuracy for the three T5 encoders. Base is mean pooling without correction; ABTT is applied without SIF weighting, with D tuned per layer on the training split. For every test file we compute $s_i = \max_{j \neq i} \cos(e_i, e_j)$ against the rest of the test set and compare to the learned threshold τ (fit on the training split via an F_1-optimal cut on same-directory vs. different-directory pairs). The decision is binary first (existing if $s_i \geq \tau$, else new); only files predicted existing are then routed to their top-1 neighbor's directory. We report **existing accuracy** (restricted to files whose true partner is in the test set), **new accuracy** (files that should be flagged as novel), and **overall assignment accuracy**. Accuracies are fractions (0.885 is 88.5%). Bold marks the layer chosen on the training split, the layer with the highest training-set DirAcc@1 under ABTT, and its ABTT values; the other columns of that row stay plain, because the layer is selected for ABTT only. It is the layer behind the ABTT cell of Table 16 (listed in Table 21), and not always the layer with the highest test score in this table.

		Routing: DirAcc@1 (T5 encoders, single seed)					
Model	**Layer**	DirAcc@1 (Base)	DirAcc@1 (ABTT)	Existing (Base)	Existing (ABTT)	New (Base)	New (ABTT)
LaTa	1	0.721	0.880	0.630	0.910	0.916	0.907
	2	0.383	0.869	0.153	0.905	0.913	0.892
	3	0.350	0.882	0.256	0.882	0.842	0.950
	4	0.303	0.881	0.389	0.893	0.718	0.920
	5	0.297	0.875	0.374	0.892	0.715	0.913
	6	0.297	0.874	0.378	0.884	0.712	0.913
	7	0.302	0.868	0.372	0.875	0.715	0.916
	8	0.302	**0.861**	0.379	**0.884**	0.706	**0.885**
	9	0.309	0.858	0.366	0.886	0.724	0.873
	10	0.317	0.852	0.344	0.873	0.752	0.882
	11	0.331	0.857	0.310	0.867	0.786	0.901
	12	0.689	0.841	0.576	0.890	0.932	0.833
PhilTa	**1**	0.693	**0.883**	0.591	**0.918**	0.904	**0.901**
	2	0.603	0.869	0.479	0.895	0.867	0.885
	3	0.389	0.865	0.193	0.897	0.889	0.879
	4	0.369	0.859	0.148	0.886	0.923	0.873
	5	0.334	0.854	0.252	0.879	0.796	0.873
	6	0.310	0.845	0.338	0.905	0.700	0.817
	7	0.325	0.847	0.301	0.901	0.746	0.830
	8	0.350	0.857	0.262	0.875	0.808	0.885
	9	0.337	0.857	0.303	0.890	0.762	0.870
	10	0.344	0.833	0.299	0.908	0.780	0.793
	11	0.355	0.829	0.256	0.869	0.824	0.836
	12	0.528	0.854	0.338	0.880	0.898	0.876
mT5-base	**1**	0.451	**0.885**	0.135	**0.877**	0.981	**0.947**
	2	0.435	0.874	0.110	0.890	0.975	0.910
	3	0.445	0.830	0.161	0.794	0.926	0.941
	4	0.467	0.798	0.269	0.787	0.858	0.882
	5	0.352	0.756	0.422	0.746	0.659	0.864
	6	0.346	0.775	0.636	0.766	0.489	0.876
	7	0.352	0.775	0.594	0.793	0.517	0.851
	8	0.366	0.784	0.557	0.809	0.560	0.842
	9	0.394	0.787	0.364	0.841	0.746	0.814
	10	0.387	0.788	0.314	0.794	0.783	0.876
	11	0.378	0.765	0.297	0.763	0.780	0.854
	12	0.466	0.779	0.198	0.834	0.926	0.799

Table 24: Per-layer routing DirAcc@1, overall and for existing and new witnesses, for the three T5 encoders. Base is mean pooling without correction; ABTT is applied without SIF weighting, with D tuned per layer on the training split. This table is computed once, without the five reseedings of the query and reference files, which were run only for the SIF-conditioned variants (see Table 33 for the five-reseeding SIF+ABTT view). We report DirAcc@1 with an existing/new decomposition. Accuracies are fractions (0.885 is 88.5%). Bold marks the layer chosen on the training split, the layer with the highest training-set DirAcc@1 under ABTT, and its ABTT values; the other columns of that row stay plain, because the layer is selected for ABTT only. It is the layer behind the ABTT cell of Table 16 (listed in Table 21), and not always the layer with the highest test score in this table.

Model	Layer	Ranking (embedding-trained models)			
		AUROC (Base)	AUROC (ABTT)	Cosine gap (Base)	Cosine gap (ABTT)
LaBSE	1	0.806	0.977	0.060	0.520
	2	0.833	0.979	0.037	0.531
	3	0.853	0.978	0.062	0.543
	4	0.862	0.977	0.073	0.525
	5	0.864	0.968	0.066	0.501
	6	0.894	0.976	0.078	0.532
	7	0.887	0.975	0.088	0.524
	8	0.879	0.975	0.080	0.522
	9	0.916	0.979	0.090	0.565
	10	0.934	0.982	0.103	0.586
	11	0.960	**0.987**	0.061	**0.611**
	12	0.956	0.981	0.183	0.592
Qwen3-0.6B	1	0.858	0.978	0.033	0.564
	2	0.863	**0.973**	0.024	**0.553**
	3	0.862	0.980	0.037	0.556
	4	0.860	0.986	0.036	0.547
	5	0.861	0.981	0.030	0.545
	6	0.867	0.983	0.035	0.545
	7	0.876	0.980	0.033	0.544
	8	0.882	0.972	0.037	0.532
	9	0.881	0.977	0.034	0.532
	10	0.887	0.976	0.031	0.529
	11	0.895	0.977	0.030	0.530
	12	0.901	0.973	0.027	0.530
	13	0.907	0.974	0.026	0.523
	14	0.905	0.974	0.024	0.519
	15	0.901	0.971	0.022	0.514
	16	0.910	0.975	0.026	0.509
	17	0.916	0.979	0.032	0.521
	18	0.933	0.983	0.032	0.537
	19	0.936	0.980	0.035	0.541
	20	0.943	0.982	0.041	0.559
	21	0.953	0.982	0.047	0.562
	22	0.963	0.983	0.044	0.583
	23	0.961	0.984	0.038	0.578
	24	0.963	0.984	0.032	0.576
	25	0.961	0.983	0.027	0.578
	26	0.966	0.985	0.024	0.568
	27	0.967	0.982	0.017	0.559
	28	0.958	0.975	0.098	0.556
KaLM-mini	**1**	0.887	**0.981**	0.063	**0.568**
	2	0.906	0.983	0.044	0.566
	3	0.881	0.981	0.062	0.551
	4	0.871	0.983	0.060	0.553
	5	0.861	0.983	0.062	0.541
	6	0.872	0.983	0.063	0.529
	7	0.888	0.979	0.072	0.524
	8	0.890	0.976	0.072	0.524
	9	0.892	0.974	0.066	0.507
	10	0.893	0.971	0.067	0.494
	11	0.894	0.970	0.061	0.475
	12	0.905	0.970	0.071	0.475
	13	0.900	0.968	0.066	0.467
	14	0.904	0.971	0.063	0.471
	15	0.907	0.973	0.067	0.468
	16	0.917	0.972	0.071	0.489
	17	0.926	0.975	0.075	0.501
	18	0.945	0.977	0.060	0.519
	19	0.956	0.979	0.061	0.531
	20	0.960	0.980	0.054	0.555
	21	0.962	0.981	0.043	0.555
	22	0.971	0.985	0.050	0.555
	23	0.972	0.986	0.056	0.552

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Model	Layer	Ranking (embedding-trained models)			
		AUROC (Base)	AUROC (ABTT)	Cosine gap (Base)	Cosine gap (ABTT)
	24	0.956	0.981	0.243	0.520

Table 25: Per-layer ranking metrics for the embedding-trained models (LaBSE, Qwen3-0.6B, KaLM-mini), under the same two-method comparison as Table 22. Base is mean pooling without correction; ABTT is applied without SIF weighting, with D tuned per layer on the training split. Both AUROC and cosine gap are computed over the test n×n cosine matrix with no threshold and no directory routing: AUROC treats same-directory test pairs as positives, and cosine gap is the difference between mean same-directory and mean different-directory cosine. Bold marks the layer chosen on the training split, the layer with the highest training-set AUROC under ABTT, and its ABTT values; the other columns of that row stay plain, because the layer is selected for ABTT only. It is the layer behind the ABTT cell of Table 16 (listed in Table 21), and not always the layer with the highest test score in this table.

Model	Layer	Routing (embedding-trained models)					
		Existing (Base)	Existing (ABTT)	New (Base)	New (ABTT)	Overall (Base)	Overall (ABTT)
LaBSE	1	0.222	0.879	0.898	0.873	0.477	0.876
	2	0.381	0.875	0.811	0.827	0.543	0.857
	3	0.277	0.886	0.907	0.867	0.514	0.879
	4	0.372	0.897	0.885	0.836	0.565	0.874
	5	0.327	0.845	0.901	0.851	0.543	0.847
	6	0.548	0.860	0.882	0.851	0.674	0.857
	7	0.398	0.849	0.892	0.830	0.584	0.841
	8	0.385	0.869	0.870	0.743	0.568	0.822
	9	0.421	0.871	0.929	0.817	0.612	0.851
	10	0.606	0.892	0.892	0.805	0.713	0.859
	11	0.785	**0.923**	0.913	**0.882**	0.833	**0.908**
	12	0.729	0.914	0.938	0.864	0.808	0.895
Qwen3-0.6B	1	0.264	0.921	0.960	0.885	0.526	0.908
	2	0.437	0.869	0.814	0.916	0.579	0.887
	3	0.250	0.914	0.947	0.907	0.513	0.911
	4	0.411	0.936	0.848	0.842	0.576	0.901
	5	0.181	**0.910**	0.932	**0.923**	0.464	**0.915**
	6	0.422	0.935	0.830	0.796	0.576	0.882
	7	0.250	0.890	0.947	0.876	0.513	0.885
	8	0.409	0.931	0.895	0.799	0.592	0.881
	9	0.460	0.901	0.851	0.851	0.607	0.882
	10	0.533	0.892	0.836	0.864	0.647	0.881
	11	0.293	0.903	0.978	0.808	0.551	0.867
	12	0.407	0.912	0.938	0.817	0.607	0.876
	13	0.432	0.908	0.944	0.836	0.625	0.881
	14	0.456	0.907	0.926	0.811	0.633	0.871
	15	0.469	0.901	0.920	0.845	0.639	0.880
	16	0.639	0.897	0.830	0.808	0.711	0.864
	17	0.480	0.886	0.935	0.851	0.652	0.873
	18	0.553	0.882	0.944	0.839	0.700	0.866
	19	0.514	0.871	0.966	0.858	0.684	0.866
	20	0.654	0.865	0.898	0.892	0.746	0.875
	21	0.647	0.912	0.950	0.898	0.761	0.907
	22	0.731	0.888	0.938	0.916	0.809	0.899
	23	0.622	0.914	0.966	0.895	0.752	0.907
	24	0.748	0.901	0.935	0.935	0.818	0.914
	25	0.574	0.893	0.975	0.923	0.725	0.904
	26	0.613	0.899	0.975	0.895	0.749	0.897
	27	0.793	0.877	0.833	0.923	0.808	0.894
	28	0.796	0.877	0.867	0.913	0.823	0.890
KaLM-mini	1	0.391	0.921	0.907	0.920	0.585	0.921
	2	0.493	0.916	0.848	0.879	0.627	0.902
	3	0.443	**0.923**	0.870	**0.907**	0.604	**0.917**
	4	0.308	0.931	0.935	0.895	0.544	0.917
	5	0.338	0.905	0.882	0.923	0.543	0.911

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Model	Layer	Routing (embedding-trained models)					
		Existing (Base)	Existing (ABTT)	New (Base)	New (ABTT)	Overall (Base)	Overall (ABTT)
	6	0.256	0.895	0.910	0.920	0.502	0.904
	7	0.475	0.864	0.854	0.944	0.618	0.894
	8	0.467	0.862	0.870	0.913	0.619	0.881
	9	0.469	0.873	0.885	0.870	0.626	0.872
	10	0.449	0.875	0.885	0.870	0.613	0.873
	11	0.458	0.862	0.885	0.864	0.619	0.862
	12	0.507	0.862	0.895	0.904	0.653	0.878
	13	0.540	0.860	0.895	0.870	0.674	0.864
	14	0.593	0.836	0.845	0.885	0.688	0.854
	15	0.566	0.864	0.898	0.827	0.691	0.850
	16	0.550	0.849	0.935	0.830	0.695	0.841
	17	0.675	0.864	0.920	0.889	0.767	0.873
	18	0.697	0.875	0.960	0.898	0.796	0.883
	19	0.721	0.873	0.966	0.892	0.814	0.880
	20	0.705	0.791	0.966	0.913	0.803	0.837
	21	0.727	0.811	0.966	0.926	0.817	0.854
	22	0.850	0.879	0.916	0.923	0.875	0.895
	23	0.849	0.884	0.920	0.898	0.875	0.889
	24	0.787	0.892	0.944	0.867	0.846	0.882

Table 26: Per-layer routing accuracy for the embedding-trained models. Base is mean pooling without correction; ABTT is applied without SIF weighting, with D tuned per layer on the training split. For every test file we compute $s_i = \max_{j \neq i} \cos(e_i, e_j)$ against the rest of the test set and compare to the learned threshold τ (fit on the training split via an F_1-optimal cut on same-directory vs. different-directory pairs). The decision is binary first (existing if $s_i \geq \tau$, else new); only files predicted existing are then routed to their top-1 neighbor's directory. We report **existing accuracy** (restricted to files whose true partner is in the test set), **new accuracy** (files that should be flagged as novel), and **overall assignment accuracy**. Accuracies are fractions (0.885 is 88.5%). Bold marks the layer chosen on the training split, the layer with the highest training-set DirAcc@1 under ABTT, and its ABTT values; the other columns of that row stay plain, because the layer is selected for ABTT only. It is the layer behind the ABTT cell of Table 16 (listed in Table 21), and not always the layer with the highest test score in this table.

Model	Layer	Routing: DirAcc@1 (embedding-trained models, single seed)					
		DirAcc@1 (Base)	DirAcc@1 (ABTT)	Existing (Base)	Existing (ABTT)	New (Base)	New (ABTT)
LaBSE	1	0.465	0.851	0.222	0.879	0.898	0.873
	2	0.494	0.830	0.381	0.875	0.811	0.827
	3	0.500	0.848	0.277	0.886	0.907	0.867
	4	0.542	0.841	0.372	0.897	0.885	0.836
	5	0.521	0.811	0.327	0.845	0.901	0.851
	6	0.640	0.831	0.548	0.860	0.882	0.851
	7	0.561	0.810	0.398	0.849	0.892	0.830
	8	0.544	0.779	0.385	0.869	0.870	0.743
	9	0.608	0.823	0.421	0.871	0.929	0.817
	10	0.693	0.832	0.606	0.892	0.892	0.805
	11	0.815	**0.885**	0.785	**0.923**	0.913	**0.882**
	12	0.791	0.868	0.729	0.914	0.938	0.864
Qwen3-0.6B	1	0.521	0.875	0.264	0.921	0.960	0.885
	2	0.558	0.871	0.437	0.869	0.814	0.916
	3	0.508	0.890	0.250	0.914	0.947	0.907
	4	0.559	0.874	0.411	0.936	0.848	0.842
	5	0.450	**0.894**	0.181	**0.910**	0.932	**0.923**
	6	0.551	0.858	0.422	0.935	0.830	0.796
	7	0.507	0.859	0.250	0.890	0.947	0.876
	8	0.580	0.852	0.409	0.931	0.895	0.799
	9	0.587	0.852	0.460	0.901	0.851	0.851
	10	0.622	0.848	0.533	0.892	0.836	0.864
	11	0.548	0.830	0.293	0.903	0.978	0.808
	12	0.603	0.848	0.407	0.912	0.938	0.817
	13	0.618	0.851	0.432	0.908	0.944	0.836

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Model	Layer	Routing: DirAcc@1 (embedding-trained models, single seed)					
		DirAcc@1 (Base)	DirAcc@1 (ABTT)	Existing (Base)	Existing (ABTT)	New (Base)	New (ABTT)
	14	0.621	0.840	0.456	0.907	0.926	0.811
	15	0.627	0.848	0.469	0.901	0.920	0.845
	16	0.682	0.830	0.639	0.897	0.830	0.808
	17	0.641	0.840	0.480	0.886	0.935	0.851
	18	0.691	0.837	0.553	0.882	0.944	0.839
	19	0.676	0.837	0.514	0.871	0.966	0.858
	20	0.733	0.852	0.654	0.865	0.898	0.892
	21	0.752	0.881	0.647	0.912	0.950	0.898
	22	0.795	0.876	0.731	0.888	0.938	0.916
	23	0.742	0.886	0.622	0.914	0.966	0.895
	24	0.808	0.896	0.748	0.901	0.935	0.935
	25	0.718	0.886	0.574	0.893	0.975	0.923
	26	0.740	0.881	0.613	0.899	0.975	0.895
	27	0.787	0.876	0.793	0.877	0.833	0.923
	28	0.803	0.872	0.796	0.877	0.867	0.913
KaLM-mini	1	0.577	0.902	0.391	0.921	0.907	0.920
	2	0.607	0.879	0.493	0.916	0.848	0.879
	3	0.587	**0.894**	0.443	**0.923**	0.870	**0.907**
	4	0.538	0.887	0.308	0.931	0.935	0.895
	5	0.524	0.882	0.338	0.905	0.882	0.923
	6	0.490	0.876	0.256	0.895	0.910	0.920
	7	0.592	0.873	0.475	0.864	0.854	0.944
	8	0.597	0.860	0.467	0.862	0.870	0.913
	9	0.601	0.843	0.469	0.873	0.885	0.870
	10	0.584	0.844	0.449	0.875	0.885	0.870
	11	0.586	0.836	0.458	0.862	0.885	0.864
	12	0.622	0.850	0.507	0.862	0.895	0.904
	13	0.633	0.840	0.540	0.860	0.895	0.870
	14	0.636	0.829	0.593	0.836	0.845	0.885
	15	0.650	0.817	0.566	0.864	0.898	0.827
	16	0.676	0.819	0.550	0.849	0.935	0.830
	17	0.747	0.852	0.675	0.864	0.920	0.889
	18	0.777	0.865	0.697	0.875	0.960	0.898
	19	0.801	0.861	0.721	0.873	0.966	0.892
	20	0.789	0.817	0.705	0.791	0.966	0.913
	21	0.802	0.831	0.727	0.811	0.966	0.926
	22	0.857	0.875	0.850	0.879	0.916	0.923
	23	0.859	0.872	0.849	0.884	0.920	0.898
	24	0.833	0.858	0.787	0.892	0.944	0.867

Table 27: Per-layer routing DirAcc@1, overall and for existing and new witnesses, for the embedding-trained models. Base is mean pooling without correction; ABTT is applied without SIF weighting, with D tuned per layer on the training split. This table is computed once, without the five reseedings of the query and reference files, which were run only for the SIF-conditioned variants (see Table 33 for the five-reseeding SIF+ABTT view). We report DirAcc@1 with an existing/new decomposition. Accuracies are fractions (0.885 is 88.5%). Bold marks the layer chosen on the training split, the layer with the highest training-set DirAcc@1 under ABTT, and its ABTT values; the other columns of that row stay plain, because the layer is selected for ABTT only. It is the layer behind the ABTT cell of Table 16 (listed in Table 21), and not always the layer with the highest test score in this table.

J Last-Token versus Mean Pooling

Decoder-only embedding models often use last-token pooling, so this appendix checks that mean pooling, which every main-text result uses, does not handicap them. We compare the two poolings under the same ABTT pipeline, training and test split, and ABTT choices. Table 28 reports, for each model and pooling, the layer chosen on the training split by DirAcc@1, the rule behind the routing cells of Table 2, with test assignment accuracy, DirAcc@1 and cosine gap at that layer. Mean pooling routes better for each of the five models in this run, the two decoders included; mT5-base was not part of it.

Model	Pooling	Layer	D	Assignment acc.	DirAcc@1	Cosine gap
LaTa	Mean	8	10	0.885	0.861	0.558
	Last token	7	10	0.664	0.466	0.308
PhilTa	Mean	1	10	0.911	0.883	0.573
	Last token	11	10	0.589	0.406	0.205
LaBSE	Mean	11	10	0.908	0.885	0.610
	Last token	12	10	0.895	0.871	0.586
Qwen3-0.6B	Mean	5	10	0.915	0.894	0.545
	Last token	26	5	0.881	0.864	0.580
KaLM-mini	Mean	3	10	0.917	0.894	0.552
	Last token	24	10	0.875	0.847	0.483

Table 28: Mean versus last-token pooling under ABTT with D tuned per layer on the training split, for the five models whose last-token embeddings were extracted (mT5-base was not part of this run). For each model and pooling the layer is the one with the highest training-set DirAcc@1, the rule behind the routing cells of Table 2 (their layers are listed in Table 21); D is the number of principal components removed at that layer. The three score columns are test-set routing assignment accuracy, routing DirAcc@1, and ranking cosine gap at that layer; the two accuracies are fractions (0.885 is 88.5%).

K SIF and Whitening Baselines

Section 3 compares SIF frequency weighting and reduced-rank whitening with ABTT; this appendix gives both. Tables 31, 32 and 33 report SIF alone, SIF followed by ABTT with $D = 10$, and SIF followed by ABTT with D tuned per layer, at every layer of the six panel models, with one primary metric per task.

SIF. SIF acts before pooling. It can down-weight the frequent tokens that feed the dominant direction (Appendix C), but it removes no direction from the pooled vector. At every collapsed layer it stays below ABTT.

Shortlist accuracy. In the shortlist form of routing, each directory receives its best witness match score, a pseudo-option `NEW` receives score τ, and the simulated expert succeeds when the labeled directory appears in the shortlist. Shortlist accuracy, DirAcc@K, is the fraction of test queries whose labeled directory is among the top K options. Table 29 gives this shortlist view under SIF+ABTT, over five reseedings (Appendix H). DirAcc@2 ranges from 95.1 to 96.1 percent across the six models, and little is gained after DirAcc@3.

Whitening. Full-rank PCA whitening (Su et al., 2021) is ill-conditioned here, since it estimates 768 to 1,024 dimensions from 847 training vectors (condition numbers 3×10^5 to 10^{16}). It ranks below ABTT at every layer: its test AUROC is 0.820–0.876 for the T5 encoders and LaBSE, and 0.552–0.643 for the two decoders. It also breaks the routing threshold. Full-rank whitening leaves the different-directory cosine mean at or below 0.015, so across all 100 whitening cells every file is routed to an existing directory, existing accuracy is 1.000, and new accuracy never exceeds one file in 323. Assignment accuracy then sits at the 62.4 percent class prior. That prior beats the baseline in 60 of 100 cells while DirAcc@1 never exceeds 41.5 percent, so the apparent gain is a thresholding artifact. Whitening to a reduced dimension, as Su et al. (2021) also do, drops the near-null directions and is the fairer comparison (Table 30). Fit on training embeddings with $r = 128$ or 256 components, it ranks on par with ABTT. At each model's train-selected layer, $r = 128$ reaches test AUROC 0.973–0.984, against 0.971–0.987 for ABTT. Over all 100 panel layers, collapsed layers included, both $r = 128$ and $r = 256$ stay within 0.021 of ABTT; $r = 64$ falls slightly below. On ranking, the repair therefore does not depend on ABTT in particular. On routing it does. At each model's train-selected routing layer, $r = 128$ reaches DirAcc@1 1.9 to 4.5 points below ABTT, and $r = 256$ up to 15 points below. We therefore keep ABTT as the reference correction.

Model	DirAcc@1	DirAcc@2	DirAcc@3	DirAcc@4	DirAcc@5
LaTa	88.9 ± 0.5	96.1 ± 0.5	97.4 ± 0.6	97.9 ± 0.3	98.2 ± 0.2
PhilTa	89.8 ± 0.8	95.8 ± 0.6	97.4 ± 0.3	97.7 ± 0.3	97.8 ± 0.4
mT5-base	89.3 ± 0.5	95.1 ± 0.2	96.7 ± 0.7	97.1 ± 0.4	97.2 ± 0.4
LaBSE	90.6 ± 1.0	95.5 ± 0.5	96.4 ± 0.5	97.2 ± 0.6	97.8 ± 0.5
Qwen3-0.6B	89.3 ± 0.7	95.7 ± 0.4	97.1 ± 0.3	97.5 ± 0.4	97.6 ± 0.4
KaLM-mini	89.3 ± 0.5	95.1 ± 0.4	96.6 ± 0.7	97.2 ± 0.7	97.6 ± 0.5

Table 29: Routing shortlist accuracy under SIF+ABTT: DirAcc@K, as mean ± standard deviation over five reseedings of the query and reference files, with the training/test split fixed. For each model the reported layer is chosen on the training split: the layer with the highest training-set DirAcc@1 for SIF+ABTT in the single-seed run, the layer behind the SIF+ABTT cell of Table 16, listed in Table 21. D is tuned per layer on the training split. This is not always the layer with the highest mean over the five reseedings. Each row is the bold row of Table 33. Each cell is the percentage of test queries whose labeled directory appears within the top K options. Per-layer shortlist accuracy: Tables 24, 27, and 33.

	Ranking AUROC						Routing DirAcc@1					Routing assignment				
			whitening, r					whitening, r					whitening, r			
Model	Base	ABTT	64	128	256	full	ABTT	64	128	256	full	ABTT	64	128	256	full
LaTa	0.938	0.971	0.969	0.973	0.983	0.872	86.1	81.2	83.1	77.0	41.1	88.5	84.7	86.2	80.7	62.5
PhilTa	0.939	0.983	0.974	0.977	0.979	0.854	88.3	83.0	86.4	83.7	41.0	91.1	86.5	88.8	86.8	62.5
mT5-base	0.838	0.975	0.971	0.979	0.974	0.845	88.5	78.9	85.8	83.8	41.0	90.3	82.6	88.6	87.2	62.5
LaBSE	0.956	0.987	0.979	0.984	0.976	0.866	88.5	86.1	84.0	73.5	40.3	90.8	88.8	86.7	78.0	62.5
Qwen3-0.6B	0.966	0.973	0.973	0.978	0.983	0.594	89.4	84.8	85.0	75.9	20.5	91.5	87.2	88.0	79.3	62.4
KaLM-mini	0.972	0.981	0.976	0.981	0.981	0.562	89.4	82.6	86.7	84.5	16.8	91.7	85.4	89.2	87.8	62.4

Table 30: PCA whitening to r components against the baseline and ABTT. Whitening keeps the top r principal components of the centered training embeddings and rescales each to unit variance. Full keeps every component: 768 for the T5 encoders and LaBSE, and 847, the number of training passages, for Qwen3-0.6B and KaLM-mini, whose last component then has numerically zero variance. Ranking cells are read at each setting's train-selected layer (highest training AUROC), routing cells at the layer with the highest training DirAcc@1. Routing values are single-split test DirAcc@1 and assignment accuracy in percent, with the threshold learned on training pairs. Every transform is fit on training embeddings only. Whitening ranks on par with ABTT, but routes below it.

Model	Layer	Ranking: SIF-suite AUROC (six models)			
		AUROC (Base)	AUROC (SIF)	AUROC (SIF+ABTT$_{10}$)	AUROC (SIF+ABTT)
LaTa	1	0.934	0.948	0.972	0.976
	2	0.563	0.938	0.964	0.971
	3	0.513	0.921	0.967	0.966
	4	0.500	0.890	0.970	0.968
	5	0.496	0.878	0.970	0.970
	6	0.496	0.865	0.965	0.965
	7	0.498	0.858	0.966	**0.966**
	8	0.502	0.842	0.964	0.964
	9	0.503	0.854	0.964	0.964
	10	0.505	0.859	0.966	0.966
	11	0.508	0.867	0.965	0.965
	12	0.938	0.948	0.963	0.963
PhilTa	1	0.939	0.955	0.975	0.975
	2	0.911	0.941	0.973	0.973
	3	0.601	0.928	0.968	0.968
	4	0.551	0.916	0.972	0.972
	5	0.542	0.904	0.972	0.972
	6	0.541	0.884	0.981	0.981
	7	0.542	0.891	0.981	0.981
	8	0.542	0.895	0.982	**0.982**
	9	0.539	0.881	0.977	0.977
	10	0.538	0.874	0.977	0.977
	11	0.539	0.872	0.979	0.979
	12	0.911	0.929	0.969	0.969

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Model	Layer	Ranking: SIF-suite AUROC (six models)			
		AUROC (Base)	AUROC (SIF)	AUROC (SIF+ABTT$_{10}$)	AUROC (SIF+ABTT)
mT5-base	**1**	0.822	0.845	0.974	**0.974**
	2	0.822	0.835	0.966	0.966
	3	0.804	0.808	0.958	0.958
	4	0.799	0.809	0.966	0.966
	5	0.654	0.672	0.974	0.974
	6	0.657	0.678	0.971	0.971
	7	0.656	0.680	0.967	0.967
	8	0.656	0.681	0.968	0.968
	9	0.660	0.685	0.967	0.967
	10	0.659	0.684	0.966	0.966
	11	0.659	0.689	0.966	0.966
	12	0.838	0.833	0.970	0.970
LaBSE	1	0.806	0.818	0.974	0.974
	2	0.833	0.840	0.975	0.975
	3	0.853	0.862	0.968	0.968
	4	0.862	0.869	0.966	0.966
	5	0.864	0.869	0.964	0.964
	6	0.894	0.895	0.965	0.965
	7	0.887	0.892	0.964	0.964
	8	0.879	0.883	0.968	0.968
	9	0.916	0.915	0.973	0.973
	10	0.934	0.934	0.976	**0.976**
	11	0.960	0.959	0.983	0.983
	12	0.956	0.956	0.980	0.984
Qwen3-0.6B	1	0.858	0.852	0.982	0.981
	2	0.863	0.853	0.972	**0.972**
	3	0.862	0.858	0.977	0.975
	4	0.860	0.860	0.976	0.977
	5	0.861	0.858	0.971	0.971
	6	0.867	0.861	0.974	0.974
	7	0.876	0.866	0.971	0.971
	8	0.882	0.868	0.969	0.969
	9	0.881	0.866	0.968	0.968
	10	0.887	0.870	0.966	0.966
	11	0.895	0.875	0.964	0.964
	12	0.901	0.881	0.966	0.966
	13	0.907	0.887	0.967	0.967
	14	0.905	0.885	0.966	0.966
	15	0.901	0.883	0.968	0.968
	16	0.910	0.894	0.965	0.965
	17	0.916	0.899	0.967	0.974
	18	0.933	0.914	0.966	0.973
	19	0.936	0.915	0.966	0.976
	20	0.943	0.918	0.971	0.977
	21	0.953	0.931	0.971	0.971
	22	0.963	0.942	0.973	0.973
	23	0.961	0.941	0.973	0.973
	24	0.963	0.943	0.976	0.974
	25	0.961	0.938	0.978	0.975
	26	0.966	0.942	0.980	0.980
	27	0.967	0.936	0.978	0.978
	28	0.958	0.925	0.968	0.968
KaLM-mini	1	0.887	0.880	0.982	0.980
	2	0.906	0.899	0.981	**0.981**
	3	0.881	0.882	0.977	0.977
	4	0.871	0.873	0.974	0.974
	5	0.861	0.862	0.971	0.971
	6	0.872	0.867	0.967	0.967
	7	0.888	0.879	0.972	0.972
	8	0.890	0.880	0.973	0.975
	9	0.892	0.882	0.968	0.968
	10	0.893	0.882	0.964	0.964
	11	0.894	0.885	0.969	0.969

(continued on next page)

Model	Layer	Ranking: SIF-suite AUROC (six models)			
		AUROC (Base)	AUROC (SIF)	AUROC (SIF+ABTT$_{10}$)	AUROC (SIF+ABTT)
	12	0.905	0.896	0.966	0.966
	13	0.900	0.894	0.964	0.964
	14	0.904	0.898	0.970	0.970
	15	0.907	0.901	0.970	0.970
	16	0.917	0.913	0.967	0.967
	17	0.926	0.923	0.968	0.968
	18	0.945	0.939	0.977	0.977
	19	0.956	0.950	0.978	0.978
	20	0.960	0.956	0.978	0.978
	21	0.962	0.958	0.980	0.983
	22	0.971	0.967	0.983	0.984
	23	0.972	0.964	0.986	0.985
	24	0.956	0.956	0.976	0.976

Table 31: Per-layer ranking AUROC across the SIF-conditioned post-processing suite for all six models. SIF replaces mean pooling with SIF-weighted pooling; SIF+ABTT$_{10}$ adds ABTT with fixed D=10; SIF+ABTT tunes D per layer on the training split. Gap is omitted to keep the table narrow; the pure-ABTT comparison (not SIF-conditioned) is in Tables 22 and 25. Bold marks the layer chosen on the training split, the layer with the highest training-set AUROC under SIF+ABTT, and its SIF+ABTT values; the other columns of that row stay plain, because the layer is selected for SIF+ABTT only. It is the layer behind the SIF+ABTT cell of Table 16 (listed in Table 21), and not always the layer with the highest test score in this table.

Model	Layer	Routing: SIF-suite assignment accuracy (six models)			
		Overall (Base)	Overall (SIF)	Overall (SIF+ABTT$_{10}$)	Overall (SIF+ABTT)
LaTa	1	0.738	0.832	0.920	0.899
	2	0.439	0.791	0.904	0.900
	3	0.477	0.720	0.910	0.907
	4	0.513	0.668	0.900	0.902
	5	0.502	0.648	0.906	0.906
	6	0.503	0.668	0.895	0.895
	7	0.501	0.642	0.903	0.903
	8	0.502	0.614	0.892	0.892
	9	0.501	0.618	0.889	0.889
	10	0.498	0.655	0.890	0.890
	11	0.490	0.621	0.880	0.880
	12	0.710	0.804	0.902	**0.902**
PhilTa	**1**	0.709	0.790	0.914	**0.914**
	2	0.625	0.738	0.892	0.892
	3	0.455	0.713	0.892	0.892
	4	0.439	0.663	0.886	0.886
	5	0.457	0.626	0.899	0.899
	6	0.474	0.612	0.906	0.906
	7	0.469	0.625	0.906	0.906
	8	0.467	0.614	0.886	0.886
	9	0.476	0.590	0.895	0.895
	10	0.480	0.580	0.888	0.888
	11	0.470	0.565	0.886	0.886
	12	0.549	0.667	0.899	0.899
mT5-base	1	0.453	0.545	0.907	0.907
	2	0.436	0.537	0.903	**0.903**
	3	0.449	0.568	0.885	0.885
	4	0.491	0.488	0.857	0.857
	5	0.512	0.483	0.826	0.826
	6	0.580	0.558	0.817	0.817
	7	0.565	0.554	0.832	0.832
	8	0.558	0.548	0.834	0.834
	9	0.508	0.495	0.845	0.845
	10	0.491	0.495	0.832	0.832

(continued on next page)

Model	Layer	Routing: SIF-suite assignment accuracy (six models)			
		Overall (Base)	Overall (SIF)	Overall (SIF+ABTT$_{10}$)	Overall (SIF+ABTT)
	11	0.479	0.484	0.816	0.816
	12	0.472	0.541	0.858	0.858
LaBSE	1	0.477	0.521	0.886	0.886
	2	0.543	0.508	0.875	0.875
	3	0.514	0.577	0.860	0.860
	4	0.565	0.587	0.869	0.869
	5	0.543	0.598	0.857	0.857
	6	0.674	0.648	0.872	0.872
	7	0.584	0.634	0.858	0.858
	8	0.568	0.571	0.854	0.854
	9	0.612	0.613	0.859	0.859
	10	0.713	0.698	0.864	0.864
	11	0.833	0.807	0.896	**0.896**
	12	0.808	0.834	0.890	0.880
Qwen3-0.6B	1	0.526	0.566	0.914	0.909
	2	0.579	0.524	0.894	0.894
	3	0.513	0.537	0.904	0.901
	4	0.576	0.531	0.899	0.904
	5	0.464	0.537	0.908	**0.908**
	6	0.576	0.526	0.906	0.906
	7	0.513	0.573	0.911	0.911
	8	0.592	0.627	0.911	0.911
	9	0.607	0.576	0.910	0.910
	10	0.647	0.599	0.900	0.900
	11	0.551	0.606	0.893	0.893
	12	0.607	0.663	0.900	0.900
	13	0.625	0.675	0.897	0.897
	14	0.633	0.571	0.893	0.893
	15	0.639	0.600	0.893	0.893
	16	0.711	0.679	0.880	0.880
	17	0.652	0.686	0.900	0.876
	18	0.700	0.731	0.899	0.882
	19	0.684	0.700	0.886	0.850
	20	0.746	0.676	0.882	0.866
	21	0.761	0.739	0.892	0.892
	22	0.809	0.719	0.893	0.893
	23	0.752	0.759	0.893	0.893
	24	0.818	0.706	0.900	0.899
	25	0.725	0.748	0.897	0.893
	26	0.749	0.772	0.903	0.903
	27	0.808	0.760	0.899	0.899
	28	0.823	0.774	0.890	0.890
KaLM-mini	**1**	0.585	0.604	0.906	**0.908**
	2	0.627	0.594	0.908	0.908
	3	0.604	0.576	0.907	0.907
	4	0.544	0.577	0.901	0.901
	5	0.543	0.533	0.897	0.897
	6	0.502	0.558	0.900	0.900
	7	0.618	0.593	0.901	0.901
	8	0.619	0.607	0.892	0.878
	9	0.626	0.663	0.876	0.876
	10	0.613	0.603	0.866	0.866
	11	0.619	0.627	0.874	0.874
	12	0.653	0.642	0.874	0.874
	13	0.674	0.666	0.864	0.864
	14	0.688	0.648	0.861	0.861
	15	0.691	0.698	0.866	0.866
	16	0.695	0.738	0.859	0.859
	17	0.767	0.763	0.882	0.882
	18	0.796	0.796	0.879	0.879
	19	0.814	0.804	0.878	0.878
	20	0.803	0.821	0.876	0.876
	21	0.817	0.832	0.874	0.876

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Model	Layer	Routing: SIF-suite assignment accuracy (six models)			
		Overall (Base)	Overall (SIF)	Overall (SIF+ABTT$_{10}$)	Overall (SIF+ABTT)
	22	0.875	0.857	0.876	0.876
	23	0.875	0.862	0.880	0.881
	24	0.846	0.847	0.869	0.869

Table 32: Per-layer routing assignment accuracy across the SIF-conditioned suite for all six models. Restricted to overall routing accuracy (existing/new decomposition omitted) to keep the table compact; see Tables 23 and 26 for the pure-ABTT pairwise comparison with existing/new broken out. Accuracies are fractions (0.885 is 88.5%). Bold marks the layer chosen on the training split, the layer with the highest training-set DirAcc@1 under SIF+ABTT, and its SIF+ABTT values; the other columns of that row stay plain, because the layer is selected for SIF+ABTT only. It is the layer behind the SIF+ABTT cell of Table 16 (listed in Table 21), and not always the layer with the highest test score in this table.

Model	Layer	Routing: DirAcc@1 (five reseedings, mean ± std, SIF+ABTT)					
		DirAcc@1 (Base)	DirAcc@1 (SIF+ABTT)	Existing (Base)	Existing (SIF+ABTT)	New (Base)	New (SIF+ABTT)
LaTa	1	0.731 ± 0.010	0.889 ± 0.010	0.563 ± 0.007	0.844 ± 0.011	0.949 ± 0.011	0.948 ± 0.010
	2	0.429 ± 0.011	0.888 ± 0.007	0.049 ± 0.006	0.841 ± 0.007	0.920 ± 0.008	0.948 ± 0.007
	3	0.393 ± 0.011	0.891 ± 0.010	0.047 ± 0.006	0.847 ± 0.010	0.840 ± 0.020	0.947 ± 0.012
	4	0.331 ± 0.019	0.889 ± 0.009	0.048 ± 0.009	0.860 ± 0.009	0.696 ± 0.030	0.925 ± 0.013
	5	0.326 ± 0.021	0.891 ± 0.009	0.038 ± 0.005	0.858 ± 0.008	0.698 ± 0.028	0.934 ± 0.014
	6	0.322 ± 0.018	0.891 ± 0.009	0.038 ± 0.007	0.863 ± 0.010	0.688 ± 0.026	0.927 ± 0.013
	7	0.328 ± 0.020	0.896 ± 0.008	0.045 ± 0.007	0.848 ± 0.009	0.693 ± 0.028	0.957 ± 0.010
	8	0.325 ± 0.022	0.884 ± 0.008	0.046 ± 0.010	0.834 ± 0.008	0.685 ± 0.032	0.949 ± 0.011
	9	0.332 ± 0.019	0.886 ± 0.011	0.045 ± 0.010	0.848 ± 0.009	0.701 ± 0.024	0.935 ± 0.015
	10	0.346 ± 0.017	0.889 ± 0.004	0.045 ± 0.010	0.846 ± 0.008	0.735 ± 0.021	0.944 ± 0.012
	11	0.361 ± 0.018	0.882 ± 0.009	0.045 ± 0.009	0.862 ± 0.010	0.767 ± 0.029	0.907 ± 0.014
	12	0.707 ± 0.014	**0.889 ± 0.005**	0.511 ± 0.016	**0.834 ± 0.005**	0.961 ± 0.008	**0.961 ± 0.010**
PhilTa	**1**	0.705 ± 0.015	**0.898 ± 0.008**	0.531 ± 0.016	**0.858 ± 0.010**	0.931 ± 0.012	**0.951 ± 0.011**
	2	0.626 ± 0.019	0.888 ± 0.010	0.410 ± 0.017	0.866 ± 0.010	0.906 ± 0.018	0.917 ± 0.017
	3	0.440 ± 0.013	0.893 ± 0.009	0.071 ± 0.014	0.858 ± 0.010	0.917 ± 0.013	0.939 ± 0.011
	4	0.421 ± 0.016	0.891 ± 0.009	0.027 ± 0.007	0.858 ± 0.008	0.928 ± 0.015	0.935 ± 0.013
	5	0.375 ± 0.016	0.895 ± 0.010	0.042 ± 0.013	0.852 ± 0.010	0.804 ± 0.018	0.950 ± 0.014
	6	0.337 ± 0.015	0.897 ± 0.009	0.060 ± 0.012	0.861 ± 0.010	0.696 ± 0.026	0.944 ± 0.013
	7	0.361 ± 0.017	0.900 ± 0.009	0.056 ± 0.012	0.860 ± 0.008	0.755 ± 0.020	0.952 ± 0.013
	8	0.389 ± 0.016	0.891 ± 0.010	0.058 ± 0.014	0.859 ± 0.009	0.818 ± 0.018	0.932 ± 0.015
	9	0.371 ± 0.016	0.893 ± 0.008	0.066 ± 0.014	0.861 ± 0.009	0.766 ± 0.027	0.934 ± 0.016
	10	0.377 ± 0.014	0.892 ± 0.009	0.066 ± 0.014	0.855 ± 0.005	0.778 ± 0.023	0.940 ± 0.017
	11	0.394 ± 0.014	0.888 ± 0.008	0.059 ± 0.010	0.851 ± 0.010	0.827 ± 0.021	0.935 ± 0.013
	12	0.566 ± 0.013	0.897 ± 0.010	0.282 ± 0.014	0.863 ± 0.010	0.934 ± 0.007	0.941 ± 0.018
mT5-base	1	0.487 ± 0.012	0.899 ± 0.005	0.100 ± 0.009	0.856 ± 0.007	0.987 ± 0.007	0.954 ± 0.011
	2	0.476 ± 0.012	**0.893 ± 0.005**	0.083 ± 0.008	**0.851 ± 0.012**	0.984 ± 0.008	**0.947 ± 0.013**
	3	0.483 ± 0.014	0.883 ± 0.011	0.124 ± 0.009	0.817 ± 0.014	0.947 ± 0.008	0.970 ± 0.007
	4	0.500 ± 0.015	0.852 ± 0.007	0.206 ± 0.006	0.772 ± 0.010	0.879 ± 0.020	0.956 ± 0.010
	5	0.371 ± 0.014	0.825 ± 0.008	0.155 ± 0.006	0.724 ± 0.007	0.651 ± 0.023	0.956 ± 0.006
	6	0.361 ± 0.012	0.810 ± 0.005	0.263 ± 0.021	0.765 ± 0.012	0.488 ± 0.018	0.868 ± 0.016
	7	0.370 ± 0.010	0.827 ± 0.008	0.255 ± 0.014	0.760 ± 0.011	0.519 ± 0.017	0.914 ± 0.013
	8	0.382 ± 0.008	0.824 ± 0.012	0.244 ± 0.007	0.773 ± 0.012	0.561 ± 0.017	0.891 ± 0.016
	9	0.428 ± 0.014	0.838 ± 0.010	0.171 ± 0.008	0.783 ± 0.013	0.759 ± 0.024	0.909 ± 0.006
	10	0.421 ± 0.010	0.821 ± 0.017	0.140 ± 0.011	0.767 ± 0.015	0.783 ± 0.021	0.892 ± 0.019
	11	0.420 ± 0.015	0.803 ± 0.012	0.131 ± 0.012	0.721 ± 0.013	0.792 ± 0.016	0.910 ± 0.015
	12	0.507 ± 0.009	0.841 ± 0.009	0.169 ± 0.007	0.772 ± 0.016	0.942 ± 0.011	0.931 ± 0.011
LaBSE	1	0.502 ± 0.012	0.858 ± 0.007	0.170 ± 0.003	0.838 ± 0.007	0.929 ± 0.016	0.885 ± 0.009
	2	0.523 ± 0.011	0.860 ± 0.004	0.274 ± 0.005	0.817 ± 0.003	0.844 ± 0.022	0.915 ± 0.008
	3	0.535 ± 0.010	0.847 ± 0.006	0.218 ± 0.009	0.831 ± 0.009	0.945 ± 0.008	0.867 ± 0.007
	4	0.569 ± 0.008	0.850 ± 0.006	0.299 ± 0.004	0.837 ± 0.006	0.917 ± 0.014	0.866 ± 0.011
	5	0.551 ± 0.012	0.845 ± 0.005	0.255 ± 0.012	0.794 ± 0.005	0.933 ± 0.013	0.911 ± 0.010
	6	0.661 ± 0.011	0.859 ± 0.004	0.461 ± 0.019	0.817 ± 0.008	0.919 ± 0.018	0.914 ± 0.015
	7	0.596 ± 0.010	0.849 ± 0.004	0.333 ± 0.016	0.808 ± 0.006	0.935 ± 0.016	0.902 ± 0.010

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Model	Layer	Routing: DirAcc@1 (five reseedings, mean ± std, SIF+ABTT)					
		DirAcc@1 (Base)	DirAcc@1 (SIF+ABTT)	Existing (Base)	Existing (SIF+ABTT)	New (Base)	New (SIF+ABTT)
	8	0.572 ± 0.012	0.847 ± 0.006	0.314 ± 0.019	0.802 ± 0.009	0.904 ± 0.016	0.905 ± 0.008
	9	0.631 ± 0.008	0.854 ± 0.007	0.381 ± 0.015	0.828 ± 0.002	0.954 ± 0.008	0.887 ± 0.012
	10	0.712 ± 0.016	0.858 ± 0.005	0.551 ± 0.019	0.836 ± 0.005	0.920 ± 0.015	0.887 ± 0.010
	11	0.820 ± 0.010	**0.906 ± 0.010**	0.731 ± 0.012	**0.875 ± 0.013**	0.935 ± 0.012	**0.944 ± 0.017**
	12	0.793 ± 0.015	0.884 ± 0.006	0.670 ± 0.019	0.850 ± 0.012	0.952 ± 0.013	0.927 ± 0.015
Qwen3-0.6B	1	0.550 ± 0.008	0.894 ± 0.005	0.223 ± 0.006	0.840 ± 0.010	0.972 ± 0.003	0.965 ± 0.011
	2	0.572 ± 0.015	0.882 ± 0.004	0.363 ± 0.014	0.828 ± 0.005	0.841 ± 0.032	0.952 ± 0.007
	3	0.541 ± 0.007	0.889 ± 0.006	0.216 ± 0.007	0.827 ± 0.008	0.962 ± 0.007	0.968 ± 0.011
	4	0.580 ± 0.012	0.891 ± 0.004	0.353 ± 0.008	0.845 ± 0.007	0.874 ± 0.028	0.950 ± 0.006
	5	0.487 ± 0.017	**0.893 ± 0.007**	0.133 ± 0.012	**0.839 ± 0.011**	0.944 ± 0.014	**0.964 ± 0.009**
	6	0.568 ± 0.012	0.897 ± 0.005	0.350 ± 0.010	0.857 ± 0.008	0.850 ± 0.023	0.948 ± 0.012
	7	0.541 ± 0.007	0.902 ± 0.006	0.211 ± 0.009	0.862 ± 0.008	0.967 ± 0.012	0.953 ± 0.012
	8	0.609 ± 0.009	0.903 ± 0.006	0.362 ± 0.013	0.861 ± 0.010	0.929 ± 0.017	0.956 ± 0.005
	9	0.612 ± 0.013	0.900 ± 0.004	0.393 ± 0.016	0.868 ± 0.008	0.894 ± 0.021	0.943 ± 0.006
	10	0.638 ± 0.013	0.890 ± 0.009	0.452 ± 0.009	0.847 ± 0.014	0.878 ± 0.022	0.944 ± 0.010
	11	0.578 ± 0.006	0.885 ± 0.005	0.257 ± 0.010	0.838 ± 0.014	0.991 ± 0.005	0.947 ± 0.012
	12	0.627 ± 0.009	0.892 ± 0.005	0.366 ± 0.016	0.844 ± 0.010	0.964 ± 0.013	0.955 ± 0.007
	13	0.639 ± 0.010	0.887 ± 0.006	0.383 ± 0.014	0.849 ± 0.011	0.970 ± 0.015	0.937 ± 0.011
	14	0.646 ± 0.009	0.885 ± 0.006	0.402 ± 0.017	0.852 ± 0.009	0.960 ± 0.016	0.927 ± 0.007
	15	0.648 ± 0.012	0.885 ± 0.007	0.410 ± 0.017	0.852 ± 0.010	0.954 ± 0.017	0.929 ± 0.018
	16	0.691 ± 0.016	0.873 ± 0.005	0.546 ± 0.017	0.837 ± 0.009	0.878 ± 0.017	0.920 ± 0.016
	17	0.658 ± 0.010	0.862 ± 0.006	0.420 ± 0.018	0.807 ± 0.012	0.966 ± 0.013	0.933 ± 0.013
	18	0.709 ± 0.012	0.868 ± 0.002	0.503 ± 0.018	0.833 ± 0.013	0.975 ± 0.006	0.914 ± 0.017
	19	0.691 ± 0.010	0.874 ± 0.002	0.462 ± 0.018	0.860 ± 0.007	0.986 ± 0.006	0.893 ± 0.012
	20	0.743 ± 0.020	0.860 ± 0.004	0.588 ± 0.023	0.797 ± 0.011	0.943 ± 0.018	0.941 ± 0.010
	21	0.758 ± 0.017	0.879 ± 0.006	0.585 ± 0.021	0.836 ± 0.008	0.981 ± 0.010	0.935 ± 0.010
	22	0.799 ± 0.014	0.882 ± 0.006	0.671 ± 0.019	0.855 ± 0.002	0.965 ± 0.006	0.918 ± 0.015
	23	0.757 ± 0.012	0.887 ± 0.006	0.577 ± 0.014	0.857 ± 0.007	0.989 ± 0.004	0.927 ± 0.011
	24	0.810 ± 0.014	0.893 ± 0.005	0.695 ± 0.021	0.861 ± 0.004	0.959 ± 0.003	0.935 ± 0.011
	25	0.732 ± 0.011	0.890 ± 0.009	0.527 ± 0.016	0.866 ± 0.006	0.997 ± 0.003	0.920 ± 0.014
	26	0.749 ± 0.013	0.895 ± 0.008	0.562 ± 0.017	0.858 ± 0.004	0.991 ± 0.003	0.943 ± 0.015
	27	0.812 ± 0.004	0.881 ± 0.005	0.741 ± 0.008	0.824 ± 0.006	0.903 ± 0.011	0.954 ± 0.011
	28	0.824 ± 0.010	0.871 ± 0.006	0.736 ± 0.015	0.802 ± 0.010	0.939 ± 0.005	0.961 ± 0.007
KaLM-mini	**1**	0.596 ± 0.012	**0.893 ± 0.005**	0.344 ± 0.003	**0.846 ± 0.005**	0.922 ± 0.022	**0.953 ± 0.006**
	2	0.620 ± 0.018	0.893 ± 0.003	0.422 ± 0.014	0.845 ± 0.005	0.876 ± 0.023	0.955 ± 0.008
	3	0.605 ± 0.011	0.893 ± 0.004	0.379 ± 0.008	0.843 ± 0.000	0.896 ± 0.018	0.959 ± 0.010
	4	0.562 ± 0.015	0.889 ± 0.005	0.265 ± 0.009	0.842 ± 0.008	0.945 ± 0.016	0.949 ± 0.008
	5	0.550 ± 0.013	0.890 ± 0.004	0.270 ± 0.005	0.841 ± 0.004	0.910 ± 0.019	0.952 ± 0.009
	6	0.522 ± 0.016	0.893 ± 0.004	0.203 ± 0.012	0.850 ± 0.005	0.934 ± 0.014	0.949 ± 0.014
	7	0.604 ± 0.018	0.893 ± 0.003	0.393 ± 0.018	0.840 ± 0.008	0.876 ± 0.023	0.962 ± 0.008
	8	0.607 ± 0.014	0.866 ± 0.003	0.388 ± 0.011	0.798 ± 0.008	0.891 ± 0.017	0.954 ± 0.007
	9	0.614 ± 0.011	0.874 ± 0.005	0.387 ± 0.008	0.821 ± 0.008	0.908 ± 0.013	0.943 ± 0.008
	10	0.602 ± 0.013	0.859 ± 0.008	0.360 ± 0.005	0.774 ± 0.010	0.916 ± 0.017	0.969 ± 0.010
	11	0.601 ± 0.012	0.864 ± 0.008	0.361 ± 0.007	0.792 ± 0.010	0.911 ± 0.020	0.956 ± 0.008
	12	0.640 ± 0.010	0.868 ± 0.006	0.416 ± 0.006	0.801 ± 0.007	0.929 ± 0.019	0.954 ± 0.006
	13	0.645 ± 0.011	0.862 ± 0.005	0.436 ± 0.005	0.793 ± 0.004	0.915 ± 0.018	0.951 ± 0.012
	14	0.654 ± 0.010	0.859 ± 0.006	0.479 ± 0.004	0.795 ± 0.008	0.880 ± 0.017	0.941 ± 0.011
	15	0.661 ± 0.009	0.870 ± 0.006	0.460 ± 0.009	0.814 ± 0.009	0.921 ± 0.014	0.941 ± 0.006
	16	0.685 ± 0.008	0.870 ± 0.007	0.477 ± 0.009	0.824 ± 0.008	0.953 ± 0.015	0.930 ± 0.008
	17	0.757 ± 0.006	0.886 ± 0.010	0.612 ± 0.016	0.841 ± 0.005	0.944 ± 0.010	0.943 ± 0.016
	18	0.781 ± 0.008	0.880 ± 0.008	0.636 ± 0.013	0.840 ± 0.002	0.967 ± 0.007	0.931 ± 0.016
	19	0.805 ± 0.012	0.877 ± 0.008	0.673 ± 0.020	0.831 ± 0.007	0.976 ± 0.003	0.935 ± 0.011
	20	0.794 ± 0.010	0.878 ± 0.006	0.650 ± 0.018	0.831 ± 0.003	0.979 ± 0.004	0.937 ± 0.011
	21	0.805 ± 0.010	0.871 ± 0.007	0.673 ± 0.016	0.831 ± 0.004	0.975 ± 0.006	0.924 ± 0.012
	22	0.871 ± 0.006	0.873 ± 0.006	0.811 ± 0.005	0.813 ± 0.007	0.949 ± 0.006	0.951 ± 0.007
	23	0.873 ± 0.007	0.885 ± 0.005	0.811 ± 0.006	0.835 ± 0.007	0.953 ± 0.007	0.950 ± 0.006

24	0.848 ± 0.009	0.856 ± 0.004	0.749 ± 0.017	0.807 ± 0.006	0.975 ± 0.007	0.918 ± 0.009

Table 33: Per-layer routing DirAcc@1, overall and for existing and new witnesses, across all six models, averaged over five reseedings of the query and reference files, with the training/test split fixed (mean ± std). SIF+ABTT applies SIF weighting plus ABTT with D tuned per layer on the training split. Accuracies are fractions (0.885 is 88.5%). This table covers SIF+ABTT; five-reseeding ABTT-only values for LaTa, Qwen3-0.6B and KaLM-mini, at their train-selected layers, are in Table 19 (Appendix H). Overall assignment accuracy coincides with DirAcc@1 under the five reseedings, so we report the existing/new decomposition instead. Bold marks the layer reported per model and its SIF+ABTT values, the row printed in Table 29; the Base values in that row stay plain. For each model the reported layer is chosen on the training split: the layer with the highest training-set DirAcc@1 for SIF+ABTT in the single-seed run, the layer behind the SIF+ABTT cell of Table 16, listed in Table 21. D is tuned per layer on the training split. This is not always the layer with the highest mean over the five reseedings.

L Cluster Geometry

This appendix shows the effect of ABTT on the whole corpus at the routing layers of Table 2. Figures 10 and 11 give the t-SNE (van der Maaten and Hinton, 2008) and UMAP (McInnes et al., 2018) panels for LaTa, PhilTa, and mT5-base, and Figures 12 and 13 extend them to LaBSE, Qwen3-0.6B, and KaLM-mini. All four draw all 1,705 witnesses, color the same six largest directories, and compare baseline mean pooling with ABTT-only at $D = 10$. Each model sits at the layer chosen on the training split by DirAcc@1 under ABTT, the routing ABTT layer of Table 21. In both maps, ABTT draws the witnesses of each colored directory into tighter groups than baseline mean pooling does. The maps are two-dimensional, so we read them as illustrations and compute no statistic on them.

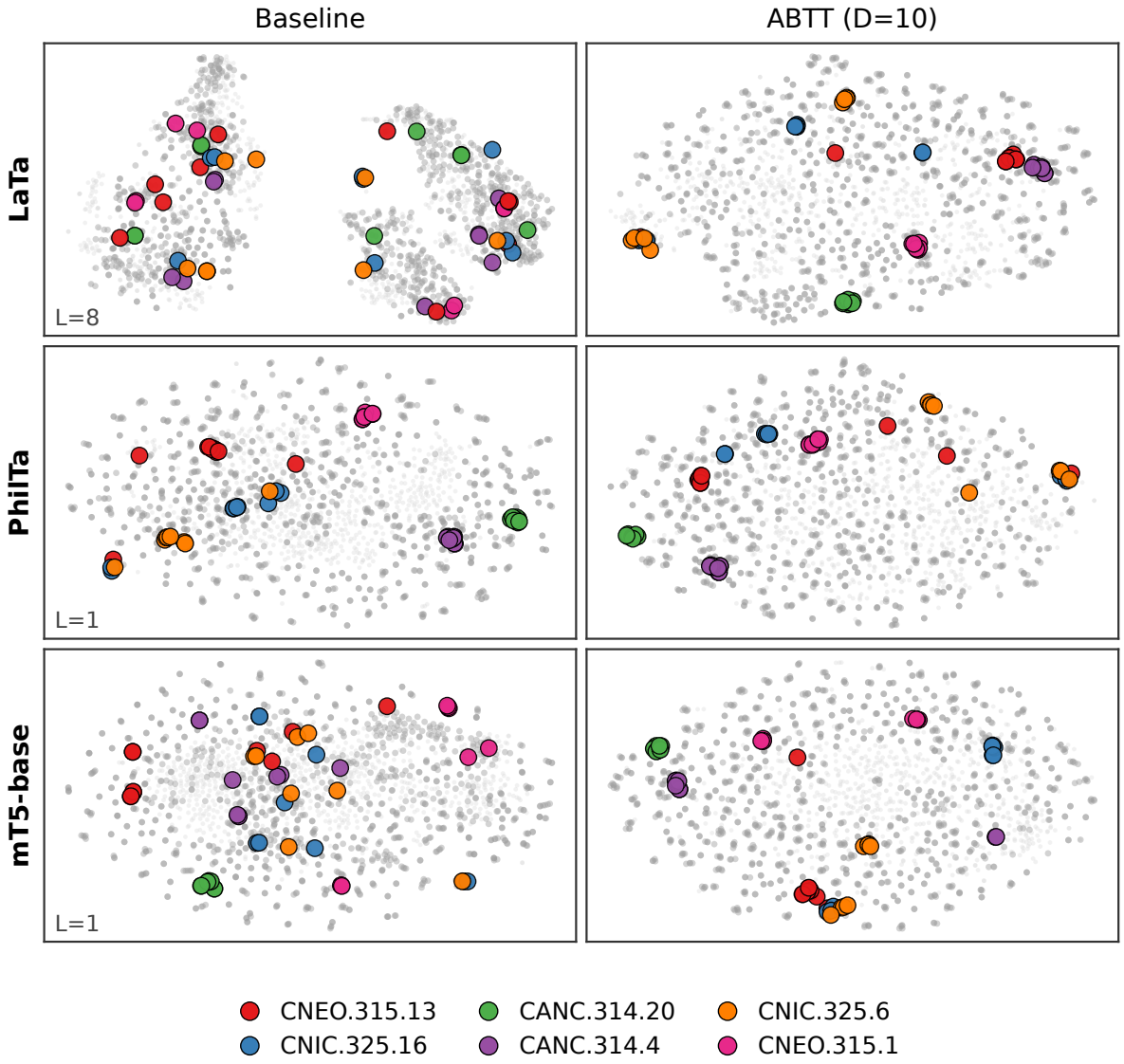


Figure 10: t-SNE maps of all 1,705 manuscript witnesses for LaTa, PhilTa, and mT5-base, each at the layer chosen on the training split: the layer with the highest training-set DirAcc@1 under ABTT, printed in each panel as L and equal to the routing ABTT layer in Table 21. Columns compare baseline mean pooling with ABTT-only using $D = 10$, fit on the training embeddings. Colors mark the six largest directories; other witnesses of directories with at least two witnesses are mid gray, and singletons light gray.

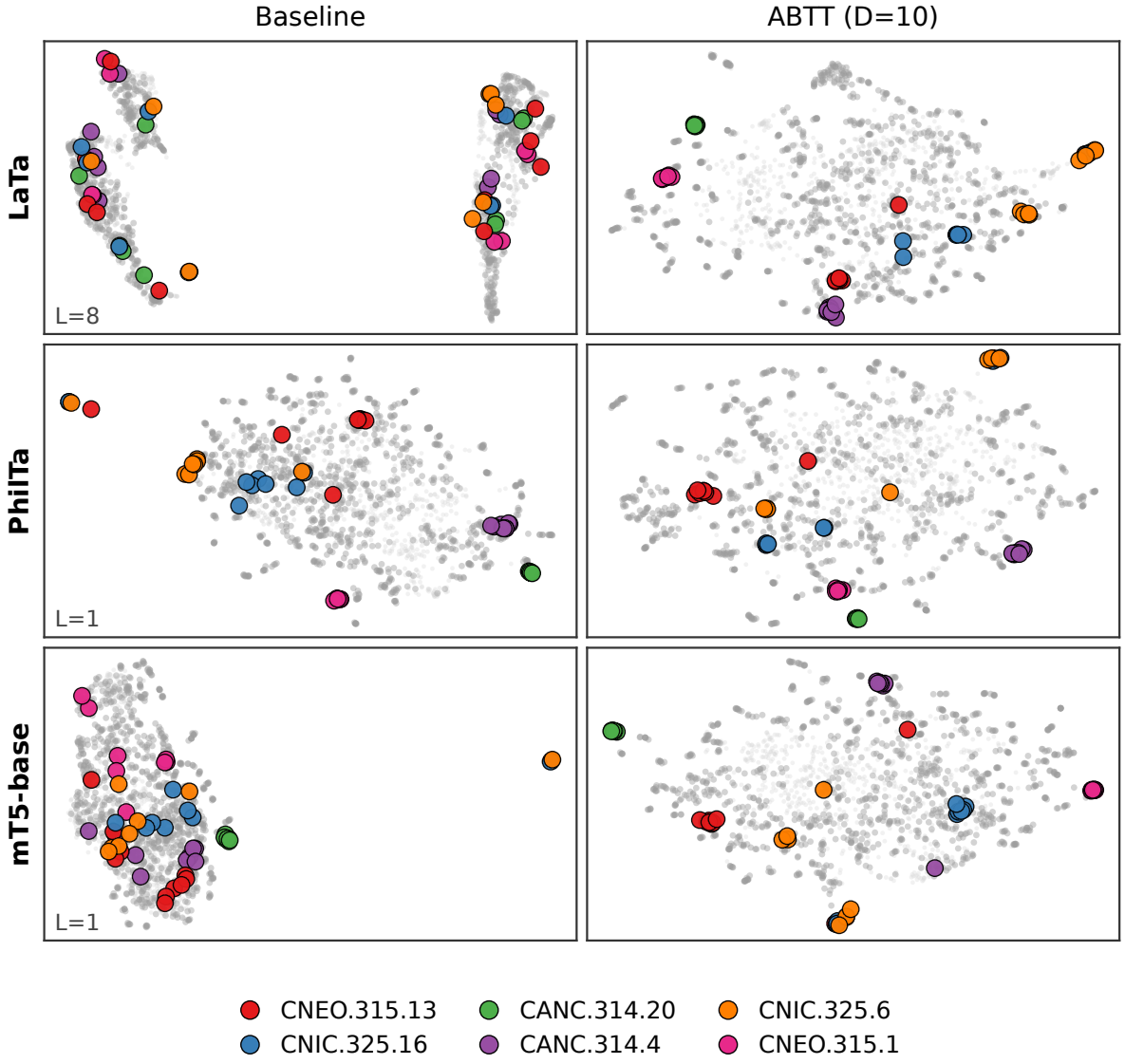


Figure 11: UMAP maps with the same configuration as Figure 10. The same cluster-tightening pattern appears under a different mapping method.

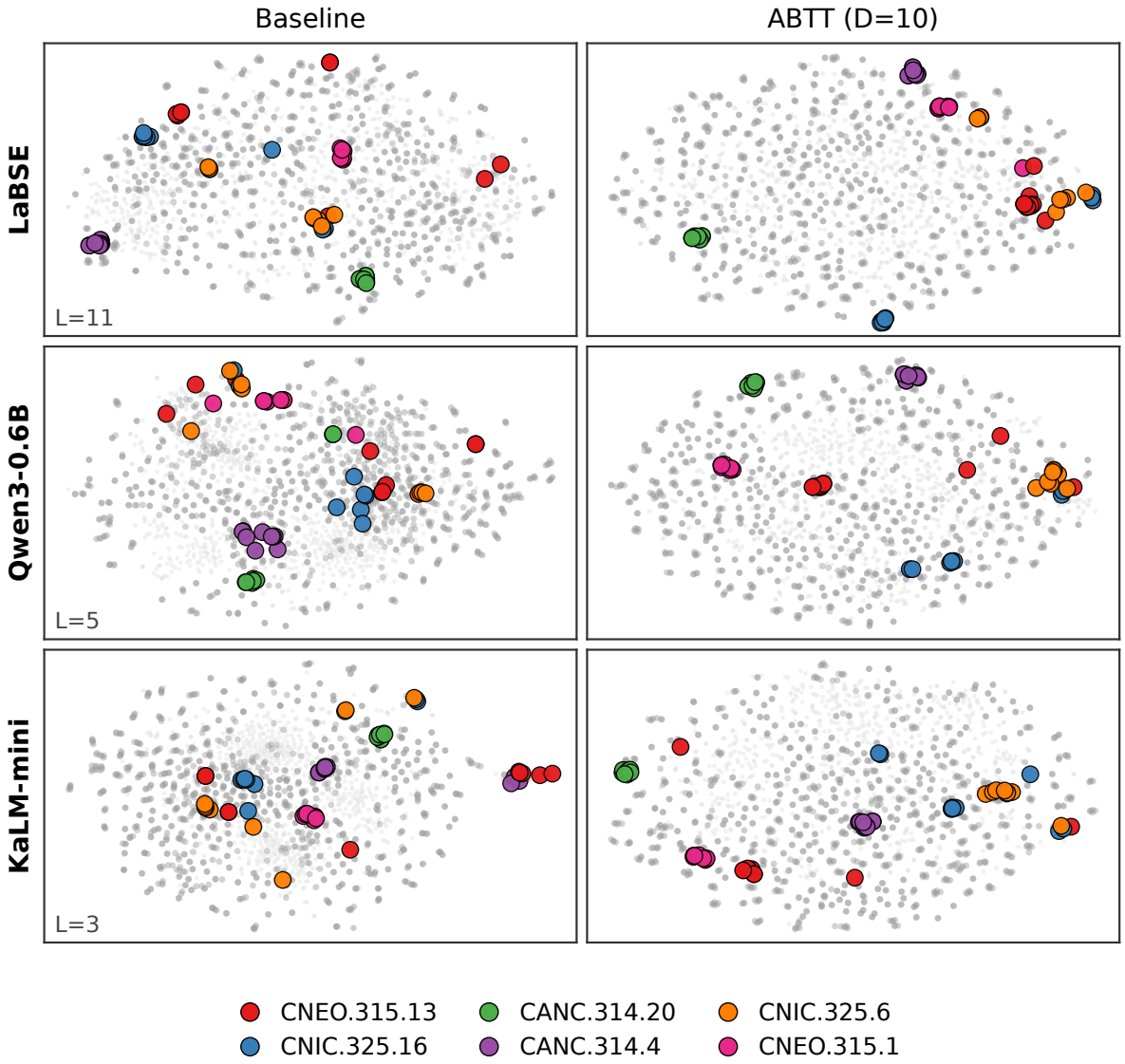


Figure 12: t-SNE maps of all 1,705 manuscript witnesses for LaBSE, Qwen3-0.6B, and KaLM-mini under the layer rule of Figure 10: the layer with the highest training-set DirAcc@1 under ABTT, printed in each panel and equal to the routing ABTT layer in Table 21. Columns compare baseline mean pooling with ABTT-only using $D = 10$. Colors mark the same six largest directories as Figure 10.

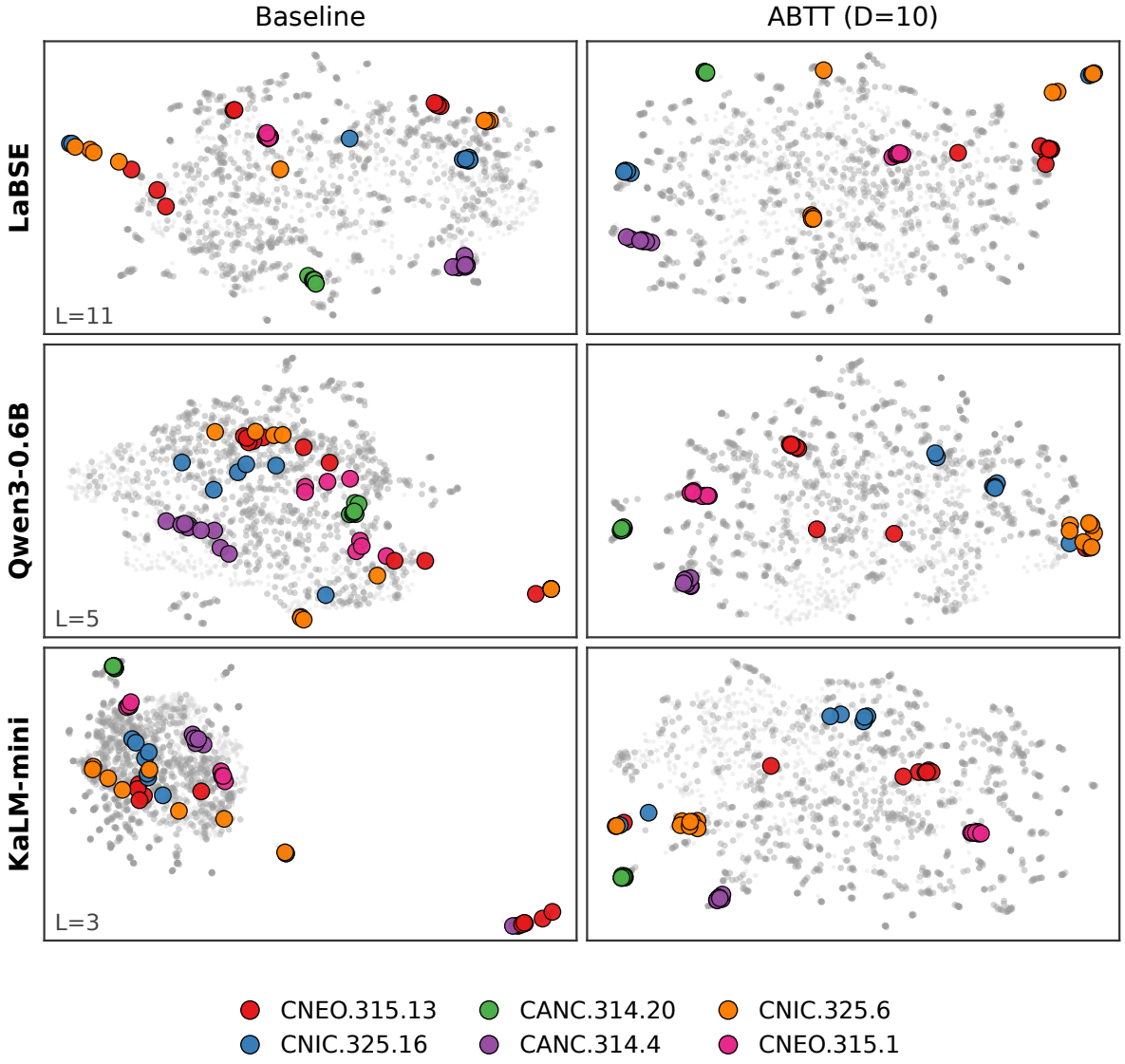


Figure 13: UMAP maps matching Figure 12. The cluster-tightening pattern appears under a second mapping method.